

The Homotopy Theory of Operator Algebras

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July 18, 2026

Abstract

In this thesis, we undertake a systematic study of the homotopy theory of operator algebras, focusing on the category of equivariant C^* -algebras and Banach algebras. The first part of the thesis (Section 2) mainly focuses on the classical aspects of homotopy theory in these categories, setting the stage for the later development of abstract homotopy theory. We also address several open problems in the homotopy theory of C^* -algebras. Firstly, we introduce a new class of C^* -algebras, the Schochet C^* -algebras, in relation to a conjecture of Schochet about the Calkin projection on an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space & we manage to resolve this conjecture in the positive. Secondly, we address a question of Schochet & Kaminker on the homotopy invariance of the Brown-Douglas-Fillmore Ext groups and resolve this in the negative. Thirdly, we also managed to address a question of Conway's with regards to the Schochet property being satisfied by certain spaces. In the process, we also proved some other new results which shed light on related problems, including those related to semiprojectivity. Finally, we were able to construct an obstruction to a conjecture of Blackadar's on approximately semiprojective C^* -algebras.

In the second part of the thesis (Section 3 and Section 4), we move on to abstract homotopy theory and introduce structures, known as categories of fibrant objects, which model homotopy theory on the category of C^* -algebras and Banach algebras. Such structures were already known in the literature but we explicitly use them to address another question of Schochet's regarding an appropriate abstract framework for studying (co)homology theories on C^* -algebras. In particular, we use these structures to refine Cuntz's proof of the Bott Periodicity Theorem, proving it for some interesting subcategories in a unified way. We also extend the theory of categories of fibrant objects, explicitly proving some results which are analogues of known results for model categories. Finally, we also highlight an important result about categories of fibrant objects, the Spanier-Whitehead construction, which says that there exists a stable homotopy category associated to every category of fibrant objects and that it has a triangulated structure. This result was used without formal proof in the literature and we close this gap by using tools from higher category theory.

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1 Introduction

In this introduction, we will provide information relevant to this thesis in the following order:

1. The abstract.
2. Our overarching goals.
3. Prerequisites and reading recommendations.
4. Conventions & general structure.
5. Acknowledgements.

We will try to be detailed and as complete as possible.

1.1 Overarching Goals

1.2 Prerequisites & Reading Recommendations

Due to the fact that some parts of this thesis involve new results, it will not always be possible to write in a self-contained way. For full appreciation of everything written here, we will need basic background in functional analysis, operator algebras, abstract harmonic analysis and category theory. Here are some reading recommendations for these topics:

Functional Analysis: One can certainly use [56] because it is extremely well-written.

Operator Algebras: [51] is a good reference but we think that [36] is a better resource to learn from as a beginner. It also contains a nice introduction to the necessary concepts from functional analysis.

Harmonic Analysis: The resource [23] is very well-written.

Category Theory: [62] is accessible but lacks examples from functional analysis and these might be very interesting for readers who would like an early glimpse into the relevance of category theory towards analytic topics. For that, it is a good idea to look into the rather accessible resource [38]; this also doubles up as an accessible treatment of functional analysis.

At some points in the text, we rely on some prior exposure to homotopy theory and, in particular, ∞ -category theory. However, this is not strictly necessary since we do not use any major results from classical homotopy theory on spaces and much of the ∞ -category theory we need can be understood synthetically. We will provide reading directions as needed.

1.3 Conventions & General Structure

Most of the structure of the thesis is clear/self-explanatory. We have tried to explain notation and conventions where necessary, even if some of it is standard. There are some conventions that may be used throughout the thesis and are included in the tables below for the sake of clarity. Most of our choices have been made so that these conventions are “obvious”. Nevertheless, we have them here for easy reference in the future.

General Mathematical Notation

$\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

$\mathbb{F}$ denotes any field.

$\mathbb{N}$ denotes the set of natural numbers (excluding 0)

$\mathbb{N}_0$ is the set of natural numbers including 0

$\mathbb{R}_0^+$ denotes the set of all non-negative real numbers.

$\mathbb{R}^+$ denotes the set of all positive real numbers.

$\mathcal{K}$ denotes the C^* -algebra of compact operators on an infinite-dimensional separable $\mathbb{C}$ -Hilbert space.

If X, Y are sets, then $\text{Maps}(X, Y)$ is the set of all functions from X to Y .

If G is a topological group, then G is a LCH group iff it is a locally compact Hausdorff group.

If A, B are mathematical objects, then $A \stackrel{\text{def}}{=} B$ means that A is defined to be equal to B .

If A, B are statements, then $A \stackrel{\text{def}}{\Leftrightarrow} B$ means that A is defined to be equivalent to B .

If A is a C^* -algebra, then $\preceq_A$ denotes the natural order on A . We will usually omit the subscript.

If $G \in \text{Grp}$ and $g \in G$, then u_g is the notation for the canonical unitaries in $C_{\max}^*(G)$ or $C_{\text{red}}^*(G)$.

Here are some common acronyms we'll use.

Common Acronyms

WLOG - Without loss of generality.

TFAE - The following are equivalent.

TFSH - The following statements hold.

NCG - Noncommutative Geometry

nbhd - neighborhood.

SOT - strong-operator topology (particularly on $\mathbb{C}$ -Hilbert spaces).

WOT - weak-operator topology (particularly on $\mathbb{C}$ -Hilbert spaces)

Here are some of the categories we'll use.

Some Categories

If $\mathbb{F}$ is a field, then $\text{Vec}_{\mathbb{F}}$ is the category of $\mathbb{F}$ -vector spaces.

If $\mathbb{F}$ is a field, then $\text{Alg}_{\mathbb{F}}$ is the category of $\mathbb{F}$ -algebras.

Top is the category of topological spaces.

${}^{\text{CH}}\text{Top}$ is the category of compact Hausdorff spaces.

TopGrp is the category of topological groups.

$C^*\text{Alg}$ is the category of C^* -algebras.

${}_{\omega}C^*\text{Alg}$ is the category of separable C^* -algebras.

If G is a group (topological or otherwise), then BG is its associated groupoid.

Here are some bits of categorical notation.

Categorical Notation

If $\mathcal{C}$ is a category and x is an object in $\mathcal{C}$, then we denote that by writing $x \in \mathcal{C}$.

If $\mathcal{C}$ is a category, then $\text{Mor}(\mathcal{C})$ refers to the set^a of all morphisms of $\mathcal{C}$.

If $\mathcal{C}$ is a category and $x, y \in \mathcal{C}$, then $\mathcal{C}(x, y)$ is the set of all morphisms from x to y . Alternatively, we may also sometimes use $\mathcal{C}(x, y)$ when it is more convenient. The only exception is when $\mathcal{C}$ is Set .

If $\mathcal{C}$ is a category and $f \in \text{Mor}(\mathcal{C})$, then $\text{dom}(f)$ is the domain of f and $\text{codom}(f)$ is its codomain.

If $\mathcal{C}$ is a category and $x \in \mathcal{C}$, then $\text{Aut}(x)$ is the set of all isomorphisms with (co)domain x .

If $\mathcal{C}$ is a category and $x \in \mathcal{C}$, then $x/\mathcal{C}$ is the slice category under x .

If $\mathcal{C}$ is a category and $x \in \mathcal{C}$, then $\mathcal{C}/x$ is the slice category over x .

If $\mathcal{C}$ is a category, then $Y^0 : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$ is the covariant Yoneda embedding.

If $\mathcal{C}$ is a category, then $Y_0 : \mathcal{C}^{\text{op}} \rightarrow \text{Fun}(\mathcal{C}, \text{Set})$ is the contravariant Yoneda embedding.

If $\mathcal{C}$ is a pointed category, then the zero object will typically be denoted by 0 unless otherwise stated.

^aThis will make sense through the notion of a Grothendieck universe.

Here are a few further points that are worth noting:

1. Whenever we define a new category, we will either present it as an example or as a construction.

2. Also, regarding the slice categories specifically, we will sometimes use that notation to simply talk about the morphisms with (co)domain x , without actually assuming further categorical structure.
3. Unless we say otherwise, diagrams with solid arrows represent given data while the dotted/dashed arrows will always represent data whose existence we can prove.
4. We will use the word “quantization” many times in the thesis. This is not to be interpreted physically. It is simply an indication that we are taking a standardized notion from topology, measure theory or geometry & we’re bringing it into the realm of C^* -algebras in the spirit of noncommutative geometry.

Finally, we also mention that there are several places where we have labeled certain definitions/results with the name of the author. This has only been done just for the sake of clarity about our contributions. In other places where we reprove results or simply extend what is in the literature, we do state that very clearly.

1.4 Acknowledgements

My deepest gratitude goes to Prof. Dr. Sven Raum for his constant guidance and support. His knowledge, open-mindedness and willingness to discuss anything & everything of a mathematical nature has been extremely inspirational to me. I also want to extend my thanks to the operator algebras workgroup in Potsdam, who have been nothing but gracious & welcoming. I find myself incredibly lucky to have interacted with academics of such diverse interests over the past two years and look forward to continuing to do so in the years to come.

To my best friends, for their unconditional love & care. To *Bogdan*, for the beautiful times we have spent arguing about mathematics, philosophy and really, anything under (or over) the Sun. I will forever treasure our most intense conversations, having learned much about myself from them. To *Vlad*, for the care & love you have shown to me, even when I have not shown it to myself. Your honesty and directness have always been a blessing to me. To *Kevin*, for being the oil to my water. Thank you for the constant tug-of-war that has me constantly re-evaluating my beliefs & adjusting them to become a better person. To my friends from Lund and Singapore, both of whom were instrumental in shaping my views and who serve as a constant tempering presence in my life today. Special thanks goes to Carl-Fredrik Lidgren for assisting me with questions about stable homotopy theory & higher categories, some of which were helpful for work in this thesis. I also thank Senan Sekhon for the beautiful template used in this thesis.

Finally, I should express my infinite gratitude to my family. None of this would have been possible without them. To my mother and father, for their infinite willpower and constant efforts to assist me in living my dreams. To my brother, Ashish, who has accompanied me through life & is the most important person in the world to me.

1.5 A Word About Foundations

We do not want to make a big deal out of size issues but we do need to make a big deal out of them since we will eventually be dealing with accessibility and presentability of ∞ -categories. The discussion here is of independent interest to set theorists, of course, but it is applicable for our work in the aforementioned direction. Even though this section is short, we would recommend coming back to it later when it actually is referenced; we assume basic familiarity with cardinals and cardinal arithmetic. The key tool that is used in the formalization of category theory is the notion of a *Grothendieck universe*. We provide the definition for the convenience of the reader.

Definition 1.1

A Grothendieck universe is a set U such that:

1. If $S \in U$ and $x \in S$, then $x \in U$.
2. If $S \in U$, then $\mathcal{P}(S) \in U$.
3. $\mathbb{N} \in U$.
4. If $I \in U$ and $f : I \rightarrow U$ is a map, then $\bigcup_{i \in I} f(i) \in U$.

If U is a Grothendieck universe, then its elements are known as U -small sets.

Here is a remark about this definition.

Remark 1.2. Note that the use of $\mathbb{N}$ presumes some assumption about the existence of Peano arithmetic. So, one would need to postulate that the Peano axioms hold or one would have to postulate the Axiom of Infinity in some form (from which we can prove the existence of $\mathbb{N}$ and the arithmetic that comes with it).

Now, one can state an additional axiom of interest for the purposes of foundations.

Axiom 1.3. The Existence of Grothendieck Universes

Let x be a set. Then, there exists a Grothendieck universe that contains x .

This axiom is a powerful one and effectively allows us to engage with categories (or mathematical structures, in general) of arbitrary size. Indeed, the connection with size issues needs to be further expanded upon and for this purpose, we introduce some terminology involving cardinal numbers.

Terminology 1.4. Let κ be a cardinal and let S be a set. S is κ -finite iff:

$$|S| < \kappa.$$

κ is regular iff it is infinite and for any κ -finite indexed family $(S_i)_{i \in I}$ of κ -finite sets, we have that $\bigcup_{i \in I} S_i$ is κ -finite.

Indeed, the countable cardinal is clearly regular; this follows from the definition.

2 Homotopical Constructions of Equivariant Operator Algebras

Summary & Motivation

In Section 2.1, we will introduce the basic language of homotopy theory on categories of equivariant operator algebras; we will also introduce these categories explicitly. This is a, more or less, straightforward extension of known constructions in the non-equivariant situation, though there are some original observations towards the end of Section 2.1. In Section 2.2, we will focus our attention towards resolving two problems in the literature, introducing a new class of interesting C^* -algebras in the process and proving some new facts about this class.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

G will denote a fixed topological group.

$\mathcal{A}$ will denote a fixed abelian category.

2.1 Homotopies & Fibrations of Equivariant C^* -Algebras & Banach Algebras

Let's start with the definition of an equivariant C^* -algebra.

Definition 2.1

Let $G \in \text{TopGrp}$. A G - C^* -algebra is a pair (A, α) consisting of:

1. A C^* -algebra A .
2. A continuous group action $\alpha : G \times A \rightarrow A$ such that for each $g \in G$, the map:

$$\alpha(g, \cdot) : A \rightarrow A,$$

is a $\star$ -automorphism. We refer to this as a continuous action by G on A via automorphisms.

If $(A, \alpha), (B, \beta)$ are G - C^* -algebras, then a morphism:

$$\phi : (A, \alpha) \rightarrow (B, \beta),$$

is a $\star$ -homomorphism $\phi : A \rightarrow B$ such that for all $g \in G$, the following diagram:

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \alpha_g \downarrow & & \downarrow \beta_g \\ A & \xrightarrow{\phi} & B \end{array}$$

commutes. ϕ is, then, said to be a G -equivariant $\star$ -homomorphism.

Let us obtain a separate characterization of (2) as stipulated in the definition.

Proposition 2.2

Let $G \in \text{TopGrp}$, let $A \in C^*\text{Alg}$ and let $\alpha : G \times A \rightarrow A$ be a map. For each $g \in G$, define:

$$\beta_g : A \rightarrow A, \quad a \mapsto \beta_g(a) \stackrel{\text{def}}{=} g \cdot a.$$

This defines a map $\beta : G \rightarrow \text{Maps}(A, A)$. TFAE:

1. (A, α) is a G - C^* -algebra.
2. β is a group homomorphism into $\text{Aut}(A)$ and for each $a \in A$, the map:

$$G \ni g \mapsto \beta_g(a) \in A,$$

is continuous at the identity.

3. β is a group homomorphism into $\text{Aut}(A)$ and for each $a \in A$, the map:

$$G \ni g \mapsto \beta_g(a) \in A,$$

is continuous.

Proof.

(1) $\Rightarrow$ (2): The fact that β is a group homomorphism follows from the fact that α is an action. Next, let $(g_\delta)_{\delta \in D}$ be a net in G that converges to 1 and let $a \in A$. Then, $g_\delta a \rightarrow a$ means that $\alpha(g_\delta, a) \rightarrow \alpha(1, a)$ but this holds because α is continuous.

(2) $\Rightarrow$ (3): It's sufficient to prove continuity; let $(g_\delta)_{\delta \in D}$ be a net in G that converges to some $g \in G$. Then, $g^{-1}g_\delta \rightarrow 1$ so we see that:

$$\beta_{g^{-1}g_\delta}(a) \rightarrow a.$$

Now, we have the following computation:

$$\begin{aligned} \forall \delta \in D : \|\beta_{g_\delta}(a) - \beta_g(a)\| &= \|\beta_g(\beta_{g^{-1}g_\delta}(a)) - \beta_g(a)\| \\ &\leq \|\beta_{g^{-1}g_\delta}(a) - a\| \\ &= \|\beta_{g^{-1}g_\delta}(a) - a\|. \end{aligned}$$

So, the last term goes to 0 in the limit and we're done.

(3) $\Rightarrow$ (1): Let $((g_\delta, a_\delta)_{\delta \in D})$ be a net in $G \times A$ that converges to some $(g, a) \in G \times A$. Then, we have a calculation:

$$\begin{aligned} \forall \delta \in D : \|\alpha(g_\delta, a_\delta) - \alpha(g, a)\| &= \|\alpha(g_\delta, a_\delta) - \alpha(g_\delta, a) + \alpha(g_\delta, a) - \alpha(g, a)\| \\ &\leq \|\alpha(g_\delta, a_\delta) - \alpha(g_\delta, a)\| + \|\alpha(g_\delta, a) - \alpha(g, a)\| \\ &= \|\beta_{g_\delta}(a_\delta) - \beta_{g_\delta}(a)\| + \|\beta_{g_\delta}(a) - \beta_g(a)\| \\ &= \|\beta_{g_\delta}(a_\delta - a)\| + \|(\beta_{g_\delta} - \beta_g)(a)\| \\ &\leq \|\beta_{g_\delta g^{-1}}\beta_g(a_\delta - a)\| + \|\beta_{g_\delta}(a) - \beta_g(a)\| \\ &\leq \|\beta_g(a_\delta - a)\| + \|\beta_{g_\delta}(a) - \beta_g(a)\|, \end{aligned}$$

where we've used the fact that $\star$ -homomorphisms are contractive a few times towards the end of the calculation above. Obviously, the last expression goes to 0 when δ is "large" enough, so we're done. ■

In the next remark, we will comment on the definition and the result for Banach algebras.

Remark 2.3. Let $G \in \text{TopGrp}$ and let $A \in \text{BanAlg}$. Then, Definition 2.1 goes through without much modification if we wish to define G -Banach algebras; we simply need to require that the map $\alpha(g, \cdot)$ is an automorphism of $\mathbb{C}$ -Banach algebras. Moreover, morphisms between G -Banach algebras are, now, just continuous algebra homomorphisms such that the exact same diagram commutes. However, Proposition 2.2 no longer goes through in general. Indeed, the argument we used for (3) $\Rightarrow$ (1) does not work since we used the contractivity of $\star$ -homomorphisms between C^* -algebras. If we, however, restrict ourselves to $\text{BanAlg}_{\leq 1}$, then the argument certainly works.

In light of this result, it is more convenient for us to adopt the following notation.

Notation 2.4. Let $G \in {}^{\text{LCH}}\text{TopGrp}$. From now on, a G - C^* -algebra is a pair:

$$(A, \alpha),$$

of a C^* -algebra A and a group homomorphism $\alpha : G \rightarrow \text{Aut}(A)$ such that for all $a \in A$, the map:

$$\beta_a : G \rightarrow A, g \mapsto \beta_a(g) \stackrel{\text{def}}{=} \alpha(g)(a),$$

is continuous.

Let us form a category out of this data.

Example 2.5. Let $G \in \text{TopGrp}$. We form a structure G - $C^*\text{Alg}$ as follows:

1. The objects are G - C^* -algebras.
2. The morphisms are G -equivariant $\star$ -homomorphisms, with the usual composition.

We have the following claim.

Claim. This structure forms a category.

The identity map on any G - C^* -algebra is an equivariant $\star$ -homomorphism. Moreover, equivariance is preserved under composition and since $\star$ -homomorphisms compose to form $\star$ -homomorphisms, it follows that the composition of two equivariant $\star$ -homomorphisms is an equivariant $\star$ -homomorphism.

This structure forms a category; it is the category of G - C^* -algebras. We can also define the equivariant categories:

$$G\text{-BanAlg}, G\text{-BanAlg}_{\leq 1},$$

of G -Banach algebras, with the main difference between them being the morphisms.

Let us present a number of standard homotopical constructions which will be useful in the next section.

Construction 2.6. Let $A \in G\text{-}C^*\text{Alg}$. Define:

$$\text{Cyl}_G(A) \stackrel{\text{def}}{=} \mathcal{C}([0, 1], A),$$

$$\text{Cone}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(0) = 0\},$$

$$\text{Sus}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(1) = \phi(0) = 0\}.$$

These constructions also make sense when $A \in G\text{-BanAlg}$. In all of these cases, the various operations of interest are induced point-wise on elements of each of these sets. Moreover, we define:

$$\forall g \in G : \forall f \in \text{Cyl}_G(A) : \forall t \in [0, 1] : (g \cdot f)(t) \stackrel{\text{def}}{=} gf(t).$$

This action is also naturally induced on Cone_G and Sus_G . We name these constructs as follows:

1. $\text{Cyl}_G(A)$ is the (equivariant) cylinder of A .
2. $\text{Cone}_G(A)$ is the (equivariant) cone of A .
3. $\text{Sus}_G(A)$ is the (equivariant) suspension of A .

We have the following claim.

Claim 2.1. Let $A \in G\text{-}C^*\text{Alg}$. Then:

$$\text{Cyl}_G(A), \text{Cone}_G(A), \text{Sus}_G(A),$$

are G - C^* -algebras. The same proof goes through when $A \in G\text{-BanAlg}$.

See Page 95 for the proof. Moreover, we define for each $t \in [0, 1]$ a map:

$$p_t^A : \text{Cyl}_G(A) \rightarrow A, \phi \mapsto p_t^A(\phi) \stackrel{\text{def}}{=} \phi(t).$$

This is known as the t -th canonical inclusion^a or the evaluation map at^b t .

Claim 2.2. For each $t \in [0, 1]$ and $A \in G\text{-}C^*\text{Alg}$, the map p_t^A is an equivariant $\star$ -homomorphism. When $A \in G\text{-BanAlg}$, then this is an equivariant continuous algebra homomorphism.

See Page 95 for the proof. The evaluation map is useful specifically in providing an alternative characterization of two equivariant $\star$ -homomorphisms being homotopic; see Proposition 2.19.

^aWe'll assume that the cylinder of A is always equipped with this family of maps.

^bThis terminology makes sense when $A \in G\text{-BanAlg}$ without any change.

In the next few examples, we will construct functors associated with the cylinder, cone & suspension.

Example 2.7. Define a map of categories:

$$\text{Cyl}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cyl}_G(A)$ is just the cylinder of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Cyl}_G(f) : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B), \quad \phi \mapsto \text{Cyl}_G(f)(\phi) \stackrel{\text{def}}{=} f \circ \phi.$$

It's easily checked that this is well-defined.

Let us show that this is a functor:

1. Let $A \in G\text{-C}^*\text{Alg}$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(\text{Id}_A)(\phi) &= \text{Id}_A \circ \phi \\ &= \phi \\ &= \text{Id}_{\text{Cyl}_G(A)}(\phi), \end{aligned}$$

so $\text{Cyl}_G(\text{Id}_A) = \text{Id}_{\text{Cyl}_G(A)}$.

2. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $f \in G\text{-C}^*\text{Alg}(A, B)$ and $g \in G\text{-C}^*\text{Alg}(B, C)$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(g \circ f)(\phi) &= (g \circ f) \circ \phi \\ &= g \circ (f \circ \phi) \\ &= g \circ \text{Cyl}_G(f)(\phi) \\ &= \text{Cyl}_G(g)(\text{Cyl}_G(f)(\phi)) \\ &= (\text{Cyl}_G(g) \circ \text{Cyl}_G(f))(\phi), \end{aligned}$$

from which we conclude that $\text{Cyl}_G(g \circ f) = \text{Cyl}_G(g) \circ \text{Cyl}_G(f)$.

This proves that this is a functor; we will refer to it as the cylinder functor.

Example 2.8. Define a map of categories:

$$\text{Cone}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cone}_G(A)$ is the cone of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f \in G\text{-C}^*\text{Alg}(A, B)$, then we define:

$$\text{Cone}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Cone}_G(A)}.$$

Now, we need to check that $\text{Cone}_G(f)$ actually maps into $\text{Cone}_G(B)$; let $\phi \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \text{Cone}_G(f)(\phi)(0) &= (f \circ \phi)(0) \\ &= f(\phi(0)) \\ &= f(0) \\ &= 0, \end{aligned}$$

as was desired.

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the cone (functor).

Example 2.9. Define a map of categories:

$$\text{Sus}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Sus}_G(A)$ is the suspension of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Sus}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Sus}_G(A)}.$$

We need to show that $\text{Sus}_G(f)$ maps suspensions into suspensions and that is similar to [Example 2.8](#).

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the suspension (functor).

Now, we can finally note that the canonical inclusions defined in [Construction 2.6](#) assemble into a natural transformation.

Claim 2.3. For each $t \in [0, 1]$, define a map of functors:

$$\alpha_t : \text{Cyl}_G \Rightarrow \text{Id},$$

on $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$ as follows; $\alpha_t^A \stackrel{\text{def}}{=} p_t^A$. This map of functors is a natural transformation^a.

^aThe proof can be found on Page 95.

In the next construction, we will place the cone and suspension into exact sequences.

Construction 2.10. Let $A \in G\text{-}C^*\text{Alg}$. Using the notation of [Construction 2.6](#), we have two sequences:

$$0 \longrightarrow \text{Cone}_G(A) \xrightarrow{\iota} \text{Cyl}_G(A) \xrightarrow{p_1^A} A \longrightarrow 0$$

$$0 \longrightarrow \text{Sus}_G(A) \xrightarrow{\iota} \text{Cone}_G(A) \xrightarrow{p_0^A} A \longrightarrow 0$$

where the maps ι are two different inclusion maps which simply include an element of the cone/suspension into the cylinder. These sequences are defined even when A is a G -Banach algebra. We have the following notable result.

Claim 2.4. The sequences above are exact, even when $A \in G\text{-BanAlg}$.

See Page 96 for the proof of this claim. These facts are useful to have on standby.

As we mentioned in the introduction, our interest in the homotopy theory of operator algebras mainly exists because it is useful for understanding various (co)homology theories on admissible subcategories of $G\text{-}C^*\text{Alg}$ and $G\text{-BanAlg}$. Regarding the former, this is a legitimate “problem” that we describe formally as a question.

Question 2.11. Is there an abstract framework which describes the homotopy theory of the following categories:

1. $C^*\text{Alg}$,
 2. ${}_{\omega}C^*\text{Alg}$, the full subcategory of separable C^* -algebras,
 3. ${}_{\omega}\text{Nuc}C^*\text{Alg}$, the full subcategory of separable nuclear C^* -algebras,
- all at once?

We have a remark about this question.

Remark 2.12. This question was implicitly posed as a remark in [\[67, p.408\]](#). In the same paper, it is suggested that one should use the closed model categories of Quillen to define the notion of an admissible subcategory and even provides some explicit conditions that one could impose on a full subcategory to make it admissible in [\[67, Remark 2.12, p.420-421\]](#). Nevertheless, the matter is essentially left unresolved. In Section 4.1, we will propose a working definition of an admissible subcategory of $C^*\text{Alg}$ which addresses [Question 2.11](#).

In spite of the fact that we do not yet have the definition of an admissible subcategory ([Definition 4.5](#)), we will still use it for the purposes of explaining the notions that will be needed to motivate our work. This is not detrimental to our exposition since we are not going to use this notion in a significant way. Whenever this notion is used, it should temporarily be understood to mean that we are working with any of the three categories listed in [Question 2.11](#). With that said, let us introduce some standard properties of functors from an admissible category of equivariant C^* -algebras.

Terminology 2.13. Let $F : G\text{-}C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor.

1. F is exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C) \longrightarrow 0$$

is exact in $\mathcal{A}$.

2. F is split-exact iff for any split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightleftharpoons[F(\alpha)]{F(\psi)} F(C) \longrightarrow 0$$

is split exact in $\mathcal{A}$.

F is half-exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C)$$

is exact in $\mathcal{A}$.

We have a remark about [Terminology 2.13](#).

Remark 2.14. The given terminology makes sense when $\mathcal{A}$ is replaced with any category for which we have a reasonable notion of “exact sequence”. There are many such categories which are not abelian (or even pre-additive) and are, yet, very useful in practice. It also makes sense when $G\text{-}C^*\text{Alg}$ is replaced by $G\text{-BanAlg}$; it is, indeed, interesting to study homology theories on the equivariant category of Banach algebras. However, unlike the case with C^* -algebras, we are not aware of any interesting (co)homology theories that are defined on proper, full subcategories of $G\text{-BanAlg}$.

Moreover, such homology theories are not the topic of this thesis, so we omit a lengthy discussion of them. In our work with the Spanier-Whitehead construction in Section 4.2, we will briefly mention them again as motivation.

An important (and independently interesting) property of the suspension functor is that it is exact. This is sometimes used in technical arguments, so we will include a proof of this fact. To do this, we need some notation.

Notation 2.15. Let X be a LCH space and let $A \in \text{Ban}_{\mathbb{C}}$ with $a \in A$ and $f \in \mathcal{C}_0(X)$. Define:

$$fa : X \rightarrow A, \quad x \mapsto (fa)(x) \stackrel{\text{def}}{=} f(x)a.$$

Note that this actually turns A into a $\mathcal{C}_0(X)$ -module, though this structure isn't of particular interest to us at this stage. Rather, it's more interesting that $\text{Span}\{fa : f \in \mathcal{C}_0(X) \wedge a \in A\}$ has nice behavior relative to $\mathcal{C}_0(X, A)$.

The following technical lemma tells us that the span above generates $\mathcal{C}_0(X, A)$ with the given assumptions on X and A .

Lemma 2.16. [66, Lemma 10.1.1, p.175-176] Let $X \in {}^{\text{LCH}}\text{Top}$ and let $A \in \text{Ban}_{\mathbb{C}}$. Then:

$$\text{Span}\{fa : f \in \mathcal{C}_0(X) \wedge a \in A\},$$

is dense in $\mathcal{C}_0(X, A)$.

The proof of the lemma in the reference is for the case where A is a C^* -algebra but the result holds in the stated generality by the same proof. The difficulty in showing exactness is proving the surjectivity of the last map in the image sequence. The lemma above allows us to obtain an approximate solution which is upgraded to a full solution.

Proposition 2.17

The suspension is an exact functor.

Proof. Consider the following exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $G\text{-}C^*\text{Alg}$. Then, we obtain the following sequence:

$$0 \longrightarrow \text{Sus}_G(A) \xrightarrow{\text{Sus}_G(\phi)} \text{Sus}_G(B) \xrightarrow{\text{Sus}_G(\psi)} \text{Sus}_G(C) \longrightarrow 0$$

whose exactness needs to be checked. We'll do this in three separate steps.

1. Let $f \in \text{Sus}_G(A)$ so that $\text{Sus}_G(\phi)(f) = 0$. This means that $\phi \circ f = 0$ so $\text{ran}(f) \subseteq \ker(\phi) = 0$. That is, $f = 0$.
2. We need to show that $\text{Sus}_G(\psi)$ is surjective. Note that the following set:

$$\text{Span}\{fc : f \in \mathcal{C}_0(0, 1) \wedge c \in C\},$$

approximates every element in $\text{Sus}_G(C)$ so it remains to show that the elements in the dense subspace above belong to $\text{ran}(\text{Sus}_G(\psi))$. Indeed, let $c \in C$. Since ψ is surjective, there is a $b \in B$ such that:

$$\psi(b) = c.$$

So, we see that if $f \in \mathcal{C}_0(0, 1)$, then:

$$\begin{aligned} \text{Sus}_G(\psi)(fb) &= \psi \circ fb \\ &= f\psi(b) \\ &= fc, \end{aligned}$$

which proves that $\text{Sus}_G(\psi)$ is surjective.

3. Let's show that the second sequence is exact at $\text{Sus}_G(B)$. By functoriality, we have that:

$$\begin{aligned} 0 &= \text{Sus}_G(0) \\ &= \text{Sus}_G(\psi \circ \phi) \\ &= \text{Sus}_G(\psi) \circ \text{Sus}_G(\phi), \end{aligned}$$

which implies that $\text{ran}(\text{Sus}_G(\phi)) \subseteq \ker(\text{Sus}_G(\psi))$.

This completes the proof. ■

One property that functors of interest should have is *homotopy invariance*. For this to be defined, we need the notion of homotopy and luckily, the two categories of interest have enough structure to accommodate such a definition.

Terminology 2.18. Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$.

1. A homotopy from f to g is a family $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ in $G\text{-}C^*\text{Alg}(A, B)$ such that:

$$H_a : [0, 1] \rightarrow B, \quad t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, $\phi_0 = f$ and $\phi_1 = g$.

2. f is homotopic to g iff there exists a homotopy from f to g . We denote this by writing $f \simeq_h g$.
3. A and B are homotopy equivalent iff there exist $f \in G\text{-}C^*\text{Alg}(A, B)$ and $g \in G\text{-}C^*\text{Alg}(B, A)$ so that:

$$g \circ f \simeq_h \text{Id}_A,$$

$$f \circ g \simeq_h \text{Id}_B.$$

In this case, we refer to g as the homotopy inverse of f and g as the homotopy inverse of f . Finally, A and B are homotopically equivalent iff there is an equivariant $\star$ -homomorphism:

$$f : A \rightarrow B,$$

which is also a homotopy equivalence. The same definition makes sense in $G\text{-BanAlg}$.

There is an alternative, useful characterization of homotopy that we present in the next result.

Proposition 2.19

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.
2. There is an equivariant $\star$ -homomorphism $h : A \rightarrow \text{Cyl}_G(B)$ such that $p_0^B \circ h = f$ and $p_1^B \circ h = g$.

The morphism h in (2) is also called a homotopy from f to g . The result also holds when $A, B \in G\text{-BanAlg}$.

Proof. Let's prove this for equivariant C^* -algebras first.

(1) $\Rightarrow$ (2): By definition, for each $t \in [0, 1]$, we have a $\star$ -homomorphism $\phi_t : A \rightarrow B$ such that:

$$H_a : [0, 1] \rightarrow B, \quad t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, we have that $\phi_0 = f$ and $\phi_1 = g$. Now, we define:

$$h : A \rightarrow \text{Cyl}_G(B), \quad a \mapsto h(a) \stackrel{\text{def}}{=} H_a.$$

Let's show that h is a $\star$ -homomorphism. We'll check linearity first.

$$\begin{aligned} \forall a, b \in A : \forall \lambda \in \mathbb{C} : \forall t \in [0, 1] : h(\lambda a + b)(t) &= H_{\lambda a + b}(t) \\ &= \phi_t(\lambda a + b) \\ &= \lambda \phi_t(a) + \phi_t(b) \\ &= \lambda H_a(t) + H_b(t) \\ &= \lambda h(a)(t) + h(b)(t) \\ &= (\lambda h(a) + h(b))(t). \end{aligned}$$

Doing similar calculations with the preservation of the involution & products, we obtain that h is a $\star$ -homomorphism. Next, we'll show that $p_0^B \circ h = f$. This is a calculation:

$$\begin{aligned} \forall a \in A : (p_0^B \circ h)(a) &= p_0^B(h(a)) \\ &= p_0^B(H_a) \\ &= H_a(0) \\ &= \phi_0 \\ &= f. \end{aligned}$$

A similar calculation with p_1^A yields the implication.

(2) $\Rightarrow$ (1): Let $t \in [0, 1]$. Define:

$$\phi_t : A \rightarrow B, \quad a \mapsto \phi_t(a) \stackrel{\text{def}}{=} h(a)(t).$$

Let's check that $\phi_0 = f$. We have the following computation:

$$\begin{aligned} \forall a \in A : \phi_0(a) &= h(a)(0) \\ &= (p_0^B \circ h)(a) \\ &= f(a), \end{aligned}$$

as was desired. A similar calculation gives us that $\phi_1 = g$. Finally, for each $a \in A$, the following map:

$$H_a : [0, 1] \rightarrow B, \quad t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous because h maps into the cylinder of B .

For equivariant Banach algebras, the only technical point lies in (1) $\Rightarrow$ (2). In that case, we would've had to explicitly prove that h is continuous since that would not follow from its algebraic structure. For that, we note that $(\phi_t)_{t \in [0, 1]}$ is a family of continuous linear maps which is pointwise bounded since each H_a is a continuous map on a compact space. By the Uniform Boundedness Principle, we see that there exists a $c > 0$ such that:

$$\forall t \in [0, 1] : \|\phi_t\| \leq c.$$

So, this means that:

$$\begin{aligned} \forall a \in A : \|H_a\| &= \sup_{t \in [0, 1]} \|H_a(t)\| \\ &= \sup_{t \in [0, 1]} \|\phi_t(a)\| \\ &\leq c\|a\|, \end{aligned}$$

which is exactly the statement that h is bounded. ■

For instance, it is slightly better to have a proof of the fact that homotopy is an equivalence relation using the second characterization of homotopy as presented in Proposition 2.19. We will present this in the next claim.

Claim 2.5. Equivariant homotopy between parallel equivariant $\star$ -homomorphisms is an equivalence relation^a.

^aThe same result holds for homotopy in G -BanAlg. See Page 96 for the proof.

Of course, equivariant $\star$ -isomorphisms are homotopy equivalences.

Claim 2.6. Isomorphisms in G -C $\star$ Alg/ G -BanAlg are homotopy equivalences^a.

^aSee Page 96 for the proof.

In light of the facts above, we can now introduce homotopy categories.

Construction 2.20. We define:

$$\mathbf{Ho}(G\text{-C}\star\text{Alg}) \stackrel{\text{def}}{=} G\text{-C}\star\text{Alg} / \sim,$$

where $\sim$ is the equivalence relation above on morphism classes. This is known as the homotopy category of G -C $\star$ Alg. The same construction can be adapted for G -BanAlg without change.

Equipped with the notion of homotopy, we can also define what it means for a functor to be homotopy-invariant. We will be concerned with the basic study of homotopy-invariant functors later in Section 4.1.

Terminology 2.21. Let $\mathcal{C}$ be a full subcategory of G -C $\star$ Alg, let $\mathcal{D}$ be a category and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. F is homotopy-invariant iff for two parallel morphisms $\phi, \psi \in \mathcal{C}$, we have that:

$$\phi \sim_h \psi \Rightarrow F(\phi) = F(\psi).$$

This also works for G -BanAlg.

This is a reasonable place to introduce the definition of contractibility of a $C^\star$ -algebra.

Terminology 2.22. Let $A \in G\text{-C}\star\text{Alg} / G\text{-BanAlg}$. A is contractible iff Id_A is homotopic to 0.

In the next claim, we will establish the contractibility of the cone over any equivariant operator algebra.

Claim 2.7. Let $A \in G\text{-C}\star\text{Alg} / G\text{-BanAlg}$. Then, $\text{Cone}_G(A)$ is contractible^a.

^aThe proof can be found on Page 96.

One should note that contractibility of Banach algebras/ $C^\star$ -algebras is not the same as contractibility of spaces.

Example 2.23. Let $A \in {}^1\text{BanAlg} / {}^1\text{C}\star\text{Alg}$.

Claim. A is not contractible.

If it was contractible, then $K_0(A) \simeq 0$ since K_0 is homotopy-invariant. In particular:

$$[1_A]_{K_0} = [0]_{K_0},$$

which implies that $1_A = 0$ and this is impossible.

In particular, $\mathbb{C}$ is **not** contractible as a Banach algebra.

The main result about homotopies between morphisms of equivariant operator algebras is given by the next proposition.

Proposition 2.24

Let $A, B, C \in G\text{-C}\star\text{Alg}$, let $f_1, f_2 \in G\text{-C}\star\text{Alg}(A, B)$ and let $g_1, g_2 \in G\text{-C}\star\text{Alg}(B, C)$ so that:

$$f_1 \simeq_h f_2,$$

$$g_1 \simeq_h g_2.$$

Then, $g_1 \circ f_1 \simeq_h g_2 \circ f_2$.

Proof. We will proceed in three steps.

1. Show that the result holds when $f_1 = f_2$.
2. Show that the result holds when $g_1 = g_2$.
3. Deduce the general case from the two specific situations above.

With this plan, we proceed with the proof.

1. Let $f_1 = f_2 = f$. Then, there is a homotopy $h : B \rightarrow \text{Cyl}_G(C)$ such that:

$$g_1 = p_0^C \circ h,$$

$$g_2 = p_1^C \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} h \circ f.$$

This is an equivariant $\star$ -homomorphism $A \rightarrow \text{Cyl}_G(C)$ and it is clearly a homotopy from $g_1 \circ f$ to $g_2 \circ f$.

2. Let $g_1 = g_2 = g$. Then, there is a homotopy $h : A \rightarrow \text{Cyl}_G(B)$ such that:

$$\begin{aligned} f_1 &= p_0^B \circ h, \\ f_2 &= p_1^B \circ h. \end{aligned}$$

Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(g) \circ h.$$

This is an equivariant $\star$ -homomorphism. Let's show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$:

$$\begin{aligned} p_0^C \circ H &= p_0^C \circ (\text{Cyl}_G(g) \circ h) \\ &= (p_0^C \circ \text{Cyl}_G(g)) \circ h \\ &= (g \circ p_0^B) \circ h \\ &= g \circ (p_0^B \circ h) \\ &= g \circ f_1, \end{aligned}$$

where we used the naturality of the time evaluation maps as proved in **Claim 2.2**. Showing that:

$$p_1^C \circ H = g \circ f_2,$$

is a similar computation.

3. We have that:

$$\begin{aligned} g_1 \circ f_1 &\simeq_h g_2 \circ f_1 \\ &\simeq_h g_2 \circ f_2, \end{aligned}$$

as was desired.

This completes the proof. ■

With the proposition above, we can establish three other formal properties of homotopy on categories of operator algebras.

Claim 2.8. The following statements hold:

1. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, B)$ with g_1, g_2 being homotopy inverses of f . Then:

$$g_1 \simeq_h g_2.$$

That is, homotopy inverses are “unique”.

2. Let $A, B \in G\text{-C}^*\text{Alg}$, let $f \in G\text{-C}^*\text{Alg}(A, B)$ and let $g_1, g_2 \in G\text{-C}^*\text{Alg}(B, A)$ such that:

$$\begin{aligned} f \circ g_1 &\simeq_h \text{Id}_B, \\ g_2 \circ f &\simeq_h \text{Id}_A. \end{aligned}$$

Then, $g_1 \simeq_h g_2$. In particular, f is a homotopy equivalence.

3. Let f, g be morphisms in $G\text{-C}^*\text{Alg}$ with $g \circ f$ being well-defined. If any two of these:

$$f, g, g \circ f,$$

are homotopy equivalences, then the third is also a homotopy equivalence.

The same result holds in $G\text{-BanAlg}$. The proof is given on Page 97.

Let us introduce the analogue of a cofibration into our setting; these are the Schochet fibrations.

Definition 2.25

Let $A, B \in G\text{-C}^*\text{Alg}$ and let $p \in G\text{-C}^*\text{Alg}(A, B)$. p is a Schochet fibration iff for any $D \in G\text{-C}^*\text{Alg}$, for any equivariant homotopy $(h_t : D \rightarrow B)_{t \in [0,1]}$ and for any $f \in G\text{-C}^*\text{Alg}(D, A)$ such that the diagram:

$$\begin{array}{ccc} & & A \\ & \nearrow f & \downarrow p \\ D & \xrightarrow{h_t} & B \end{array}$$

commutes when $t = 0$, then there exists an equivariant homotopy $(H_t : D \rightarrow A)_{t \in [0,1]}$ such that the following diagram:

$$\begin{array}{ccc} & & A \\ & \nearrow f & \downarrow p \\ D & \xrightarrow{h_t} & B \end{array}$$

H_t (dashed arrow from D to A)

commutes for each $t \in [0,1]$ and $H_0 = f$. This definition also makes sense in $G\text{-BanAlg}$.

Our eventual goal is to use the notion above to show that $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ are categories of fibrant objects; see Proposition 4.1. For this to happen, we need to establish some key permanence properties of Schochet fibrations. For this, it will be helpful to have a characterization of Schochet fibrations which was mentioned in [67, Definition 1.2, p.406] for $\star$ -homomorphisms between C^* -algebras but was not explicitly established. We will fill in those details.

Proposition 2.26

Let $A, B \in G\text{-C}^*\text{Alg}$ and let $p \in G\text{-C}^*\text{Alg}(A, B)$. TFAE:

1. p is a Schochet fibration.
2. If $D \in G\text{-C}^*\text{Alg}$, $f : D \rightarrow A$ is a $\star$ -homomorphism and $h : D \rightarrow \text{Cyl}_G(B)$ is a homotopy of the composite $p \circ f$, then there exists a homotopy $H : D \rightarrow \text{Cyl}_G(A)$ of f such that the following diagram:

$$\begin{array}{ccc}
 & & A \\
 & \nearrow f & \uparrow p_0 \\
 D & \xrightarrow{H} & \text{Cyl}_G(A) \\
 & \searrow h & \downarrow \text{Cyl}_G(p) \\
 & & \text{Cyl}_G(B)
 \end{array}$$

commutes.

The same result holds in $G\text{-BanAlg}$. The proof is given on Page ??.

Proof.

(1) $\Rightarrow$ (2): Indeed, we have the following diagram:

$$\begin{array}{ccc}
 & & A \\
 & \nearrow f & \downarrow p \\
 D & \xrightarrow{h_t} & B
 \end{array}$$

where $(F_t : D \rightarrow A)_{t \in [0,1]}$ is a homotopy with $F_0 = f$. Now, we define:

$$\forall x \in D : \forall t \in [0, 1] : H(x)(t) \stackrel{\text{def}}{=} F_t(x).$$

Clearly, $H : D \rightarrow \text{Cyl}_G(A)$ is a homotopy such that:

$$\begin{aligned}
 \forall x \in D : (p_0 \circ H)(x) &= p_0(H(x)) \\
 &= H(x)(0) \\
 &= F_0(x) \\
 &= f(x),
 \end{aligned}$$

which means that $p_0 \circ H = f$. On the other hand:

$$\begin{aligned}
 \forall x \in D : \forall t \in [0, 1] : [(\text{Cyl}_G(p) \circ H)(x)](t) &= \text{Cyl}_G(p)(H(x))(t) \\
 &= (p \circ H(x))(t) \\
 &= p(H(x)(t)) \\
 &= p(F_t(x)) \\
 &= (p \circ F_t)(x) \\
 &= h_t(x),
 \end{aligned}$$

as was desired.

(2) $\Rightarrow$ (1): This follows simply by reversing the argument above. ■

Let us mention an important example of a Schochet fibration.

Example 2.27. Let $A \in G\text{-C}^*\text{Alg}$. Here's the main claim.

Claim. For each $t \in [0, 1]$, the canonical inclusion:

$$p_t : \text{Cyl}_G(A) \rightarrow A,$$

as defined in [Construction 2.6](#), is a Schochet fibration. This result also holds when $A \in G\text{-BanAlg}$.

This is proved in the non-equivariant case in [\[67, Lemma 1.3, p.407\]](#). To apply that to the equivariant case, one needs to check that the calculated lift is equivariant and it is, because the morphisms we start with are equivariant.

Let us collect some notable properties of Schochet fibrations; these were “known” in the non-equivariant case within [\[67\]](#), though they were not always given complete proofs so we will fill in the gaps. Our approach will be to use the characterization of Schochet fibrations in statement (2) of Proposition 2.26. This might seem like it complicates our exposition but we will see that such an approach is useful for generating similar results in categories of fibrant objects. One reason for this is that we are just using the “formal” properties of Schochet fibrations and the homotopical structures we have introduced thus far. In particular, the functoriality of the cylinder and the naturality of time evaluation maps will play a significant role in the proof.

Proposition 2.28

The following statements hold:

1. Schochet fibrations in $G\text{-C}^*\text{Alg}$ are closed under composition.
2. Let $A \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, 0)$. Then, f is a Schochet fibration.
3. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, B)$ be an isomorphism. f is a Schochet fibration.
4. The cylinder functor preserves pullbacks.
5. Schochet fibrations are preserved by base extensions.

These results also hold in $G\text{-BanAlg}$.

Let us recall the meaning of some property being preserved by base extensions.

Terminology 2.29. Let $\mathcal{C}$ be a category and let P be a property of morphisms in $\mathcal{C}$. We say that P is preserved by base extension iff for any pullback diagram in $\mathcal{C}$ of the following form:

$$\begin{array}{ccc} p & \longrightarrow & x \\ \downarrow & & \downarrow \\ z & \longrightarrow & y \end{array}$$

if $x \rightarrow y$ has property P , then so does $p \rightarrow z$.

Now, we can prove our result.

Proof of Proposition 2.28. We will prove each statement one-by-one, mainly using Proposition 2.26.

1. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $p \in G\text{-C}^*\text{Alg}(A, B)$ and $q \in G\text{-C}^*\text{Alg}(B, C)$ be Schochet fibrations. This part is just a two-step lifting and we approach by starting with the following diagram which we will explain afterwards:

$$\begin{array}{ccccc} & & & & A \\ & & & & \uparrow p_0^A \\ & & & & \text{Cyl}_G(A) \\ & f \nearrow & & \downarrow \text{Cyl}_G(p) & \\ D & \xrightarrow{H} & & & \text{Cyl}_G(B) \\ & \xleftarrow{\tilde{H}} & & \downarrow \text{Cyl}_G(q) & \\ & & & & \text{Cyl}_G(C) \\ & h \searrow & & & \end{array}$$

Let us ignore the dotted arrows for the moment. We note that h is a homotopy of $(q \circ p) \circ f$ and that this is the same diagram as in statement (2) of Proposition 2.26, except that we have used the functoriality of Cyl_G to separate p and q . Now, we want to fill in the diagram by the dotted arrows and let us explain why this is possible. So, h is, in particular, a homotopy of $q \circ (p \circ f)$. Since q is a Schochet fibration, we can find a homotopy lift $\tilde{H}$ to $\text{Cyl}_G(B)$ and this is a homotopy of $p \circ f$. Since p is a Schochet fibration, we can find a homotopy lift H to $\text{Cyl}_G(A)$ and this is a homotopy of f . But this was the desired lift for the original lifting problem.

2. Let $D \in G\text{-C}^*\text{Alg}$ and suppose we have a commutative diagram:

$$\begin{array}{ccc} & & A \\ & p \nearrow & \uparrow p_0^A \\ D & & \text{Cyl}_G(A) \\ & h \searrow & \downarrow \text{Cyl}_G(f) \\ & & \text{Cyl}_G(0) \end{array}$$

where h is a homotopy of $f \circ p$. We define:

$$s : D \rightarrow \text{Cyl}_G(D), \quad d \mapsto s(d),$$

where $s(d)$ is the constant function from $[0, 1]$ into D with value d . Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(p) \circ s.$$

This is a homotopy of p by the following computation:

$$\begin{aligned} p_0^A \circ H &= p_0^A \circ \text{Cyl}_G(p) \circ s \\ &= p \circ p_0^D \circ s \\ &= p \circ \text{Id}_D \\ &= p. \end{aligned}$$

Next, we observe that this is the required homotopy lift:

$$\begin{aligned}
\text{Cyl}_G(f) \circ H &= \text{Cyl}_G(f) \circ \text{Cyl}_G(p) \circ s \\
&= \text{Cyl}_G(f \circ p) \circ s \\
&= \text{Cyl}_G(p_0^0 \circ h) \circ s \\
&= \text{Cyl}_G(p_0^0) \circ \text{Cyl}_G(h) \circ s \\
&= 0 \\
&= h,
\end{aligned}$$

as was desired.

3. Consider the following homotopy lifting problem:

$$\begin{array}{ccc}
& & A \\
& \nearrow g & \uparrow p_0^A \\
D & & \text{Cyl}_G(A) \\
& \searrow h & \downarrow \text{Cyl}_G(f) \\
& & \text{Cyl}_G(B)
\end{array}$$

where h is a homotopy of $f \circ g$. Since f is an isomorphism, we see that $\text{Cyl}_G(f)$ is an isomorphism and:

$$\text{Cyl}_G(f)^{-1} = \text{Cyl}_G(f^{-1}).$$

Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(f^{-1}) \circ h.$$

Let us check that this is a homotopy of g :

$$\begin{aligned}
p_0^A \circ H &= p_0^A \circ \text{Cyl}_G(f^{-1}) \circ h \\
&= f^{-1} \circ p_0^B \circ h \\
&= f^{-1} \circ (f \circ g) \\
&= g.
\end{aligned}$$

Next, we show that this is our desired homotopy lift:

$$\begin{aligned}
\text{Cyl}_G(f) \circ H &= \text{Cyl}_G(f) \circ (\text{Cyl}_G(f^{-1}) \circ h) \\
&= \text{Cyl}_G(\text{Id}_B) \circ h \\
&= \text{Id}_{\text{Cyl}_G(B)} \circ h \\
&= h,
\end{aligned}$$

as was desired.

4. The cylinder functor is a representable functor and such functors preserve limits, so we're done.

5. Consider the following pullback diagram in $G\text{-}C^*\text{Alg}$ or $G\text{-}Ban\text{Alg}$:

$$\begin{array}{ccc}
P & \xrightarrow{\bar{p}} & C \\
\bar{q} \downarrow & & \downarrow q \\
A & \xrightarrow{p} & B
\end{array}$$

and suppose that p is a Schochet fibration. We will first apply the functor Cyl_G to the diagram above. Then, we will extend the resulting diagram from the top by copying and pasting the diagram in (2) of Proposition 2.26. This gives us the following diagram that we will analyze:

$$\begin{array}{ccccc}
& & D & & \\
& \swarrow f & & \searrow h & \\
P & \xleftarrow{p_0^P} & \text{Cyl}_G(P) & \xrightarrow{\text{Cyl}_G(\bar{p})} & \text{Cyl}_G(C) \\
\bar{q} \downarrow & & \downarrow \text{Cyl}_G(\bar{q}) & & \downarrow \text{Cyl}_G(q) \\
A & \xleftarrow{p_0^A} & \text{Cyl}_G(A) & \xrightarrow{\text{Cyl}_G(p)} & \text{Cyl}_G(B)
\end{array}$$

With this diagram, the proof proceeds as follows:

- Compute $\text{Cyl}_G(q) \circ h$ as a homotopy of the composite $p \circ (\bar{q} \circ f)$.
- Since p is a Schochet fibration, this induces a morphism from D to $\text{Cyl}_G(A)$ which is equal to $\text{Cyl}_G(q) \circ h$.

- (c) The above creates a pullback cone for the pullback diagram of cylinders and since the pullback of this diagram is $\text{Cyl}_G(P)$, this induces a morphism from $D \rightarrow \text{Cyl}_G(P)$ which is the morphism we wanted.

With this outline, let us proceed with filling in details.

- (a) Let's show that:

$$p_0^B \circ \text{Cyl}_G(q) \circ h = p \circ \bar{q} \circ f.$$

Indeed, this is just a computation as follows:

$$\begin{aligned} p_0^B \circ \text{Cyl}_G(q) \circ h &= q \circ p_0^C \circ h \\ &= q \circ (\bar{p} \circ f) \\ &= (q \circ \bar{p}) \circ f \\ &= (p \circ \bar{q}) \circ f \\ &= p \circ (\bar{q} \circ f), \end{aligned}$$

as was desired.

- (b) As we mentioned above, p being a Schochet fibration gives us a diagram:

$$\begin{array}{ccccc} & & D & & \\ & \swarrow f & & \searrow h & \\ P & \xleftarrow{p_0^P} & \text{Cyl}_G(P) & \xrightarrow{\text{Cyl}_G(\bar{p})} & \text{Cyl}_G(C) \\ \downarrow \bar{q} & & \downarrow \text{Cyl}_G(\bar{q}) & \swarrow k & \downarrow \text{Cyl}_G(q) \\ A & \xleftarrow{p_0^A} & \text{Cyl}_G(A) & \xrightarrow{\text{Cyl}_G(p)} & \text{Cyl}_G(B) \end{array}$$

with $k : D \rightarrow \text{Cyl}_G(A)$ being a homotopy of $\bar{q} \circ f$. Moreover, we have that:

$$k \circ \text{Cyl}_G(p) = \text{Cyl}_G(q) \circ h.$$

by statement (2) of our characterization of Schochet fibrations.

- (c) There is a pullback cone formed from $D \rightarrow \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B)$ and $D \rightarrow \text{Cyl}_G(C) \rightarrow \text{Cyl}_G(B)$. By the universal property of the pullback, there is a unique morphism from $D \rightarrow \text{Cyl}_G(P)$ such that all sub-diagrams formed commute. So, this morphism is a homotopy for f and $\bar{p}$ is a Schochet fibration.

This completes the proof. ■

Here is one more notable result which allows us to conclude split surjectivity.

Proposition 2.30. A. Vats, 2026

Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, B)$ be a Schochet fibration such that:

$$\exists g \in G\text{-C}^*\text{Alg}(B, A) : f \circ g \simeq_h \text{Id}_B.$$

Then, f is split epi in $G\text{-C}^*\text{Alg}$ and any left homotopy inverse of f is homotopic to its section.

Proof. Let us start with the following lifting problem:

$$\begin{array}{ccc} & & A \\ & \nearrow g & \uparrow p_0^A \\ B & \xrightarrow{\quad H \quad} & \text{Cyl}_G(A) \\ & \searrow h & \downarrow \text{Cyl}_G(f) \\ & & \text{Cyl}_G(B) \end{array}$$

where h is a homotopy from $f \circ g$ to Id_B . Since f is a Schochet fibration, we have a homotopy lift H and this is a homotopy of g . Now, we have the following computation:

$$\begin{aligned} f \circ (p_1^A \circ H) &= (f \circ p_1^A) \circ H \\ &= (p_1^B \circ \text{Cyl}_G(f)) \circ H \\ &= p_1^B \circ (\text{Cyl}_G(f) \circ H) \\ &= p_1^B \circ h \\ &= \text{Id}_B \end{aligned}$$

so f is split surjective. Now, let p be a left homotopy inverse of f and let g be the section of f . Then:

$$\begin{aligned} p &= p \circ \text{Id}_B \\ &\simeq_h p \circ (f \circ g) \\ &= (p \circ f) \circ g \\ &\simeq_h g, \end{aligned}$$

| as was desired. ■

This result is interesting and we have a remark about it that we will provide momentarily. Before that, we wish to introduce another class of morphisms which can be used as the “category of fibrations” over the category of C^* -algebras.

Terminology 2.31. Let $G \in {}^{\text{LCH}}\text{TopGrp}$, let $A, B \in G\text{-}C^*\text{Alg}$ and let $\phi \in G\text{-}C^*\text{Alg}(A, B)$. ϕ is a semisplit surjection iff it is a surjection which has an equivariant contractive & completely positive (ccp) linear section.

Here is a corollary of the proposition above.

Corollary 2.32. A. Vats, 2026

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a Schochet fibration which admits a right homotopy inverse. Then, f is a semisplit surjection.

Proof. Indeed, the proposition above implies that f has a right inverse in $G\text{-}C^*\text{Alg}$ so we’re done. ■

Checking that some morphism is a Schochet fibration is difficult and one would like to construct homotopy equivalences without having to construct such fibrations. So, we introduce good deformation retracts to facilitate this process.

Definition 2.33

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be surjective. f is a good deformation retract iff there exists:

$$F \in G\text{-}C^*\text{Alg}(A, \text{Cyl}_G(A)),$$

so that:

1. $p_0^A \circ F = \text{Id}_A$.
2. $\forall t \in [0, 1] : p_t^B \circ \text{Cyl}_G(f) \circ F = f$.
3. $(p_1^A \circ F)(\ker(f)) = 0$.

We will abbreviate good deformation retracts as GDRs^a.

^aThe definition also makes sense in $G\text{-BanAlg}$.

The next result tells us that surjective GDRs give us homotopy equivalences.

Proposition 2.34

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a surjective GDR. Then, it is a homotopy equivalence.

Proof. Since f is an GDR, there is a $\star$ -homomorphism:

$$F : A \rightarrow \text{Cyl}_G(A),$$

such that the conditions of **Definition 2.33** hold. Now, we want to factorize $p_1^A \circ F$ so that it is the composition $r \circ f$. To do this, we need to define r on B so let $b \in B$. Since f is surjective, we see that:

$$\exists a \in A : b = f(a).$$

Now, we define:

$$r(f(a)) \stackrel{\text{def}}{=} (p_1^A \circ F)(a).$$

Indeed, this is well-defined because if $a' \in A$ so that $b = f(a')$, then $a - a' \in \ker(f)$ and we know that $p_1^A \circ F$ vanishes on $\ker(f)$. The fact that r is a $\star$ -homomorphism follows by direct calculation with its definition and, in particular, the fact that f, p_1^A and F are $\star$ -homomorphisms. The rest of the proof involves establishing the next claim.

Claim. $r \circ f \simeq_h \text{Id}_A$ and $f \circ r = \text{Id}_B$.

The first statement holds because the map F satisfies the following relations:

$$\begin{aligned} p_0^A \circ F &= \text{Id}_A, \\ p_1^A \circ F &= r \circ f. \end{aligned}$$

For the second relation, let $b \in B$. Then, we find $a \in A$ so that $b = f(a)$. This leads us to a computation:

$$\begin{aligned} (f \circ r)(b) &= (f \circ r \circ f)(a) \\ &= (f \circ p_1^A \circ F)(a) \\ &= (p_1^B \circ \text{Cyl}_G(f) \circ F)(a) \\ &= f(a) \\ &= b, \end{aligned}$$

as was desired. ■

We have a remark about this result.

Remark 2.35. The proposition above is interesting for two reasons:

1. So far, most of the definitions and results we’ve presented work in the category of equivariant Banach algebras. It is not clear, however, that the next result holds in that category; the problem is that we construct a candidate

homotopy inverse in the proof and there is no reason why this needs to be continuous.

2. In [67, p.412, Proposition 1.13a], it is claimed that if we have a Schochet fibration, then that morphism is a homotopy equivalence iff it is a GDR. We were not able to establish this directly ourselves and we do not think that it is true. However, we were not able to construct an explicit counterexample.

We will return to the issue in (2) within Remark 2.38.

The non-equivariant version of the next result is in [67, Proposition 1.13b, p.412]. However, a proof is not provided and a reference is given to the proof of an analogous result on spaces. That reference proves a more general result which does not seem to have an analogue in our setting. For this reason, we provide a self-contained full proof with details.

Proposition 2.36

GDRs are preserved by base extensions.

Proof. Consider the following pullback diagram:

$$\begin{array}{ccc} P & \xrightarrow{\pi_A} & A \\ \pi_C \downarrow & & \downarrow f \\ C & \xrightarrow{g} & B \end{array}$$

where f is a GDR and let $F : A \rightarrow \text{Cyl}_G(A)$ be a $\star$ -homomorphism as in Definition 2.33. Now, we want to define a $\star$ -homomorphism $G : P \rightarrow \text{Cyl}_G(P)$ which satisfies the defining conditions of an GDR. Recall that:

$$P = A \times_B C,$$

and define:

$$\forall (a, c) \in A \times_B C : \forall t \in [0, 1] : G(a, c)(t) \stackrel{\text{def}}{=} (F(a)(t), c).$$

Now, we need to check that this works.

1. Clearly, G defines a $\star$ -homomorphism; this follows from F being a $\star$ -homomorphism.
2. We need to ensure that G maps into $\text{Cyl}_G(P)$ so let $(a, c) \in P$ and let $t \in [0, 1]$. Then, we know that:

$$p_t^B \circ \text{Cyl}_G(f) \circ F = f.$$

So, now, we have the following computation:

$$\begin{aligned} f(F(a)(t)) &= (f \circ F(a))(t) \\ &= p_t^B(f \circ F(a)) \\ &= p_t^B(\text{Cyl}_G(f)(F(a))) \\ &= (p_t^B \circ \text{Cyl}_G(f) \circ F)(a) \\ &= f(a) \\ &= g(c), \end{aligned}$$

which proves that G maps into $\text{Cyl}_G(P)$.

3. Now, we need to start verifying the actual conditions of an GDR.

(a) Let $(a, c) \in P$. Then, we have the following computation:

$$\begin{aligned} (p_0^P \circ G)(a, c) &= p_0^P(G(a, c)) \\ &= G(a, c)(0) \\ &= (F(a)(0), c) \\ &= ((p_0^A \circ F)(a), c) \\ &= (\text{Id}_A(a), c) \\ &= (a, c), \end{aligned}$$

which proves that $p_0^P \circ G = \text{Id}_P$.

(b) Let $(a, c) \in P$ and let $t \in [0, 1]$. Then, we have the following computation:

$$\begin{aligned} (p_t^C \circ \text{Cyl}_G(\pi_C) \circ G)(a, c) &= p_t^C(\text{Cyl}_G(\pi_C)(G(a, c))) \\ &= p_t^C(\pi_C \circ G(a, c)) \\ &= (\pi_C \circ G(a, c))(t) \\ &= \pi_C(G(a, c)(t)) \\ &= \pi_C(F(a)(t), c) \\ &= c \\ &= \pi_C(a, c), \end{aligned}$$

as was desired.

(c) Observe that:

$$\ker(\pi_C) = \{(a, c) \in P : c = 0\}.$$

Now, we have the following computation:

$$\begin{aligned}\forall (a, c) \in \ker(\pi_C) : (p_1^P \circ G)(a, c) &= p_1^P(G(a, c)) \\ &= G(a, c)(1) \\ &= (F(a)(1), c).\end{aligned}$$

Since $(a, c) \in \ker(\pi_C)$, we have that $c = 0$. But if $c = 0$, then $g(c) = 0$ so $f(a) = 0$ and this means that $a \in \ker(f)$. Since $p_1^A \circ F$ maps $\ker(f)$ to 0, it follows that $(p_1^P \circ F)(\ker(\pi_C)) = 0$.

So, we see that π_C must be an GDR.

This completes the proof. ■

The next result will be important for our later work in addressing Question 2.11. In [67], it is established as a simple consequence of the “fact” discussed in Remark 2.38 and Proposition 2.36. In [72], this is established via a topological characterization of Schochet fibrations in terms of Serre fibrations of topological spaces. While that proof is nice, it is very clear that it depends on the structure of C^* -algebras. Our proof will generalize to other equivariant topological algebras, though we only give it explicitly for equivariant C^* -algebras and Banach algebras.

Theorem 2.37

Consider the following pullback diagram in G - $C^*\text{Alg}$ or G - BanAlg :

$$\begin{array}{ccc} P & \xrightarrow{\bar{p}} & C \\ \bar{q} \downarrow & & \downarrow q \\ A & \xrightarrow{p} & B \end{array}$$

If p is a homotopy equivalence and a Schochet fibration, then $\bar{p}$ is a homotopy equivalence and a Schochet fibration.

Proof. We will prove this for a specific pullback $P = A \times_B C$. Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration. Let $r : B \rightarrow A$ be a homotopy inverse of p ; by the strength of Proposition 2.30, we see that r can be chosen so that it is a section of p . Now, we define a map:

$$i_C : C \rightarrow P, \quad c \mapsto i_C(c) \stackrel{\text{def}}{=} ((r \circ q)(c), c).$$

Since r is a section of p , we see that i_C does, indeed, map into P and we also note that it is an equivariant $\star$ -homomorphism, since all maps involved in defining it are equivariant $\star$ -homomorphisms.

Claim. i_C is a homotopy inverse of $\bar{p}$.

Indeed, we have the following computation:

$$\begin{aligned}\forall c \in C : (\bar{p} \circ i_C)(c) &= \bar{p}((r \circ q)(c), c) \\ &= c \\ &= \text{Id}_C(c),\end{aligned}$$

which shows that i_C is a right inverse of $\bar{p}$. Moreover:

$$\begin{aligned}\forall (a, c) \in P : (i_C \circ \bar{p})(a, c) &= i_C(\bar{p}(a, c)) \\ &= i_C(c) \\ &= ((r \circ q)(c), c) \\ &= (r(q(c)), c) \\ &= (r(p(a)), c) \\ &= ((r \circ p) \times \text{Id}_C)|_P(a, c).\end{aligned}$$

So, it is sufficient for us to show that:

$$((r \circ p) \times \text{Id}_C)|_P \simeq_h \text{Id}_P.$$

If $h : A \rightarrow \text{Cyl}_G(A)$ is a homotopy from Id_A to $r \circ p$, then we define for $t \in [0, 1]$:

$$H_t : P \rightarrow A \times C, \quad (a, c) \mapsto H_t(a, c) \stackrel{\text{def}}{=} (h_t(a), c).$$

Now, the point is that each H_t may not lie in P , so this is not the homotopy we desire. What we need to do is to construct a homotopy from H to some other $\star$ -homomorphism $F : P \rightarrow \text{Cyl}_G(P)$, with the latter being the desired homotopy from Id_P to $((r \circ p) \times \text{Id}_C)|_P$. So, we need to define a family of $\star$ -homomorphisms:

$$(\Phi_{(s,t)} : P \rightarrow A \times C)_{(s,t) \in [0,1]^2},$$

where $\Phi_{(0,t)} = H_t$ for each $t \in [0, 1]$ and $\Phi_{(1,t)}$ lies in P for each $t \in [0, 1]$. We define:

$$\forall (a, c) \in P : \forall (s, t) \in [0, 1]^2 : \Phi_{(s,t)}(a, c) \stackrel{\text{def}}{=} \begin{cases} H_t(a, c) & \text{if } s = 0, \\ (a, c) & \text{if } t = 0, \\ H_{1-s}((r \circ p)(a), c) & \text{if } t = 1, \\ H_{(1-s)t}(a, c) & \text{otherwise} \end{cases}.$$

At $s = 0$, we see that this is just the family $(H_t)_{t \in [0,1]}$. Now, let $s = 1$. In that case, it is clear that every $\Phi_{(1,t)}$ takes

values in P . Moreover, when $t = 0$, this is just Id_P while when $t = 1$, we have $((r \circ p) \times \text{Id}_C)|_P$. Now, one needs to check that Φ is actually a homotopy of $\star$ -homomorphisms. This follows from a case analysis based on the way Φ is defined. In each case, we see that Φ has the appropriate continuity property because H has that property. ■

We have a remark about this result.

Remark 2.38. If it were the case that a Schochet fibration which were a homotopy equivalence was a GDR, then we would be able to provide a very easy proof of [Theorem 2.37](#) in $G\text{-C}^*\text{Alg}$. Let us present it now.

Alternative Proof of Theorem 2.37. The proof proceeds in just three steps:

1. Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration.
2. To show that $\bar{p}$ is a homotopy equivalence, it's enough to show that it's a GDR.
3. Since p is a Schochet fibration and a homotopy equivalence, it also an GDR.
4. Since GDRs are preserved by base extensions, we see that $\bar{p}$ is a GDR.

This completes the proof.

This alternative is simple but it does not immediately work in $G\text{-BanAlg}$ since [Proposition 2.34](#) may not hold.

At this stage, we have developed all of the basic results needed to show that our categories of interest do form categories of fibrant objects; this will be done in the next chapter. For the moment, we want to turn our attention towards a rather concrete conjecture which seems to have some unexpected, at first glance, applications.

2.2 Lifting Problems, Schochet C^* -Algebras & Semiprojectivity

Let us start by setting some notation.

Notation 2.39. Unless otherwise stated, H is an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Moreover:

$$\mathcal{Q}(H) \stackrel{\text{def}}{=} \mathcal{B}(H)/\mathcal{K}(H).$$

Finally, $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ denotes the Calkin projection unless otherwise stated^a.

^aThe symbol π will also be used for other surjective $\star$ -homomorphisms.

Our first goal is to provide a solution to the following conjecture due to Schochet.

Conjecture 2.40 (C. Schochet, 1984, [67, p.407]). The Calkin projection $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ is not a Schochet fibration.

We will provide further motivation for interest in this conjecture momentarily. Here is a further remark.

Remark 2.41. One can also ask the analogous question over Banach algebras; if $X \in \text{Ban}_{\mathbb{C}}$, then one can still define the Calkin algebra over X as above. However, we will concentrate on the formulation above since this seems difficult enough. In that case, we will show that the conjecture holds in [Theorem 2.59](#).

We will start by providing an extended discussion of the context behind the conjecture and its relationship with older problems in operator theory. Let us record down a useful piece of terminology that will fix language.

Terminology 2.42. Let P be a property that is satisfied by elements of a C^* -algebra and let $T \in \mathcal{B}(H)$. Then, T is essentially P iff $\pi(T)$ is P in $\mathcal{Q}(H)$. Moreover, the essential spectrum of T is defined by $\sigma_{\text{ess}}(T) \stackrel{\text{def}}{=} \sigma_{\mathcal{Q}(H)}(\pi(T))$.

We will mainly be concerned with essentially normal operators¹. It is a stunning result of Brown-Douglas-Fillmore theory that there is a classification of such operators. To state this theorem formally, we will need a construction.

Construction 2.43. Let H be a $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be Fredholm. We define:

$$\phi_T : \mathbb{C} \setminus \sigma_{\text{ess}}(T) \rightarrow \mathbb{Z}, \lambda \mapsto \phi_T(\lambda) \stackrel{\text{def}}{=} \text{Index}(T - \lambda \text{Id}_H).$$

This is known as the index function of T . This makes sense, because $\lambda \notin \sigma_{\text{ess}}(T)$ so $\pi(T) - \lambda 1_{\mathcal{Q}(H)}$ is invertible. Hence, $T - \lambda \text{Id}_H$ is a Fredholm operator and the index above is well-defined.

Now, we can state our result.

Theorem 2.44. The Brown-Douglas-Fillmore Theorem, [27, Corollary 11.2, p.118], [28]

Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be essentially normal. TFAE:

1. $T \in N_{\mathcal{B}(H)} + \mathcal{K}(H)$.
2. ϕ_T is trivial.

The key tool used to prove the theorem above (and related theorems classifying such operators) are the so-called Brown-Douglas-Fillmore (BDF) Ext groups. We will, now, provide some intuition for how these groups arise naturally; our exposition is heavily inspired by the introduction of [13]. If $T \in \mathcal{B}(H)$ is essentially normal, then T (along with $\mathcal{K}(H)$ and Id_H) generates a C^* -subalgebra A of $\mathcal{B}(H)$. With this data, we can construct a morphism $A \rightarrow C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)})$, whose kernel is $\mathcal{K}(H)$. On the other hand, the C^* -algebra on the right-hand side of the arrow is a commutative unital C^* -algebra. By Gelfand Duality and some more concrete results about functional calculus, we see that:

$$C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)}) \simeq \mathcal{C}(\sigma_{\text{ess}}(T)).$$

So, this data fits into an exact sequence:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(\sigma_{\text{ess}}(T)) \rightarrow 0.$$

In other words, essentially normal operators naturally give rise to extensions of the form given above and now, it becomes reasonable to replace $\sigma_{\text{ess}}(T)$ by some compact metrizable space X and study extensions of the following form:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(X).$$

This leads us to the following definition.

Definition 2.45

Let X be a compact, metrizable space.

1. A extension of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is a pair:

$$(A, \phi),$$

where A is a C^* -subalgebra of $\mathcal{B}(H)$ containing $\mathcal{K}(H)$ and Id_H & $\phi \in C^*\text{Alg}(A, \mathcal{C}(X))$ with $\ker(\phi) = \mathcal{K}(H)$.

2. If $(A_1, \phi_1), (A_2, \phi_2)$ are extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$, then these extensions are strongly equivalent^a iff there exists a $\star$ -isomorphism $\psi : A_2 \rightarrow A_1$ such that $\phi_2 = \phi_1 \circ \psi$.

The set of equivalence classes of extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is denoted by $\text{Ext}(X)$.

¹For an extended discussion of essentially normal operators & their history, as well as recent developments, see [22].

^aThis can be checked directly to be an equivalence relation.

We will not spend any time discussing the proofs of various results in BDF theory since that would take up too much space. The main result that needs to be known is that $\text{Ext}(X)$ can be made into a group and that Ext is functorial.

Proposition 2.46. [13, Corollary 1.18, Theorem 1.23, p. 272-273, p. 275]

We have a functor $\text{Ext}(-)$ from the category of compact metrizable spaces to AbGrp .

Now, it seems reasonable, with this functor in hand, to ask if it satisfies the Eilenberg-Steenrod axioms. For instance, it is not obvious that this functor is homotopy-invariant, though it would certainly be desirable for that to be the case. The proof of homotopy-invariance as discussed in [13, Section 2, p.275-286] is fairly simple but it would be desirable to have a different proof which “explains” why one would expect to have this property in light of the original problem. To that end, we will construct another functor for which homotopy-invariance is “obvious”.

Construction 2.47. Let X be a compact metrizable space. We define:

$$\text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

to be the set of all homotopy classes injective $\star$ -homomorphisms $\mathcal{C}(X) \rightarrow \mathcal{Q}(H)$. Using the methods of [37], this can be shown to be an abelian group. This is also a functor^a, denoted by inj^h .

^aFrom the category of compact metrizable spaces to AbGrp .

So, we have two functors from the category of compact metrizable spaces to AbGrp and it is natural to want a comparison between them. In fact, they happen to coincide on a very special class of spaces which are closely related to **Conjecture 2.40**. To define this class of spaces, we will need some terminology.

Terminology 2.48. Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. f and g are $\star$ -homotopic iff there is a continuous map^a $\phi : A \times [0, 1] \rightarrow B$ such that $\phi(\cdot, t)$ is a $\star$ -homomorphism for each $t \in [0, 1]$, with $\phi(\cdot, 0) = f$ and $\phi(\cdot, 1) = g$.

^aUnsurprisingly, we call this a $\star$ -homotopy.

This notion was introduced in [19] and [20]. Let us show that it is not different from the usual notion of homotopy. While this has not been shown explicitly in the literature, it is a very standard

Proposition 2.49

Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.
2. f and g are $\star$ -homotopic.

In fact, we get explicit ways of defining homotopies from $\star$ -homotopies and vice versa.

Proof.

(1) $\Rightarrow$ (2): Let $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ be a homotopy from f to g . We define:

$$\psi : A \times [0, 1] \rightarrow B, (a, t) \mapsto \psi(a, t) \stackrel{\text{def}}{=} \phi_t(a).$$

For each $t \in [0, 1]$, we clearly have a $\star$ -homomorphism, with the evaluation at 0 and 1 giving f & g respectively. Next, we prove that ψ is continuous; let $(b, s) \in A \times [0, 1]$. Then, we compute as follows:

$$\begin{aligned} \forall (a, t) \in A \times [0, 1] : \|\psi(a, t) - \psi(b, s)\| &= \|\phi_t(a) - \phi_s(b)\| \\ &= \|\phi_t(a) - \phi_t(b) + \phi_t(b) - \phi_s(b)\| \\ &\leq \|\phi_t(a - b)\| + \|\phi_t(b) - \phi_s(b)\| \\ &\leq \|a - b\| + \|\phi_t(b) - \phi_s(b)\|. \end{aligned}$$

So, what we get is the following estimate:

$$\forall (a, t) \in A \times [0, 1] : \|\psi(a, t) - \psi(b, s)\| \leq \|a - b\| + \|\phi_t(b) - \phi_s(b)\|.$$

If we allow for (a, t) to be close to (b, s) , then we see that the first term on the right-hand side above goes to 0 and the second term goes to zero due to the additional continuity condition on our homotopy.

(2) $\Rightarrow$ (1): Let $\psi : A \times [0, 1] \rightarrow B$ be a $\star$ -homotopy from f to g . We define, for each $t \in [0, 1]$, the map:

$$\phi_t : A \rightarrow B, a \mapsto \phi_t(a) \stackrel{\text{def}}{=} \psi(a, t).$$

This is a $\star$ -homomorphism, by the definition of a $\star$ -homotopy. Now, let $a \in A$ be fixed and define:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a) = \psi(a, t).$$

The continuity of ψ implies that H_a is continuous, so we're done. ■

Now, we can introduce the C^* -Covering Homotopy Property.

Definition 2.50. J. Conway, [19]

Let $A \in C^* \text{Alg}$. A has the C^* -Covering Homotopy Property (or satisfies the C^* -CHP or is C^* -CHP)^a iff for any $\star$ -homotopy $\phi : A \times [0, 1] \rightarrow \mathcal{Q}(H)$ and any $f \in C^* \text{Alg}(A, \mathcal{B}(H))$ such that:

$$\pi \circ f = \phi(\cdot, 0),$$

there is a $\star$ -homotopy $\Phi : A \times [0, 1] \rightarrow \mathcal{B}(H)$ such that $\pi \circ \Phi = \phi$ and $\Phi(\cdot, 0) = f$.

^aThis definition can be adapted to compact metrizable spaces by working with the algebra of continuous functions over such spaces.

The following result explains the relationship between the BDF Ext groups and the functor in [Construction 2.47](#).

Proposition 2.51. C. Schochet, J. Kaminker, [37, p.139]

If X is a compact metrizable space which is C^* -CHP, then:

$$\text{Ext}(X) \simeq \text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

and the isomorphism is natural on the full subcategory of compact metrizable C^* -CHP spaces.

In light of the fact that inj^h is homotopy-invariant, Schochet and Kaminker asked the following question.

| **Question 2.52** ([37, p.139]). Is every compact metrizable space C^* -CHP?

In the aforementioned papers [19] and [20], (John) Conway actually showed that there are a number of interesting examples of C^* -algebras which satisfy the C^* -CHP. In particular, he showed that AF algebras are C^* -CHP, $\mathcal{C}(S^1)$ is C^* -CHP and $\mathcal{C}([0, 1])$ is C^* -CHP. He also posed the following question in less formal form.

Question 2.53 (J. Conway, [19, p.136]). Are the following spaces:

1. $S^1 \vee S^1$ and;
2. $S^1 \vee [1, 2]$.
3. $[-1, 0] \cup \{2^{-n} : n \in \mathbb{N}_0\}$,

C^* -CHP?

In general, it seems difficult to construct many examples of C^* -algebras which are C^* -CHP. It is likely that the spaces in Question 2.53 were chosen since they were just out of reach using the methods of the aforementioned papers. This question will be addressed later, when we have introduced semiprojectivity into our exposition. Now, we have to relate the discussion above to [Conjecture 2.40](#). Let us introduce the first main class of C^* -algebras whose properties will concern us. The motivation for the definition is completely clear given the definition of a Schochet fibration.

Definition 2.54. A. Vats, 2026

Let $A \in C^* \text{Alg}$. A is Schochet iff for any infinite-dimensional, separable $\mathbb{C}$ -Hilbert space H and for any solid diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

(dashed arrow from A to $\mathcal{B}(H)$ labeled $(H_t)_{t \in [0, 1]}$)

which commutes for $t = 0$, there exists a homotopy $(H_t)_{t \in [0, 1]}$ such that:

$$H_0 = f,$$

$$\forall t \in [0, 1] : \pi \circ H_t = h_t.$$

Let $X \in \text{LCHTop}$ and $G \in \text{LCHTopGrp}$. X is Schochet iff $\mathcal{C}_0(X)$ is Schochet. G is Schochet iff $C^*_{\text{red}}(G)$ is Schochet.

In the next result, we will clarify the relationship between Schochet C^* -algebras and C^* -CHP C^* -algebras.

Proposition 2.55

Let $A \in C^* \text{Alg}$. TFAE:

1. A is Schochet.
2. A is C^* -CHP.

Proof.

(1) $\Rightarrow$ (2): Let $\phi : A \times [0, 1] \rightarrow \mathcal{Q}(H)$ be a $\star$ -homotopy and let $f \in C^* \text{Alg}(A, \mathcal{B}(H))$ such that:

$$\pi \circ f = \phi(\cdot, 0).$$

We can construct a homotopy $(h_t : A \rightarrow \mathcal{Q}(H))_{t \in [0,1]}$ from ϕ so that we have a lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

and since A is Schochet, we may simply lift our homotopy to some homotopy $(H_t : A \rightarrow \mathcal{B}(H))_{t \in [0,1]}$. Now, this can be turned into a $\star$ -homotopy which satisfies the requirements of the C^* -CHP property.

(2) $\Rightarrow$ (1): We simply run the argument for (1) $\Rightarrow$ (2) in reverse. ■

Having given all of this background and having posed some interesting questions, it is time that we started working towards answering them and also developing this theory a bit more systematically. Let us, before doing anything else, mention how [Conjecture 2.40](#) can be understood in the context of the material introduced previously.

Proposition 2.56. A. Vats, 2026

The Calkin projection is not a GDR.

Proof. Let us suppose that it's a GDR. By Proposition 2.34, the Calkin projection must be a homotopy equivalence so it follows that $\mathcal{B}(H)$ and $\mathcal{Q}(H)$ have isomorphic K_1 -groups. However, this is clearly not the case as:

$$K_1(\mathcal{B}(H)) \simeq 0,$$

and $K_1(\mathcal{Q}(H)) \simeq \mathbb{Z}$. ■

We have two remarks about this proof.

Remark 2.57. In fact, we can use a similar argument to show that the canonical projection cannot even have a right homotopy inverse. So, there does not seem to be any hope in utilizing Proposition 2.30 to obtain a contradiction.

Remark 2.58. The argument used above can be generalized to show that other morphisms of C^* -algebras are not GDRs. Indeed, we can do this for morphisms where the domain and codomain do not have isomorphic K -groups.

Let us resolve [Conjecture 2.40](#).

Theorem 2.59. A. Vats, 2026

The following universal C^* -algebra:

$$A \stackrel{\text{def}}{=} C^*(x | xx^* = x^*x \wedge \|x\| \leq 1).$$

is not Schochet. That is, [Conjecture 2.40](#) holds.

Proof. Let $H = \ell^2(\mathbb{N})$, let S be the unilateral shift in $\mathcal{B}(H)$ and let s be the image of S in $\mathcal{Q}(H)$ under the Calkin projection. Then, the C^* -algebra generated by S in $\mathcal{B}(H)$ is the Toeplitz C^* -algebra $\mathcal{T}$ and it contains $\mathcal{K}(H)$. Moreover, it is a known fact that $\mathcal{T}/\mathcal{K}(H) \simeq \mathcal{C}(S^1)$. Now, we will need two subclaims.

Subclaim. There is no normal element $a \in \mathcal{B}(H)$ so that $\pi(a) = s$.

This simple fact is stated without proof in [12, p.1]; we will establish it at the end of this proof, after we have dealt with the main result. Now, the second subclaim is much more standard so we will point to a reference.

Subclaim. Let $E = \{x_i : i \in I\}$ be a set of generators, let R be relations between these generators such that the universal C^* -algebra $C^*(E|R)$ exists and let $B \in C^*\text{Alg}$ contain a set $E' = \{y_i : i \in I\}$. If the elements of E' satisfy the relations R , there is a unique $\star$ -homomorphism $\phi : C^*(E|R) \rightarrow B$ sending x_i to y_i for each $i \in I$.

For the proof and further systematic discussion about universal C^* -algebras, see [40, Proposition 6.7, p.71]. By subclaim (1), s does not have a unitary preimage in $\mathcal{B}(H)$ via the Calkin projection. Now, we define a map

$$\phi(x) : [0, 1] \rightarrow \mathcal{Q}(H), \quad t \mapsto \phi(x)(t) \stackrel{\text{def}}{=} ts.$$

This is continuous since:

$$\begin{aligned} \forall t, r \in [0, 1] : \|\phi(x)(t) - \phi(x)(r)\| &= \|ts - rs\| \\ &= |t - r| \cdot \|s\| \\ &\leq |t - r|. \end{aligned}$$

So, $\phi(x) \in \text{Cyl}(\mathcal{Q}(H))$. Moreover, $\phi(x)$ actually maps into $\text{Cyl}(\mathcal{T}/\mathcal{K}(H))$ and this is commutative so it follows that $\phi(x)$ is normal in $\mathcal{Q}(H)$. Now, we also have the following estimate:

$$\|\phi(x)\| = \sup_{t \in [0,1]} \|\phi(x)(t)\| = \sup_{t \in [0,1]} \|ts\| \leq 1.$$

So, $\phi(x)$ is a normal contraction in $\text{Cyl}(\mathcal{Q}(H))$. By the universal property of universal C^* -algebras (subclaim (2)), this induces a $\star$ -homomorphism $\phi : A \rightarrow \text{Cyl}(\mathcal{Q}(H))$. That is, this is a homotopy from A to $\mathcal{Q}(H)$, from the zero $\star$ -homomorphism to the $\star$ -homomorphism which sends x to s ; the former is ϕ_0 and the latter is ϕ_1 . Now, the key point

is that ϕ_0 lifts to $\mathcal{B}(H)$ while ϕ_1 does not lift to $\mathcal{B}(H)$ due to subclaim (1). Indeed, if ϕ_1 did have a lift $\psi : A \rightarrow \mathcal{B}(H)$, then $\pi(\psi(x)) = s$ with $\psi(x)$ being a normal contraction because ψ is contractive as a $\star$ -homomorphism. So, we have constructed a homotopy from ϕ_0 to ϕ_1 with ϕ_0 being liftable and ϕ_1 not being liftable, so we're done.

Now, let us deal with the proof of subclaim (1). A very classical and direct index theory calculation reveals that:

$$\text{Index}(S) = -1.$$

If $T \in \mathcal{B}(H)$ is normal, then we know that:

$$\text{coker}(T) \simeq \ker(T^*).$$

So, we see that:

$$\begin{aligned} \text{Index}(T) &= \dim(\ker(T)) - \dim(\text{coker}(T)) \\ &= \dim(\ker(T)) - \dim(\ker(T^*)) \\ &= 0, \end{aligned}$$

where the last statement holds, again, because normality is equivalent to the assertion that:

$$\forall x \in H : \|T(x)\| = \|T^*(x)\|.$$

On the other hand, let $T \in \mathcal{B}(H)$ so that:

$$\pi(T) = s = \pi(S),$$

which implies that $T - S \in \mathcal{K}(H)$. But this means that their Fredholm indices coincide. In particular, the Fredholm index of T is -1 and this is impossible, as we have demonstrated above. ■

As a consequence, we also resolve Question 2.52.

Corollary 2.60

There exists a compact metrizable space which is not C^* -CHP. So, Question 2.52 has a negative answer.

Proof. Let us take the compact unit disk $\mathbb{D}$. Then:

$$\mathcal{C}(\mathbb{D}) \simeq C^*(x, 1 | xx^* = x^*x \wedge \|x\| \leq 1).$$

The same argument applied to this C^* -algebra shows that it is not Schochet, so $\mathbb{D}$ is not Schochet and not C^* -CHP. ■

At a later point, we will see that this corollary follows easily from a more general permanence property of Schochet C^* -algebras involving the minimal unitization. Let us describe the inspiration behind the construction above.

Remark 2.61. Our argument in Theorem 2.59 was inspired by the proof of (Bruce) Blackadar that the C^* -algebra constructed in the theorem is not l -closed^a; see [10, Example 6.5]. In fact, the C^* -algebra above, which is concretely realized as $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$, has played a rather important role in the history of semiprojectivity. It seems to have been a rather difficult open problem to establish the approximate semiprojectivity^b of this C^* -algebra. At the moment, there is even an open problem about it. Even though we do not resolve this problem, our contribution was to highlight an interesting obstruction in solving them and this relates back to some problems in operator theory.

^aWe will introduce this notion in Definition 2.65 and explain why it made sense to look at different examples of l -closed C^* -algebras.

^bSee statement (2) of Terminology 2.119.

Let us expand on Remark 2.61. Indeed, the C^* -algebras used in Theorem 2.59 and Corollary 2.60 are not extremely well-understood from a purely topological viewpoint. Indeed, we do not even know if $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed. To explain what this means, we need a bit of preparation. Let us introduce two pieces of convenient terminology and a construction.

Terminology 2.62. Let $A \in C^*\text{Alg}$ and let $I \subseteq A$ be a closed ideal. Then, the pair:

$$(A, I),$$

is known as a marked C^* -algebra.

Terminology 2.63. Let $A \in C^*\text{Alg}$, let (B, I) be a marked C^* -algebra, let $\pi_I : B \rightarrow B/I$ be the canonical projection and let $f \in C^*\text{Alg}(A, B/I)$. f is liftable iff there exists a $\star$ -homomorphism $\bar{f}$ such that the following diagram:

$$\begin{array}{ccc} & & B \\ & \nearrow \bar{f} & \downarrow \pi_I \\ A & \xrightarrow{f} & B/I \end{array}$$

commutes.

Here is our construction.

Construction 2.64. Let $A, B \in C^*\text{Alg}$. For each $a \in A$, we define a semi-metric:

$$p_a : \text{Ban}_{\mathbb{C}}(A, B) \times \text{Ban}_{\mathbb{C}}(A, B) \rightarrow [0, \infty) , (f, g) \mapsto p_a(f, g) \stackrel{\text{def}}{=} \|f(a) - g(a)\|.$$

This gives us a family of seminorms on $\text{Ban}_{\mathbb{C}}(A, B)$ and this generates a locally convex topology on $\text{Ban}_{\mathbb{C}}(A, B)$; we will refer to this as the point-norm topology. Now, we take the subspace topology on $C^*\text{Alg}(A, B)$ and this set equipped with this topology will be denoted by $\text{Hom}(A, B)$. If A and B are unital, then we denote the (topological)

subspace of unital $\star$ -homomorphisms by $\text{Hom}_1(A, B)$. Finally, if $I \subseteq B$ is a closed ideal, then we define:

$$\text{Hom}(A, B, I) \stackrel{\text{def}}{=} \pi_I \circ \text{Hom}(A, B),$$

as the topological subspace of liftable morphisms in $\text{Hom}(A, B/I)$.

Now, we can define l -openness and closedness.

Definition 2.65

Let $A \in C^*\text{Alg}$ and let (B, I) be a marked C^* -algebra.

1. A is (B, I) - l -open iff $\text{Hom}(A, B, I)$ is open in $\text{Hom}(A, B/I)$.
2. A is (B, I) - l -closed iff $\text{Hom}(A, B, I)$ is closed in $\text{Hom}(A, B/I)$.

Finally, A is l -open (closed) iff it is (B, I) - l -open (closed) for every marked C^* -algebra^a (B, I) .

^aIf we are working with the marked C^* -algebra $(\mathcal{B}(H), \mathcal{K}(H))$, then we will say that A is either H - l -open or H - l -closed.

Let us state our desired question more formally.

Question 2.66 (B. Blackadar, 2016, [10, Example 6.5, p.8-9]). Is $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ l -closed^a?

^a $\mathbb{D}$ is the compact unit disk in $\mathbb{C}$ centered at the origin.

We were not able to answer this question fully but we have managed to “explain” why one may expect it to be a difficult problem. Before we get to that, let us review what is known in the literature, starting with some terminology.

Terminology 2.67. Let (A, I) be a marked C^* -algebra and let $b \in A/I$ satisfy some property P which is satisfied by elements of C^* -algebras. b is P -liftable iff there exists an $a \in A$ such that $\pi(a) = b$ and a satisfies P .

We will also need some notation.

Notation 2.68. Let $A \in C^*\text{Alg}$. Then, the set of normal elements in A is denoted by N_A .

The following result characterizes when Question 2.66 has a positive answer.

Proposition 2.69. B. Blackadar (2016), H. Thiel (2020), [10, Example 6.5, p.8-9], [70]

TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed.
2. If (A, I) is a marked C^* -algebra and let $(x_n)_{n \in \mathbb{N}}$ be a sequence of normal elements in A/I which converges to some $x \in A/I$. If x_n is normal-liftable for each $n \in \mathbb{N}$, then x is also normal-liftable.
3. If (A, I) is a marked C^* -algebra, then $\pi(N_A)$ is closed in B/I , where $\pi : A \rightarrow A/I$ is the canonical projection.

We have a remark about this proposition.

Remark 2.70. Even though it is not proved explicitly in the reference, the equivalence of statements (1) and (2) depend on the characterization of $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ as the universal C^* -algebra generated by a normal contraction.

Now, we state our result.

Theorem 2.71. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. The sum $N_A + I$ is closed in A .
2. Any convergent sequence in $N_{A/I}$ whose terms are normal-liftable must have a normal-liftable limit^a.

^aThe lifting occurs through the canonical projection $\pi : A \rightarrow A/I$.

The proof of this result requires the C^* -Bartle-Graves Theorem. The Banach-space version of this result is an old theorem, though the version that we will provide now is a much newer result due to (Tatiana) Shulman which we hope to use later as a way of simplifying some proofs involving the Schochet property.

Theorem 2.72. The C^* -Bartle-Graves Theorem

Let $A, B \in C^*\text{Alg}$ and let $\pi \in C^*\text{Alg}(A, B)$ be surjective. Then, there exists a $s \in \mathcal{C}(B, A)$ so that:

1. s is contractive and;
2. s is unital (if A and B are unital, with π also being unital) and;
3. $\forall b \in B : s(b^*) = s(b)^*$ and;
4. $\forall b \in B^+ : s(b) \in A^+$.

Moreover, if $a_0 \in A$ and $b_0 \in B$ with $\pi(a_0) = b_0$, then s can be chosen so that:

$$s(b_0) = a_0,$$

and so that it still satisfies the above properties. We refer to s as the C^* -section of π .

Here's a remark about this result.

Remark 2.73. For the reader who wishes to see the Banach space version of this result, see [50, p.364]; there are also more modern references with cleaner proofs. For the theorem above and its proof, see [68, Theorem 5, p.7-8].

This leads us to the proof of our result.

Proof of Theorem 2.71.

(1) $\Rightarrow$ (2): We will use the C^* -Bartle-Graves Theorem as stated in Theorem 2.72; since π is a surjective $\star$ -homomorphism, it has a C^* -section $s : A/I \rightarrow A$. Now, let $(x_n + I)_{n \in \mathbb{N}}$ be a convergent sequence in A/I with each x_n being normal in A and let $x + I$ be the limit. We define:

$$\begin{aligned} \forall n \in \mathbb{N} : y_n &\stackrel{\text{def}}{=} s(x_n + I), \\ y &\stackrel{\text{def}}{=} s(x + I). \end{aligned}$$

Since s is continuous, we see that $y_n \rightarrow y$. Now, observe that:

$$\begin{aligned} \forall n \in \mathbb{N} : y_n + I &= \pi(y_n) \\ &= \pi(s(x_n + I)) \\ &= x_n + I, \end{aligned}$$

which means that $y_n - x_n \in I$ for each $n \in \mathbb{N}$. In other words, each $y_n \in N_A + I$; we use the normality of each x_n over here. Since $(y_n)_{n \in \mathbb{N}}$ is convergent and $N_A + I$ is closed, we see that $y \in N_A + I$. So, we write:

$$y = a + k,$$

for some normal a and $k \in I$. But now, we see that:

$$\begin{aligned} x + I &= (\pi \circ s)(x + I) \\ &= \pi(s(x + I)) \\ &= \pi(y) \\ &= \pi(a + k) \\ &= \pi(a) + \pi(k) \\ &= \pi(a), \end{aligned}$$

which proves the result.

(2) $\Rightarrow$ (1): Let $(a_n + k_n)_{n \in \mathbb{N}}$ be a sequence in $N_A + I$ which converges to some $x \in A$, where each a_n is normal and each $k_n \in I$. Then, we have that $(\pi(a_n + k_n))_{n \in \mathbb{N}}$ is a sequence in A/I which converges to $\pi(x)$. Moreover, this sequence in A/I consists of terms which are normal-liftable, so its limit is normal-liftable. That is:

$$\exists a \in N_A : \pi(a) = \pi(x).$$

This implies that $x - a \in I$ so $x \in N_A + I$. ■

The following is an immediate corollary.

Corollary 2.74. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is (A, I) - l -closed.
2. $N_A + I$, as defined in Theorem 2.71, is closed in A .

So, $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed iff for every marked C^* -algebra (A, I) , we have that $N_A + I$ is closed in A .

Proof. This follows by combining Proposition 2.69 and Theorem 2.71. ■

This leads us to introduce the following piece of terminology.

Terminology 2.75. Let $A \in C^* \text{Alg}$. A is normally closed iff for every closed ideal $I \subseteq A$, we have that:

$$N_A + I,$$

is closed in A .

So, we have the following corollary.

Corollary 2.76

$\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed for all normally closed C^* -algebras.

In fact, it turns out that there are many examples of normally closed C^* -algebras.

Example 2.77. Let $A \in \text{Ab}C^* \text{Alg}$. Then, A is clearly normally closed, since the normal elements of A are all elements of A . So, if $I \subseteq A$ is a closed ideal, then $N_A + I = A + I$ and this is closed.

Example 2.78. Let $A \in C^* \text{Alg}$ be simple. Then, A is clearly normally closed. Inded, let $I \subseteq A$ be a closed ideal and note that $I = 0$ or $I = A$. If I is trivial, then we see that $N_A + I = N_A$ and this is closed. If $I = A$, then $N_A + A$ is also closed; the limit x of any convergent sequence in $N_A + A$ can be written just as $0 + x$.

Example 2.79. Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Then, $\mathcal{B}(H)$ is normally closed. Of course, the main point is to simply prove that $N_{\mathcal{B}(H)} + \mathcal{K}(H)$ is closed in $\mathcal{B}(H)$. Indeed, that is true but the classical proof uses Brown-Douglas-Fillmore theory; see [27, Corollary 11.4, p.119].

Example 2.80. Let H be an infinite-dimensional, nonseparable $\mathbb{C}$ -Hilbert space and let $I \subseteq \mathcal{B}(H)$ be a closed ideal which is not the compact operators in $\mathcal{B}(H)$. The results of [76] tell us that every normal element in $\mathcal{B}(H)/I$ is normal-liftable through the canonical projection. So, $N_{\mathcal{B}(H)} + I$ is closed in $\mathcal{B}(H)$. The problem for the compact operators is still open and the proof in the aforementioned paper does not seem to generalize to other C^* -algebras.

It is, in fact, the case that the condition of being normally closed does arise naturally in the context of lifting problems. In [70], Thiel introduces the notion of a universally liftable normal element in a C^* -algebra and presents some reasonable questions about this notion. Let us refine his definition just a bit and examine this more closely.

Definition 2.81

Let $\mathcal{C}$ be a full subcategory of $C^*\text{Alg}$, let $A \in C^*\text{Alg}$ and let $a \in A$ be normal.

1. a is universally liftable for $\mathcal{C}$ iff for every $B \in \mathcal{C}$ and every surjective $\star$ -homomorphism:

$$\pi : B \rightarrow A,$$

there exists a normal $b \in B$ such that $\pi(b) = a$.

2. a is universally liftable iff it is universally liftable for $C^*\text{Alg}$.

The set of normal elements in A which are universally liftable for $\mathcal{C}$ is denoted by ${}^a\text{UnivLift}_{\mathcal{C}}(A)$.

^aWe omit the $\mathcal{C}$ when $\mathcal{C}$ is just $C^*\text{Alg}$.

There are two main questions of interest regarding universally liftable elements.

| **Question 2.82** (H. Thiel, 2020, [70]). Let $A \in C^*\text{Alg}$. Is $\text{UnivLift}(A)$ closed?

| **Question 2.83** (H. Thiel, 2020, [70]). Let $A \in C^*\text{Alg}$. Is $\text{UnivLift}(A)$ open?

The next result indicates progress on Question 2.82.

Proposition 2.84. A. Vats, 2026

Let $A \in C^*\text{Alg}$. The set of universally liftable elements for normally closed C^* -algebras is closed.

Proof. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of universally liftable normal elements^a in A which converges to some normal $a \in A$. Let $B \in C^*\text{Alg}$ be normally closed and let $\pi : B \rightarrow A$ be a surjective $\star$ -homomorphism. For each $n \in \mathbb{N}$, there is a normal $b_n \in B$ so that $\pi(b_n) = a_n$. Now, we identify A with $B/\ker(\pi)$. Since B is normally closed, we can simply use Theorem 2.71 to conclude that a is normal-liftable so we're done. ■

^aFor normally closed C^* -algebras.

If $X \subseteq \mathbb{C}$ is compact, then we see that:

$$\mathcal{C}(X) \simeq C^*(z),$$

where z is a normal element whose spectrum is all of X . In other words, $\mathcal{C}(X)$ is realized as a universal C^* -algebra in this way and this means that the Schochet property for such subsets of the plane boils down to lifting properties of normal elements in the Calkin algebra. To that end, there is an interesting topological characterization of those compact subsets of $\mathbb{C}$ which admit non-normal-liftable normal elements in the Calkin algebra.

Theorem 2.85. T. Shulman, D. Enders, 2019, [26, Proposition 5.13, p.26-27]

Let $X \subseteq \mathbb{C}$ be compact. TFAE:

1. Any normal element in $\mathcal{Q}(H)$ with spectrum in X is normal-liftable.
2. Any normal element in $\mathcal{Q}(H)$ with spectrum in X lifts to a normal operator with the same spectrum.
3. $\dim(X) \leq 1$ and $H^1(X, \mathbb{Z}) \simeq 0$.

For the sake of keeping our work somewhat self-contained, let us review the definition of the Lebesgue covering dimension and the definition of the first Čech cohomology group. We will also discuss some basic examples. The reader who is familiar with these notions may feel free to skip to Proposition 2.86.

[REVIEW IT]

Now, the proof of Theorem 2.59 and the results above provide a criterion for determining that a given compact subset of $\mathbb{C}$ is not a Schochet space. This is recorded in the proof of the next result.

Proposition 2.86. A. Vats, 2026

Let $X \subseteq \mathbb{C}$ satisfy the following conditions:

1. it is compact,

2. $0 \in X$,
3. the following topological subspace of $\mathcal{Q}(H)$:

$$N_X \stackrel{\text{def}}{=} \{a \in \mathcal{Q}(H) : a \text{ is normal} \wedge \sigma_{\mathcal{Q}(H)}(a) = X\},$$

is path-connected,

and suppose that either $\dim(X) = 2$ or $H^1(X, \mathbb{Z})$ is non-trivial. Then, $\mathcal{C}(X)$ is not Schochet.

Proof. If $\dim(X) = 2$ or $H^1(X, \mathbb{Z})$ is non-trivial, then [Theorem 2.85](#) we find that there exists a normal element $y \in \mathcal{Q}(H)$ whose spectrum is contained in X but which is not normal-liftable. Now, $y \in N_X$ and $0 \in N_X$ since $0 \in X$. Since N_X is path-connected, we can find a path $\gamma : [0, 1] \rightarrow N_X$ such that $\gamma(0) = 0$ and $\gamma(1) = y$. The existence of this path allows us to construct a homotopy of $\star$ -homomorphisms (by universal property) between the zero $\star$ -homomorphism and the one which maps the inclusion $\iota : X \rightarrow \mathbb{C}$ to y . We see that this homotopy is initially liftable but that is not the case at the end, so we conclude that $\mathcal{C}(X)$ cannot be Schochet. ■

This gives us the following corollary.

Corollary 2.87. A. Vats, 2026

Let $X \subseteq \mathbb{C}$ be compact so that:

$$\forall t \in [0, 1] : tX \subseteq X,$$

and either $\dim(X) = 2$ or $H^1(X, \mathbb{Z})$ is non-trivial. Then, $\mathcal{C}(X)$ is not Schochet.

Proof. Indeed, we just need to show that N_X is path-connected. Since 0 lies in X , it is sufficient to show that any $a \in N_X$ is connected by some path to 0 . For this, we define the following map:

$$\gamma : [0, 1] \rightarrow N_X, \quad t \mapsto \gamma(t) \stackrel{\text{def}}{=} (1-t)a.$$

So, this is a path from a to 0 . Moreover:

$$\begin{aligned} \forall t \in [0, 1] : \sigma_{\mathcal{Q}(H)}(\gamma(t)) &= \sigma_{\mathcal{Q}(H)}((1-t)a) \\ &= (1-t)\sigma_{\mathcal{Q}(H)}(a) \\ &\subseteq (1-t)X \\ &\subseteq X, \end{aligned}$$

so we're done. ■

We have a remark about the results above.

Remark 2.88. It would actually be very desirable to have an analogue of [Theorem 2.85](#) with $\mathcal{Q}(H)$ replaced by A/I for some $A \in \mathbf{C}^*\mathbf{Alg}$ and I being a closed ideal in A ; it does not seem to be the case that the methods of proving the aforementioned theorem as found in [\[26\]](#) generalize. If we had such a criterion, then that would allow us to

Now, we should also discuss some permanence properties since this will give us access to some examples.

Proposition 2.89. A. Vats, 2026

TFSH:

1. If $A, B \in \mathbf{C}^*\mathbf{Alg}$ are Schochet, then $A \star B$ is Schochet.
2. If $A, B \in \mathbf{C}^*\mathbf{Alg}$ with A being Schochet and a split epi $\phi \in \mathbf{C}^*\mathbf{Alg}(A, B)$, then B is Schochet.
3. If $A, B \in \mathbf{C}^*\mathbf{Alg}$ with A and B being $\star$ -isomorphic and A being Schochet, then B is Schochet.
4. If $A, B \in \mathbf{C}^*\mathbf{Alg}$ with A being Schochet and a Schochet fibration $f \in \mathbf{C}^*\mathbf{Alg}(A, B)$ so that:

$$\exists g \in \mathbf{C}^*\mathbf{Alg}(B, A) : f \circ g \simeq_h \text{Id}_B,$$

then B is Schochet.

5. If $A, B \in \mathbf{C}^*\mathbf{Alg}$ with $A \oplus B$ being Schochet, then A and B are Schochet.
6. If $A \in \mathbf{C}^*\mathbf{Alg}$ and $M_{n \times n}(A)$ is Schochet for some $n \in \mathbb{N}$, then A is Schochet.
7. If $A \in \mathbf{C}^*\mathbf{Alg}$ and $\text{Cyl}(A)$ is Schochet, then A is Schochet.

Proof. We'll prove each statement one-by-one. In each case, we will start with the following diagram:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ & \uparrow f & \\ D & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \\ & \downarrow \pi & \end{array} \quad (1)$$

where $(h_t)_{t \in [0, 1]}$ is a homotopy such that the diagram above commutes when $t = 0$.

1. Let $D = A \star B$ and let ι_A, ι_B be the respective inclusion maps. In the case of A , we have a diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f \circ \iota_A & \downarrow \pi \\ A & \xrightarrow{(h_t \circ \iota_A)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

which commutes at $t = 0$ because the one in (1) commutes when $t = 0$; a similar diagram exists in the case of B . Since A is Schochet, there is a homotopy lift $(H_t^A)_{t \in [0,1]}$ such that $H_0^A = f \circ \iota_A$ and we have that:

$$\forall t \in [0, 1] : \pi \circ H_t^A = h_t \circ \iota_A.$$

Similarly, we have a homotopy lift $(H_t^B)_{t \in [0,1]}$ for the corresponding diagram for B . By the universal property of the free product, we obtain a unique homotopy $(H_t : A \star B \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that each H_t restricts to H_t^A and H_t^B . Now, we need to check that this works. Indeed, we need to check that $H_0 = f$. For this, we note that:

$$H_0 \circ \iota_A = H_0^A = f \circ \iota_A,$$

$$H_0 \circ \iota_B = H_0^B = f \circ \iota_B.$$

By the uniqueness part of the universal property for the free product, we see that $H_0 = f$. Let's check that:

$$\forall t \in [0, 1] : \pi \circ H_t = h_t.$$

So, let $t \in [0, 1]$. Then:

$$(\pi \circ H_t) \circ \iota_A = \pi \circ (H_t \circ \iota_A) = \pi \circ H_t^A = h_t \circ \iota_A,$$

$$(\pi \circ H_t) \circ \iota_B = \pi \circ (H_t \circ \iota_B) = \pi \circ H_t^B = h_t \circ \iota_B,$$

which implies, by uniqueness again, that our desired equality holds.

2. Let $D = B$ and consider the following diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f \circ \phi & \downarrow \pi \\ A & \xrightarrow{(h_t \circ \phi)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Since A is Schochet, we obtain a homotopy lift $(F_t : A \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that:

$$F_0 = f \circ \phi,$$

$$\forall t \in [0, 1] : \pi \circ F_t = h_t \circ \phi.$$

Now, let ψ be a split for ϕ . Then, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t \circ \psi.$$

In that case, we see that:

$$H_0 = F_0 \circ \psi = (f \circ \phi) \circ \psi = f,$$

and we also have that:

$$\forall t \in [0, 1] : \pi \circ H_t = \pi \circ F_t \circ \psi = h_t \circ \phi \circ \psi = h_t,$$

which proves the result.

3. This follows from the previous statement, since $\star$ -isomorphisms are split epis.
4. We see that f is split epi by Proposition 2.30 so we're done.
5. Let us consider the projection $\pi_A : A \oplus B \rightarrow A$ and define:

$$\iota_A : A \rightarrow A \oplus B, \quad a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0).$$

Then, we see that:

$$\begin{aligned} \forall a \in A : (\pi_A \circ \iota_A)(a) &= \pi_A(a, 0) \\ &= a, \end{aligned}$$

which means that π_A is a split-epi since ι_A is a $\star$ -homomorphism. So, A is Schochet and a similar argument yields that B is also Schochet.

6. We consider the following homotopy lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

where $h_0 = \pi \circ f$. Now, we apply the functor $M_{n \times n}(-)$ to this diagram so we obtain:

$$\begin{array}{ccc} & & M_{n \times n}(\mathcal{B}(H)) \\ & \nearrow^{M_{n \times n}(f)} & \downarrow^{M_{n \times n}(\pi)} \\ M_{n \times n}(A) & \xrightarrow{(M_{n \times n}(h_t))_{t \in [0,1]}} & M_{n \times n}(\mathcal{Q}(H)) \end{array}$$

Next, we will make some identifications as follows:

- (a) We will identify $M_{n \times n}(\mathcal{B}(H))$ with $\mathcal{B}(H^{\oplus n})$, where we note that $H^{\oplus n}$ is still an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space so we are still in the same set-up.
- (b) We will identify $M_{n \times n}(\mathcal{Q}(H))$ with $\mathcal{Q}(H^{\oplus n})$.
- (c) We will identify $M_{n \times n}(\pi)$ with the Calkin projection on $\mathcal{B}(H^{\oplus n})$.
- (d) If we view the homotopy $(h_t)_{t \in [0,1]}$ as a map from A into $\text{Cyl}(\mathcal{Q}(H))$, then we can view $(M_{n \times n}(h_t))_{t \in [0,1]}$ as a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(M_{n \times n}(\mathcal{Q}(H)))$ and due to our identification above, this is just a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(\mathcal{Q}(H^{\oplus n}))$.

With these identifications, we see that we just have a lifting problem for $M_{n \times n}(A)$ and since this is Schochet, we find a homotopy lift $(F_t : M_{n \times n}(A) \rightarrow \mathcal{B}(H^{\oplus n}))_{t \in [0,1]}$. Now, we precompose each F_t with the left-upper corner inclusion of A into $M_{n \times n}(A)$ and postcompose it with the canonical isomorphism between $M_{n \times n}(\mathcal{B}(H))$ and $\mathcal{B}(H^{\oplus n})$. This is the homotopy lift for the lifting problem associated to A .

7. Let's consider the $\star$ -homomorphism $p_0^A : \text{Cyl}(A) \rightarrow A$ and define:

$$s : A \rightarrow \text{Cyl}(A), \quad a \mapsto s(a),$$

where $s(a)$ is the constant function on $[0, 1]$ with value a . Note that:

$$\begin{aligned} \forall a \in A : (p_0^A \circ s)(a) &= p_0^A(s(a)) \\ &= s(a)(0) \\ &= a \\ &= \text{Id}_A(a), \end{aligned}$$

which implies that A is Schochet. ■

The next corollary is due to Conway but we prove it using the proposition.

Corollary 2.90

Let $X, Y \in \text{LCHTop}$ be homeomorphic, with X being Schochet. Then, Y is Schochet.

Proof. Since X and Y are homeomorphic, we see that $\mathcal{C}_0(X)$ and $\mathcal{C}_0(Y)$ are $\star$ -isomorphic and $\mathcal{C}_0(X)$ is Schochet. Hence, it follows that $\mathcal{C}_0(Y)$ is Schochet and this means that Y is Schochet. ■

The next one gives us more examples of Schochet spaces.

Corollary 2.91. A. Vats, 2026

Let $X \in \text{CHTop}$. TFAE:

1. X is Schochet.
2. Every clopen subset Y of X is Schochet.

Proof. (2) $\Rightarrow$ (1) is clear, since X is clopen in itself. So, it remains to prove (1) $\Rightarrow$ (2). We define:

$$\phi : \mathcal{C}(X) \rightarrow \mathcal{C}(Y), \quad f \mapsto \phi(f) \stackrel{\text{def}}{=} f|_Y.$$

Note that this is a $\star$ -homomorphism. Now, we define:

$$\psi : \mathcal{C}(Y) \rightarrow \mathcal{C}(X), \quad f \mapsto \psi(f),$$

where we simply take a function and extend it to X through continuation by 0; this extension will be continuous^a and the map above is a $\star$ -homomorphism. Now, we observe that ψ is a split for ϕ and since X is Schochet, it follows from Proposition 2.89 that Y is Schochet. ■

^aWe illustrate the argument for points in $X \setminus Y$ and note that the argument for continuity at points in Y will rely on it being an open set. Let $p \in X \setminus Y$ and let $(p_\delta)_{\delta \in D}$ be a net that converges to p . Since Y is closed, we see that $X \setminus Y$ is open so the net eventually lies in $X \setminus Y$. But this means that $f(p_\delta) = 0$ for each $\delta \in D$ and this implies that $f(p) = 0$.

Due to the availability of stronger theorems and more palatable descriptions of certain C^* -algebras, it is actually possible for us to establish that the examples highlighted by Conway in [19] and [20] are actually Schochet C^* -algebras. Indeed, his methods were rather technical and we are in a position where we can give some simplified proofs. Let us illustrate that establishing this property for a C^* -algebra is a non-trivial task.

Example 2.92. The next claim is the main claim of this example.

| **Claim.** $\mathbb{C}$ is a Schochet C^* -algebra.

The proof uses a subclaim that we will not prove since it is a well-known result.

| **Subclaim.** Let $p \in \mathcal{Q}(H)$ be a projection. Then, there is a projection P such that^a $p = \pi(P)$.

^aFor the proof, see [74, Proposition 3.1, p.11-12].

We have the following lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathbb{C} & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

If $t \in [0, 1]$, then $h_t(1)$ is a projection and the subclaim allows us to find a projection $S_t \in \mathcal{B}(H)$ such that:

$$\pi(S_t) = h_t(1),$$

with S_0 being declared to be $f(1)$. In that case, we define:

$$H_t : \mathbb{C} \rightarrow \mathcal{B}(H), \quad z \mapsto H_t(z) \stackrel{\text{def}}{=} zS_t.$$

Since S_t is a projection, this is a $\star$ -homomorphism^a. Moreover:

$$\forall z \in \mathbb{C} : H_0(z) = zS_0 = zf(1) = f(z),$$

$$\forall t \in [0, 1] : \forall z \in \mathbb{C} : (\pi \circ H_t)(z) = \pi(H_t(z)) = \pi(zS_t) = z\pi(S_t) = zh_t(1) = h_t(z),$$

which shows that our constructed lift works.

So, even with a C^* -algebra as simple as $\mathbb{C}$, it does take some effort to prove the Schochet property directly.

^aLinearity is clear simply by the properties of scalar multiplication of operators. The fact that this preserves products follows from the fact that S_t is idempotent. Finally, the fact that this preserves the involution follows from the self-adjointness of S_t .

In the next example, we will show that $[0, 1]$ is a Schochet space. The lifting problem for the commutative C^* -algebra $\mathcal{C}([0, 1])$ essentially boils down to lifting paths of positive contractions. We will deal with this in the next proposition.

Proposition 2.93. A. Vats, 2026

Let $(u_t)_{t \in [0,1]}$ be a path of positive contractions in $\mathcal{Q}(H)$ such that there is a positive contraction U_0 so:

$$\pi(U_0) = u_0.$$

Then, there exists a path $(U_t)_{t \in [0,1]}$ of positive contractions in $\mathcal{B}(H)$ so that $\pi(U_t) = u_t$ for each $t \in [0, 1]$.

Proof. Let $s : \mathcal{Q}(H) \rightarrow \mathcal{B}(H)$ be the unital C^* -section of π so that $s(u_0) = U_0$. We have a claim.

| **Claim.** $(s(u_t))_{t \in [0,1]}$ is a path of positive contractions in $[0, 1]$.

Since s is positive, we see that $s(u_t) \geq 0$ for each $t \in [0, 1]$. Now, the claim that:

$$\forall t \in [0, 1] : s(u_t) \leq 1,$$

is equivalent to the claim that:

$$\forall t \in [0, 1] : \|s(u_t)\| \leq 1.$$

But this holds because s is a contraction and $\|u_t\| \leq 1$ for each $t \in [0, 1]$.

In fact, this completes the proof because $\pi(s(u_t)) = u_t$ for each $t \in [0, 1]$ so we have our desired lift. ■

Now, we can state and deal with our example.

Example 2.94. The following claim is the main one for this example.

| **Claim.** $[0, 1]$ is a Schochet space.

This follows simply by writing out the lifting problem and noting that $\mathcal{C}([0, 1])$ is the universal unital C^* -algebra generated by a positive contraction. In that case, the lifting problem boils down to finding lifts for paths of positive contractions in the Calkin algebra and such lifts are afforded to us by the proposition.

Again, the proof here is only simplified by the fact that we are using a much stronger form of the Bartle-Graves Theorem which tells us so much more about the continuous section of our surjective $\star$ -homomorphism.

The next example is quite non-trivial as well but increases our range of examples.

Example 2.95. Let A be an AF algebra.

| **Claim.** A is Schochet.

While this was proved by Conway in [20], the proof seems to not be so amenable to generalization. There is another proof in [54, Theorem 2.2.9] and it seems more reasonable to attempt to generalize this proof. The main obstruction to this is Lemma 2.2.7 in the same reference. We could not find the cited paper which proves that lemma, so we point to the proof of [57, Lemma 1.3, p.958-959] instead. Now, we also have the following claim.

| **Claim.** All totally disconnected compact Hausdorff spaces are Schochet spaces.

Indeed, continuous functions over such spaces are unital commutative AF algebras so we're done.

In fact, all commutative von Neumann algebras are Schochet C^* -algebras.

Example 2.96. Let $A \subseteq \mathcal{B}(H)$ be a commutative von Neumann algebra.

| **Claim.** A is a Schochet C^* -algebra.

Indeed, the Gelfand spectrum of any commutative von Neumann algebra is extremally disconnected and Hausdorff, so it is totally disconnected. Hence, it is a Schochet space and this implies the desired claim.

We will note that this argument works for any commutative C^* -algebra with extremally disconnected spectrum. Indeed, all such C^* -algebras are Schochet.

Since $[0, 1]$ is an example of a Schochet space, we would like to conclude that $(0, 1]$ is also Schochet. Since $[0, 1]$ is just the one-point compactification of $(0, 1]$, we will need to demonstrate some compatibility between the Schochet property and one-point compactifications. At the level of C^* -algebras, this amounts to showing some compatibility between the Schochet property and unitizations. The next piece of terminology is a first step towards such a compatibility.

Terminology 2.97. Let $A, B \in {}^1C^*\text{Alg}$.

1. If $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ is a homotopy, we say that it is unital iff ϕ_t is unital for each $t \in [0, 1]$.
2. A is 1-Schochet iff for the following lifting problem^a:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

(dashed arrow from A to $\mathcal{B}(H)$ labeled $(H_t)_{t \in [0, 1]}$)

where $(h_t)_{t \in [0, 1]}$ is unital and f is unital, we always have a homotopy lift satisfying the usual conditions.

Note that the notion of being 1-Schochet only applies to unital C^* -algebras.

^aFor convenience, we will refer to this as the unital lifting problem.

We have two remarks.

Remark 2.98. The lift above does not have to be unital but it's not far away from it. Indeed:

$$\forall t \in [0, 1] : H_t(1_A) + \mathcal{K}(H) = \pi(H_t(1_A)) = h_t(1_A) = \text{Id}_H + \mathcal{K}(H),$$

which implies that $H_t(1_A) - \text{Id}_H$ is compact for each $t \in [0, 1]$.

Even though this is very easy to check, we still need to show that the unitization of a homotopy is a homotopy.

Claim 2.9. Let $A, B \in C^*\text{Alg}$ and let $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ be a homotopy. Then, the unitization:

$$(\phi_t^\sim : A^\sim \rightarrow B^\sim)_{t \in [0, 1]},$$

is a homotopy.

The only notable permanence property of being 1-Schochet that we will need is given by the next result.

Proposition 2.99

Let $A, B \in {}^1C^*\text{Alg}$ with $A \equiv_{{}^1C^*\text{Alg}} B$ and let A be 1-Schochet. Then, B is 1-Schochet.

| **Proof.** This follows analogously as in Proposition 2.89. ■

Now, there are a few major permanence properties of the (1)-Schochet property that we would like to prove by the end of this subsection. The first of these is that this property behaves well under direct sums.

Theorem 2.100. A. Vats, 2026

Let $A, B \in {}^1C^*\text{Alg}$. TFAE:

1. A and B are 1-Schochet.
2. $A \oplus B$ is 1-Schochet.

Moreover, if $A \oplus B$ is Schochet, then so are A and B .

Proof.

(1) $\Rightarrow$ (2):

(2) $\Rightarrow$ (1):

The last statement follows by the same argument as in (2) $\Rightarrow$ (1). ■

We can use this theorem to deduce the relationship between the 1-Schochet property & Schochet property.

Corollary 2.101. A. Vats, 2026

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$. TFAE:

1. A is Schochet.
2. A is 1-Schochet.

Proof.

(1) $\Rightarrow$ (2): This follows clearly from the definition of being Schochet.

(2) $\Rightarrow$ (1): We consider the following lifting problem:

$$\begin{array}{ccc}
 & & \mathcal{B}(H) \\
 & \nearrow f & \downarrow \pi \\
 & A & \xrightarrow{(h_t)_{t \in [0,1]}} \mathcal{Q}(H) \\
 & \nwarrow i & \\
 A^\sim & &
 \end{array}$$

where we have added the unitization morphism i . Since A is unital, we see that:

$$A^\sim \equiv_{{}^1\mathbf{C}^*\mathbf{Alg}} A \oplus \mathbb{C},$$

with the right-hand side being 1-Schochet so $A^\sim$ is 1-Schochet^a. So, we can extend our diagram:

$$\begin{array}{ccc}
 & & \mathcal{B}(H) \\
 & \nearrow f^\sim & \downarrow \pi \\
 & A & \xrightarrow{(h_t)_{t \in [0,1]}} \mathcal{Q}(H) \\
 & \nwarrow i & \\
 A^\sim & &
 \end{array}$$

$(h_t^\sim)_{t \in [0,1]}$

using the universal property of the unitization and note that we still have a lifting problem with a unital $\star$ -homomorphism $f^\sim$ and a unital homotopy $(h_t^\sim)_{t \in [0,1]}$. Since $A^\sim$ is 1-Schochet, we can find a homotopy lift $(F_t : A^\sim \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ which solves the unital lifting problem. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t \circ i.$$

Let us check that this works:

$$\begin{aligned}
 H_0 &= F_0 \circ i = f^\sim \circ i = f, \\
 \forall t \in [0, 1] : \pi \circ H_t &= \pi \circ (F_t \circ i) = h_t^\sim \circ i = h_t,
 \end{aligned}$$

as was desired. ■

^aThis is where we use [Theorem 2.100](#).

As an immediate corollary, we obtain an even stronger result.

Corollary 2.102

Let $A, B \in {}^1\mathbf{C}^*\mathbf{Alg}$ be Schochet. Then, $A \oplus B$ is Schochet.

Proof. Since A, B are unital and Schochet, it is the case that they are 1-Schochet. Hence, $A \oplus B$ is 1-Schochet and since it is unital, it follows that it is Schochet. Over here, we simply apply Corollary 2.101 multiple times. ■

The corollary above is rather pivotal in proving the next result.

Corollary 2.103

Let $A \in \mathbf{C}^*\mathbf{Alg}$. TFAE:

1. A is Schochet.
2. $A^\sim$ is Schochet.

Proof.

(1) $\Rightarrow$ (2): Since $A^\sim$ is unital, we can just show that it is 1-Schochet. We start with a unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A^\sim & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

If $i : A \rightarrow A^\sim$ is the canonical inclusion, then we have:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f \circ i & \downarrow \pi \\ A & \xrightarrow{(h_t \circ i)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

where this is a lifting problem for A . Since A is Schochet, we have a homotopy lift:

$$(F_t : A \rightarrow \mathcal{B}(H))_{t \in [0,1]},$$

satisfying the required conditions of such a lift. Now, we define:

$$\forall t \in [0,1] : H_t \stackrel{\text{def}}{=} F_t^\sim.$$

By **Claim 2.9**, this is a homotopy so let's check that it works. If $t = 0$, then:

$$H_0 = F_0^\sim = (f \circ i)^\sim = f,$$

because f is already unital and defined on $A^\sim$. Next, let $t \in [0,1]$ and consider the following computation:

$$\begin{aligned} \forall a + \lambda 1_{A^\sim} \in A^\sim : (\pi \circ H_t)(a + \lambda 1_{A^\sim}) &= \pi(H_t(a)) + \lambda \pi(H_t(1_{A^\sim})) \\ &= h_t(a) + \lambda \pi(\text{Id}_H) \\ &= h_t(a) + \lambda (\text{Id}_H + \mathcal{K}(H)) \\ &= h_t(a) + \lambda h_t(1_{A^\sim}) \\ &= h_t(a + \lambda 1_{A^\sim}), \end{aligned}$$

which proves that we have the required lift for the original problem.

(2) $\Rightarrow$ (1): Consider the canonical projection:

$$p : A^\sim \rightarrow A, (a, \lambda) \mapsto p(a, \lambda) \stackrel{\text{def}}{=} a.$$

One observes that:

$$\forall a \in A : (p \circ i)(a) = p(a, 0) = a,$$

which proves that p is a split epi. Since $A^\sim$ is Schochet, we see that A is Schochet. ■

Here is a corollary of the result above.

Corollary 2.104

Let $X \in \text{LCHTop}$. TFAE:

1. X is Schochet.
2. X^+ , its one-point compactification, is Schochet.

Proof. We know that:

$$\mathcal{C}_0(X)^\sim \simeq \mathcal{C}(X^+).$$

Now, X is Schochet iff $\mathcal{C}_0(X)$ is Schochet iff $\mathcal{C}_0(X)^\sim$ is Schochet iff $\mathcal{C}(X^+)$ is Schochet iff X^+ is Schochet. ■

Moreover, the Schochet property behaves well under tensoring by matrices. The author thanks Prof. Raum for the suggestion to also prove (2) $\Rightarrow$ (1) in the following theorem.

Theorem 2.105. A. Vats, 2026

Let $A \in \text{C}^*\text{Alg}$ and let $n \in \mathbb{N}$. TFAE:

1. A is Schochet.
2. $M_{n \times n}(A)$ is Schochet.

Proof.

(1) $\Rightarrow$ (2): Suppose that this implication holds in the unital case. Then, A is Schochet iff $A^\sim$ is Schochet. Since $A^\sim$ is Schochet, we see that $M_{n \times n}(A^\sim)$ is Schochet. But this is just $\star$ -isomorphic to $M_{n \times n}(A)^\sim$ so this is Schochet and this means that $M_{n \times n}(A)$ is Schochet. So, it suffices prove this result when A is unital.

(2) $\Rightarrow$ (1): This was dealt with in statement (6) of Proposition 2.89. ■

With this corollary, we can show that the half-open unit interval is a Schochet space.

Example 2.106. We have the following main claim.

Claim. $(0, 1]$ is a Schochet space.

We note that $(0, 1]$ is Schochet iff $(0, 1]^+ \simeq [0, 1]$ is Schochet. But the latter statement is true.

At this stage, we have developed most of the tools needed to show that S^1 is a Schochet space; this is one of the major examples of Conway and we give a simplified proof. The point is that $\mathcal{C}(S^1)$ is a unital C^* -algebra so it is sufficient to show that it is 1-Schochet. By using the fact that this is the universal C^* -algebra generated by a unitary, it becomes very clear that the problem is to lift a path of Calkin unitaries to unitaries in the algebra of bounded operators on a Hilbert space. This is afforded to us by the following general lemma.

Lemma 2.107. [32, Proposition 4.3.14a, p.95-96] Let $A, B \in {}^1C^*Alg$ and let $\pi \in C^*Alg(A, B)$ be surjective, with $(u_t)_{t \in [0, 1]}$ being a path of unitaries in B so that there exists a unitary $U_0 \in A$ such that $\pi(U_0) = u_0$. Then, there exists a path $(U_t)_{t \in [0, 1]}$ of unitaries in A such that $\pi(U_t) = u_t$ for each $t \in [0, 1]$.

^aThe statement can be phrased a bit more succinctly by saying that $\pi|_{\text{Unitary}(A)}$ has the path-lifting property.

Now, we can deal with the desired example.

Example 2.108. We have the following claim.

Claim. S^1 is a Schochet space.

We consider the following unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathcal{C}(S^1) & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

Observe that $\mathcal{C}(S^1)$ is the universal C^* -algebra generated by a unitary; let u denote a generator. Since each h_t is unital, we see that $(h_t(u))_{t \in [0, 1]}$ is a path of Calkin unitaries. Using the lemma, we obtain a unitary lift $(v_t)_{t \in [0, 1]}$ through the Calkin projection, with $v_0 = f(u)$. Now, we just define for each $t \in [0, 1]$:

$$H_t(u) \stackrel{\text{def}}{=} v_t,$$

and this defines, by universal property, a $\star$ -homomorphism $H_t : \mathcal{C}(S^1) \rightarrow \text{Cyl}(\mathcal{B}(H))$ with the desired property.

Ideally, we want to extend Example 2.106 to $(0, 1)$ as well and we can do it.

Example 2.109. We have the following main claim.

Claim. $(0, 1)$ is a Schochet space.

We need to show that $\mathcal{C}_0((0, 1)) \sim$ is Schochet but this is $\star$ -isomorphic to $\mathcal{C}(S^1)$ and this is certainly Schochet.

We have discussed the Schochet property relative to spaces. One can ask what the basic situation looks like for groups and we have some results in that direction. In the next example, we see that all locally finite groups are Schochet.

Example 2.110. Let $G \in \text{Grp}$.

Claim. If G is locally finite^a, then G is a Schochet group.

^aThis means that every finitely-generated subgroup is finite.

In this case, $C_{\text{red}}^*(G)$ is an AF algebra so we're done.

Which can also characterize Schochet groups among the LCH abelian groups.

Proposition 2.111. A. Vats, 2026

Let $G \in {}^{\text{LCH}}\text{TopAbGrp}$. TFSH:

1. TFAE:
 - (a) G is a Schochet group.
 - (b) $\widehat{G}$, the Pontryagin dual of G , is a Schochet space.

In particular, $\widehat{G}$ is a Schochet group iff G is a Schochet space.
2. If G is compact, then G is Schochet.

Proof. We will prove each statement one-by-one.

1. Since G is abelian, we have the following chain of isomorphisms in C^*Alg :

$$C_{\text{red}}^*(G) \simeq C_{\text{max}}^*(G) \simeq \mathcal{C}_0(\widehat{G}),$$

where the first one holds because G is amenable and the second one holds as a general result of harmonic analysis. Now, G is Schochet iff $C_{\text{red}}^*(G)$ is Schochet iff $C_0(\widehat{G})$ is Schochet iff $\widehat{G}$ is Schochet. The last statement follows via Pontryagin duality, as well as the fact that the Schochet property for groups is preserved by isomorphisms.

2. This follows from statement (1), since $\widehat{G}$ is discrete^a so it is Schochet. ■

^aThey are, in particular, locally compact Hausdorff totally disconnected spaces.

With this corollary, we can provide more interesting examples.

Example 2.112. We have that $\mathbb{T}$ is a Schochet group since it is a compact Hausdorff abelian group.

Example 2.113. Let $G = \mathbb{Z}$, the additive group of the integers. Then, we know that:

$$\widehat{G} \simeq S^1,$$

and the latter is a Schochet space. So, we see that G is a Schochet group.

Example 2.114. Let $G = \mathbb{R}$, the additive group of $\mathbb{R}$. Then, we know that:

$$\widehat{G} \simeq \mathbb{R},$$

and the latter is a Schochet space because $(0, 1)$ is a Schochet space. So, G is a Schochet group.

We would like to extend [Example 2.113](#) above to free groups. To do this, we simply use our permanence property regarding the reduced free product. Since the reduced group C^* -algebra interacts well with the reduced free product, we have the following example in our arsenal.

Example 2.115. Let $n \in \mathbb{N}$ with $n \geq 2$. Then, we see that:

$$\text{Free}(n),$$

is Schochet. Indeed, we see that $C_{\text{red}}^*(\text{Free}(n))$ is the n -fold reduced free product of $C_{\text{red}}^*(\mathbb{Z})$. Since $\mathbb{Z}$ is Schochet, we can use the result above to conclude that $C_{\text{red}}^*(\text{Free}(n))$ is Schochet so $\text{Free}(n)$ is Schochet.

To begin with, let us define what it means for a given C^* -algebra to satisfy the homotopy lifting property.

Terminology 2.116. A pair (A, π) consisting of:

1. an $A \in C^*\text{Alg}$ and;
2. a surjective $\pi \in C^*\text{Alg}(E, B)$;

satisfies the homotopy lifting property iff the following lifting problem^a:

$$\begin{array}{ccc} & & E \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & B \end{array}$$

(H_t)_{t ∈ [0,1]}

always admits a solution in $C^*\text{Alg}$. We also say that (A, π) is HLP in this case.

^aThis is defined in exactly the same sense in which lifting problems are defined for the Schochet property.

The most important consequence of semiprojectivity, which we will introduce in a moment, is the following theorem.

Theorem 2.117. Blackadar's Homotopy Lifting Theorem, [10, Theorem 5.1, p.297-298]

Let $A \in C^*\text{Alg}$ be semiprojective. Then, every pair:

$$(A, \pi : E \rightarrow B),$$

where π is a surjective $\star$ -homomorphism, is HLP.

Before we explain (approximate) semiprojectivity, we have a remark regarding our solution to [Conjecture 2.40](#).

Remark 2.118. So, in principle, [Theorem 2.117](#) provides us a way to show that certain C^* -algebras are not semiprojective; we just have to show that they fail the HLP. However, as we have already seen, this can be non-trivial. Nevertheless, the C^* -algebra we introduced in our solution to the aforementioned conjecture could reasonably be used to show that other surjective $\star$ -homomorphisms are not Schochet fibrations. The main point is to find a normal contraction in the quotient which does not have a normal lift.

Let us explain semiprojectivity and also introduce approximate semiprojectivity simultaneously.

Terminology 2.119. Let $A \in {}_{\omega}C^*\text{Alg}$.

1. A is semiprojective iff for any $B \in C^*\text{Alg}$, any increasing sequence $(J_n)_{n \in \mathbb{N}}$ of closed ideals in B with J being the closed ideal generated by the sequence and any $\star$ -homomorphism $\phi : A \rightarrow B/J$, we have that this can be lifted to a $\star$ -homomorphism $\psi : A \rightarrow B/J_n$ for sufficiently large n .
2. A is approximately semiprojective iff it is isomorphic to a sequential inductive limit of semiprojective C^* -algebras.

In the literature, an approximately semiprojective C^* -algebra is often also said to have a strong shape system.

In the next remark, we will provide some context for these notions.

Remark 2.120. In [9], it is claimed that a paper, “Homotopy Continuity & Shape Theory for C^* -Algebras”, by (Edward) Effros and (Jerome) Kaminker was responsible for developing a noncommutative version of shape theory. This theory was originally introduced in the topological setting to study certain spaces by focusing on homotopy types of the spaces that are used to construct them; for more details on these matters and history, see [29]. We could not locate the paper due to Effros and Kaminker but it is referenced in a paper by Blackadar ([9]) which seems to be the first one which introduced the notion of semiprojectivity which is in use within modern^a research^b.

^aSee Definition 2.10, p.256 of the aforementioned paper.

^bIn fact, the paper also defines semiprojectivity for morphisms between C^* -algebras and proceeds to define semiprojectivity of a given C^* -algebra as the semiprojectivity of the identity morphism. That approach seems to be useful in establishing semiprojectivity.

At this stage, we are ready to state an important conjecture due to Blackadar.

Conjecture 2.121 (B. Blackadar, 1985, [9, 4.4, p.266]). Every separable C^* -algebra is approximately semiprojective.

Now, we can finally address Question 2.53.

Theorem 2.122. A. Vats, 2026

The spaces listed in Question 2.53 are Schochet spaces.

The proof relies on a difficult permanence property of semiprojectivity that we will not prove.

Proposition 2.123. D. Enders, 2014, [25, Corollary 3.4, p.8]

Suppose we have the following pullback square:

$$\begin{array}{ccc} A \oplus_F B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & F \end{array}$$

in $C^*\text{Alg}$, where F is finite-dimensional. If A and B are semiprojective, then $A \oplus_F B$ is semiprojective.

We have a remark about this result.

Remark 2.124. It would be very desirable to have a similar result where semiprojectivity is replaced with the Schochet property. However, the proof techniques used in this reference does not seem amenable to proving such a result.

Here is the proof of our theorem.

Proof of Theorem 2.122. We will deal with each of these spaces individually.

1. Observe that:

$$\mathcal{C}(S^1 \vee S^1) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}(S^1).$$

Since $\mathcal{C}(S^1)$ is semiprojective, it is Schochet. So, the above is a pullback of two semiprojective C^* -algebras over a finite-dimensional C^* -algebra. By the strength of Proposition 2.123, we see that the pullback is semiprojective and this means that it is Schochet.

2. Observe that:

$$\mathcal{C}(S^1 \vee [0, 1]) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}([0, 1]).$$

By the same argument as in the proof of statement (1), we can conclude that this pullback is Schochet.

3. Let $X = [-1, 0] \cup \{2^{-n} : n \in \mathbb{N}_0\}$. Then, we can write:

$$\mathcal{C}(X) \simeq \mathcal{C}([-1, 0]) \oplus_{\mathbb{C}} \mathbf{c} = \{(f, (a_n)_{n \in \mathbb{N}_0}) \in \mathcal{C}([-1, 0]) \oplus \mathbf{c} : f(0) = \lim_{n \rightarrow \infty} a_n\},$$

where $\mathbf{c}$ is the C^* -algebra of all convergent sequences indexed by $\mathbb{N}_0$. Now, $[-1, 0]$ is homeomorphic to $[0, 1]$ so it is a Schochet space and this means that $\mathcal{C}([-1, 0])$ is a Schochet C^* -algebra. Moreover, $\mathbb{N}_0$ is a discrete space and $c_0(\mathbb{N}_0)$ is just $\mathcal{C}_0(\mathbb{N}_0)$, which means that this is also a Schochet C^* -algebra. Since direct sums of unital Schochet C^* -algebras are Schochet, we are done. ■

We have a remark about our proof above.

Remark 2.125. The proof above is simple and gets the job done. However, it is difficult to find it aesthetically pleasing, for the simple reason that we always have to pass through semiprojectivity & it is hard to relate this condition to the property of being Schochet. It would be desirable to find a proof of the result above which only used permanence properties associated to the Schochet property.

Let us mention a recent result which improves our intrinsic understanding of the homotopy lifting property.

Let $A \in C^*Alg$. TFAE:

1. A is l -open.
2. The system of maps:
$$\text{Hom}(A, B) \rightarrow \text{Hom}(A, B/J),$$
for all marked C^* -algebras (B, J) , is uniformly open^a.
3. If (B, I) is a marked C^* -algebra, then $(A, \pi : B \rightarrow B/I)$ is HLP and $\text{Hom}(A, B)$ is locally path-connected.

^aWe will not define this term as it will not be needed; one can find it in the reference.

In the next result, we will collect some facts that are in the literature and some which have been proven by us.

Proposition 2.127

Let $A \in C^*Alg$. TFSH:

1. (A, Id_A) is HLP.
2. A is Schochet iff (A, π) is HLP, with π being the Calkin projection.
3. A , as defined in Theorem 2.59, is approximately semiprojective.
4. The Schochet property is not closed under sequential inductive limits.
5. The HLP is not closed under sequential inductive limits.
6. If A is l -open, then it is Schochet.
7. There exists a Schochet C^* -algebra which is not l -open.
8. There exists a Schochet C^* -algebra which is not semiprojective.
9. The full subcategories of ${}_{\omega}C^*Alg$:
 - (a) the category of Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras;
do not coincide.
10. If (b) is a subcategory of (a), then Conjecture 2.121 is false.
11. Consider the full subcategories of ${}_{\omega}C^*Alg$:
 - (a) the category of approximately Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras.
Then, (b) is a subcategory of (a).

Proof. We will prove each statement one-by-one.

1. This is clear; we just need to take the lift to be the original homotopy.
2. This is the definition of the Schochet property.
3. This was a difficult result but it has been recently shown to be the case; see [42, p. 723].
4. Since semiprojective C^* -algebras are Schochet, the C^* -algebra of Theorem 2.59 is exactly an example of a sequential inductive limit of Schochet C^* -algebras which is not Schochet.
5. Since there is a non-Schochet C^* -algebra, this C^* -algebra also cannot be HLP and one can construct such an example which is approximately semiprojective, so it is an inductive limit of HLP C^* -algebras^a.
6. This follows by Theorem 2.126.
7. In [53, Example 3.9, p.7], a unital AF algebra is given which is not l -open.
8. We simply need a Schochet C^* -algebra which is not l -open and we are done^b.
9. Since there exists a non-Schochet C^* -algebra which is approximately semiprojective, we see that the category of Schochet C^* -algebras is not closed under sequential inductive limits. On the other hand, the category of approximately semiprojective C^* -algebras is closed under sequential inductive limits; this was shown in [69, Theorem 4.9, p.13]. So, it follows that these two categories cannot coincide.
10. Indeed, if every approximately semiprojective C^* -algebra is Schochet, then we can just take a separable non-Schochet C^* -algebra and this cannot be approximately semiprojective so we're done.
11. This is clear. ■

^aThis follows by Blackadar's Homotopy Lifting Theorem.

^bThis is because semiprojectivity implies l -openness.

Here are two remarks to this result.

Remark 2.128. Statement (11) in the result above forms a possible obstruction to Blackadar's Semiprojectivity Conjecture. Indeed, if we can find an approximately Schochet C^* -algebra which is not approximately semiprojective, then the conjecture would be resolved in the negative.

Remark 2.129. It is fairly easy to generate examples of C^* -algebras which do not satisfy the homotopy lifting property; we can just take a non-cofibration between compact Hausdorff spaces and the resulting map between C^* -algebras

| will not be a Schochet fibration, see [67, p.402]. However, our example above seems to be the first noncommutative counterexample to the HLP. Even if Blackadar's conjecture holds, homotopy lifting does not generally hold.

We will discuss this question in the next section and some progress that we have been able to make on it. We will see that our exploration leads us naturally back towards trying to find a systematic way of characterizing non-Schochet C^* -algebras. This is probably not so surprising since this C^* -algebra is not Schochet, as shown earlier.

3 Introducing Categories of Fibrant Objects

Summary

In this section, we will cover the following topics:

1. A survey of model category theory.
2. Some basic properties of categories of fibrant objects.

Many of the results in this section were known previously, with the exception of a few basic results in Section 3.2.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

G will denote a fixed LCH group.

3.1 A Survey of Model Category Theory

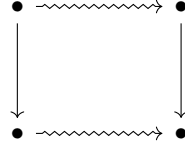
Let us start by recalling the definition of the localization of a category with respect to some set of morphisms. To state this definition, we need a bit of preparation. To be precise, we need a construction that will be used later.

Construction 3.1. Let $\mathcal{C}$ be a category. We define a structure:

$$\text{Arr}(\mathcal{C}),$$

as follows:

1. An object is an arrow^a in $\mathcal{C}$.
2. A morphism in this structure is a commutative diagram:



where the arrow on the left is the domain of the morphism, the arrow on the right is the codomain of the morphism and the two arrows on top & below form the data of the morphism. We define composition by the concatenation of commutative diagrams in the usual way through composition of morphisms in $\mathcal{C}$.

This data is easily shown to form a category; we refer to this as the arrow category of $\mathcal{C}$. In particular, if f, g are morphisms in $\mathcal{C}$ and $\alpha : f \Rightarrow g$ is a morphism in $\text{Arr}(\mathcal{C})$, then α is known as a lifting problem for f and g .

^aMore formally, it is a triple (x, y, f) with $x, y \in \mathcal{C}$ and $f \in \mathcal{C}(x, y)$ being an arrow.

We also need some terminology.

Terminology 3.2. Let $\mathcal{C}, \mathcal{D}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. We say that F inverts W iff for any $f \in W$, we have that $F(f)$ is an isomorphism. We define:

$$\text{Fun}^W(\mathcal{C}, \mathcal{D}),$$

to be the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ consisting of functors which invert W .

The universal property of the localization is easy to describe.

Definition 3.3

Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. The Gabriel-Zisman localization/1-categorical localization of $\mathcal{C}$ with respect to W is a pair $(\mathcal{C}[W^{-1}], \gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}])$ consisting of:

1. a category $\mathcal{C}[W^{-1}]$,
2. a functor $\gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ which inverts W .

Moreover, for any category $\mathcal{D}$, the following precomposition functor:

$$\text{Fun}(\mathcal{C}[W^{-1}], \mathcal{D}) \rightarrow \text{Fun}^W(\mathcal{C}, \mathcal{D}),$$

is an equivalence of categories.

We have two remarks about this notion.

Remark 3.4. The universal property above is just about all that we will ever need, though we will have reasons to talk about the calculus-of-fractions approach to understanding localizations. For a conceptual exposition of localizations, see [41, Chapter 3.3, p.26-35]. For the proof of existence, see the sketch presented in [39, Chapter 1.1, p.3-5].

Remark 3.5. We will typically refer to the Gabriel-Zisman localization as just the localization, at least until Section 4.2. In that subsection, we will introduce localizations of ∞ -categories & since one can take the ∞ -categorical localization of a category with respect to some full subcategory of morphisms, we distinguish between the two localizations by referring to the one in Definition 3.3 by the 1-categorical localization.

Let us introduce some further notation which fits in better with the perspective we will take.

Notation 3.6. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$. We define:

$$\mathbf{Ho}(\mathcal{C}) \stackrel{\text{def}}{=} \mathcal{C}[W^{-1}],$$

as the homotopy category of $\mathcal{C}$, equipped also with the localization functor $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$. If $x, y \in \mathbf{Ho}(\mathcal{C})$, then:

$$[x, y]_{\mathcal{C}} \stackrel{\text{def}}{=} \mathbf{Ho}(\mathcal{C})(x, y),$$

are known as the homotopy classes of morphisms from x to y . Sometimes, we omit the $\mathcal{C}$ on the left-hand side.

Before we introduce any of the homotopical structures we will study in this section, it is useful to have an overview of their place in history and to also have some insight into the general philosophy of homotopy theory.

Remark 3.7. It is difficult to give a full, historical account of the development of homotopy theory, so we will start with the creation of model categories. These were introduced by (Daniel) Quillen in 1967 in his book on homotopical algebra; see [59]. At that time, it was recognized that the general goal of homotopy theory was to study categories

$\mathcal{C}$ which were equipped with some collection W of morphisms of $\mathcal{C}$ which were known, abstractly, as the “weak equivalences”. Of course, it is easy to cook up examples for which we have this kind of situation:

1. Top, with the homotopy equivalences.
2. Top, with the weak homotopy equivalences.
3. Chain complexes over abelian categories, with the quasi-isomorphisms.
4. Simplicial sets, with homotopy equivalences of simplicial sets.
5. $G\text{-}\mathcal{C}^*\text{Alg}$ or $G\text{-}\text{BanAlg}$, with the equivariant homotopy equivalences.

The last example is, of course, more recent than the other 3 but it was being considered (in the non-equivariant set-up) as far back as the 1970s, with the development of the Brown-Douglas-Fillmore Ext groups. At the time, it was realized that the goal of homotopy theory was to have some measure of control over the *localization*:

$$\mathcal{C}[W^{-1}],$$

of $\mathcal{C}$ with respect to W . This is a category which is equipped with a functor $i : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ such that the objects in both categories are the same and i takes morphisms in W to isomorphisms. Now, the main point is that the structure of a model category allows us to have some measure of control over the localization and this is afforded to us by various structure theorems about the mapping spaces. This seems to be somewhat close to the speculated reason for why Quillen introduced these structures; see [34]. Even though Quillen’s work on model categories seems to not have directly influenced his work on higher algebraic K-theory ([58]), it is clear that later work done on the homotopical foundations of algebraic K-theory, such as in [35], was influenced by the development of model categories. Moreover, modern approaches to algebraic K-theory via the theory of ∞ -categories certainly do rely on model structures on categories of (bi)simplicial sets, particularly for explicit descriptions of the underlying K-theory space.

As we will see later in Section 3.1, the definition of a model category is rather complex. This is rather unfortunate, as much of the difficulty lies in verifying that some choice of category has a model structure. So, even though model categories do provide a fairly satisfactory setting in which we can “do” homotopy theory, they do not capture all examples sufficiently well. Indeed, this was first recognized in the study of the homotopy theory of sheaves of simplicial sets and sheaves of spectra by (Ken) Brown in [11]. In the same paper, he proposed a significant weakening of a model category and studied the so-called categories of fibrant objects, proving that their structure is sufficient in controlling the localization. On the other hand, developments in algebraic K-theory due to (Friedhelm) Waldhausen in [73] indicated that categories with cofibrations and weak equivalences are the right input for algebraic K-theory. In fact, it is true that there were many intermediate structures between a relative category (a category equipped with some collection W of morphisms) and a model category which were proposed to deal with various problems in stable homotopy theory; we will see one example of this in Section 3.2. It is not possible for us to give a systematic breakdown of their individual influences on modern homotopy theory. What is clear, however, is that the model categories of Quillen did have a substantial influence on later work done by homotopy theorists.

We are aware that we have not spoken about the influence of model categories on operator algebras; we will speak about this towards the end of this subsection, after we have given the definition of a model category. For now, we should proceed towards our survey of this topic and we will mainly be following [17, Chapter 2.1-2.2, p.28-41]. Our starting point is actually an abstract description of lifting problems in categories. As it turns out, this is not exactly the most natural approach towards the study of model categories. However, it is the fastest way to get to the point and highlight the main features of the theory which will come up again later in our study of categories of fibrant objects. In the next definition, we will state what it means for a lifting problem, within some arbitrary category, to have a solution.

Definition 3.8

Let $\mathcal{C}$ be a category, let $x, y, x', y' \in \mathcal{C}$ and let $f : x \rightarrow y, g : x' \rightarrow y'$ be morphisms in $\mathcal{C}$. f has the left-lifting property (LLP) with respect to g (or g has the right-lifting property (RLP) with respect to f) iff any lifting problem:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & x' \\ f \downarrow & & \downarrow g \\ y & \xrightarrow{\quad} & y' \end{array}$$

has a solution $\phi : y \rightarrow x'$. That is, the following diagram:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & x' \\ f \downarrow & \nearrow \phi & \downarrow g \\ y & \xrightarrow{\quad} & y' \end{array}$$

commutes. We denote this by writing $f \square g$.

We have a remark about this definition.

Remark 3.9. The solution ϕ must work for all possible choices of morphisms from x to x' and y to y' . So, even though lifting problems do cover a wide range of phenomenon, the requirements of a solution are rather stringent.

We introduce two pieces of convenient terminology associated to the definition.

Terminology 3.10. Let $\mathcal{C}$ be a category and let f, g be morphisms such that:

$$f \sqsupseteq g.$$

Then, we say that f left-lifts g or g right-lifts f .

Terminology 3.11. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$. We say that S has the LLP with respect to T (or T has the RLP with respect to S) iff for every $f \in S$ and $g \in T$, we have that $f \sqsupseteq g$. We denote this by writing $S \sqsupseteq T$. Also, if $T = \{g\}$, then we'll often just write $S \sqsupseteq g$ instead of $S \sqsupseteq \{g\}$; the same is true when S is a singleton.

Here are some basic examples of lifting problems.

Example 3.12. Let $A, B \in \text{Set}$ and let $f \in \text{Set}(A, B)$.

Claim. TFAE:

1. f is surjective.
2. $(\emptyset \rightarrow *) \sqsupseteq f$.

We did not label the map from $\emptyset$ to $*$ because there is only one such map.

Before we prove this equivalence, here's the diagram lurking in the background.

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ \downarrow & & \downarrow f \\ * & \longrightarrow & B \end{array}$$

Now, we prove each implication separately.

- (1) $\Rightarrow$ (2): Take the map from $*$ to B and let b be in its range. Then, there is an $a \in A$ such that $f(a) = b$. Now, define a map from $*$ to A whose value is a ; this is a solution to the lifting problem.
- (2) $\Rightarrow$ (1): Let $b \in B$. Then, we pick a map from $*$ to B with value b and we select the unique map from $\emptyset$ to A . This, of course, gives us a lifting problem which has a solution $\phi : * \rightarrow A$. The solution is literally the condition that $f(a) = b$, where the solution has value $a \in A$.

This completes the proof.

Example 3.13. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. We recall that M is projective iff:

$${}_R\text{Mod}(M, -),$$

is an exact functor.

Claim. TFAE:

1. M is projective.
2. $(0 \rightarrow M) \sqsupseteq \{\text{epimorphisms in } \mathcal{A}\}$.

This is essentially a standard characterization of projective modules.

Example 3.14. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$.

Claim. TFAE:

1. F is surjective on objects.
2. F has the RLP with respect to $\emptyset \rightarrow [0]$.

We will prove each implication separately.

- (1) $\Rightarrow$ (2): Consider the following commutative diagram:

$$\begin{array}{ccc} \emptyset & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow F \\ [0] & \xrightarrow{d} & \mathcal{D} \end{array}$$

where d represents the constant functor which selects an object $d \in \mathcal{D}$. Since F is surjective on objects, there is a $c \in \mathcal{C}$ such that $F(c) = d$. Now, we define a lift $[0] \rightarrow \mathcal{C}$ which simply selects the object c in $\mathcal{C}$.

- (2) $\Rightarrow$ (1): Let $d \in \mathcal{D}$. Then, the commutative diagram above has a lift; this lift just selects an object $c \in \mathcal{C}$. Since the lift causes the diagram to commute, we see that $F(c) = d$.

Our earlier purpose in defining the arrow category was to specify retracts of morphisms in categories.

Terminology 3.15. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$. x is a retract of y iff there is a commutative diagram:

$$\begin{array}{ccccc} & & \text{Id}_x & & \\ & \nearrow & & \searrow & \\ x & \xrightarrow{f} & y & \xrightarrow{g} & x \end{array}$$

Also, if $f, g \in \text{Mor}(\mathcal{C})$, then f is a retract of g iff f is a retract of g in $\text{Arr}(\mathcal{C})$. Finally, if $S \subseteq \text{Mor}(\mathcal{C})$, then S is closed/stable under retracts iff every morphism in $\mathcal{C}$ which is a retract of some morphism in S must also lie in S .

In fact, an object being a retract of another object does not change under dualization.

Claim 3.1. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$ with x being a retract of y . Then, x is a retract of y in $\mathcal{C}^{\text{op}}$.

The following result is standard and easy to prove.

Proposition 3.16. The Retract Lemma

Let $\mathcal{C}$ be a category, let $f : x \rightarrow y$ be a morphism and suppose we have a factorization:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow i \quad \nearrow p & \\ & z & \end{array}$$

Then, TFSH:

1. If $i \sqsupseteq f$, then f is a retract of p .
2. If $f \sqsupseteq p$, then f is a retract of i .

Proof. We'll prove each statement one-by-one.

1. We need it to be the case that f is a retract of p in $\text{Arr}(p)$. Consider the diagram:

$$\begin{array}{ccccc} x & \xrightarrow{i} & z & \xrightarrow{\quad} & x \\ \downarrow f & & \downarrow p & & \downarrow f \\ y & \xrightarrow{\text{Id}_y} & y & \xrightarrow{\text{Id}_y} & y \end{array}$$

This is a morphism from $f \Rightarrow p \Rightarrow f$ in $\text{Arr}(\mathcal{C})$ and we just need to find a morphism from z to x such that the two parts compose to the identity. We start by re-working the given triangle as follows:

$$\begin{array}{ccc} \begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow i \quad \nearrow p & \\ & z & \end{array} & \Longrightarrow & \begin{array}{ccc} x & & y \\ \downarrow i & \searrow f & \\ z & \xrightarrow{p} & y \end{array} \end{array} \Longrightarrow \begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ \downarrow i & & \downarrow f \\ z & \xrightarrow{p} & y \end{array}$$

Since $i \sqsupseteq f$, it follows that there is a solution $\phi : z \rightarrow x$. But this just means that $\phi \circ i = \text{Id}_x$.

2. If $f \sqsupseteq p$ in $\mathcal{C}$, then $p \sqsupseteq f$ in $\mathcal{C}^{\text{op}}$. So, f is a retract of i in $\mathcal{C}^{\text{op}}$ but that means that f is a retract of i in $\mathcal{C}$. ■

With retracts out of the way, we can specify weak factorization systems.

Terminology 3.17. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$ so that:

1. S and T are closed under retracts.
2. $S \sqsupseteq T$.
3. If $f \in \text{Mor}(\mathcal{C})$, then it has a factorization $f = p \circ i$ with $p \in T$ and $i \in S$.

The pair (S, T) is known as a weak factorization system (WFS).

With this, we can finally define model structures; this is different from the one given by Quillen in [59].

Definition 3.18

Let $\mathcal{C}$ be a category. A model structure on $\mathcal{C}$ consists of a triple:

$$(\text{Cof}, W, \text{Fib}),$$

where W, Fib are subcategories of $\text{Arr}(\mathcal{C})$ such that:

1. W has the 2-out-of-3 property. Namely, if we have a commutative triangle:

$$\begin{array}{ccc} & \bullet & \\ \nearrow & & \searrow \\ \bullet & \xrightarrow{\quad} & \bullet \end{array}$$

with 2-out-of-3 morphisms in this triangle being in W , then the third is also in W .

2. The pairs $(\text{Cof}, W \cap \text{Fib})$ and $(W \cap \text{Cof}, \text{Fib})$ are WFSs.

A model category is the datum of a finitely bicomplete category $\mathcal{C}$ equipped with a fixed model structure.

We have some terminology that goes along well with the definition of a model category.

Terminology 3.19. Let $\mathcal{C}$ be a category with some model structure $(\text{Cof}, W, \text{Fib})$. Then:

1. Fib is the set of fibrations.
2. Cof is the set of cofibrations.
3. W is the set of weak equivalences.
4. $W \cap \text{Fib}$ is the set of all trivial fibrations.
5. $W \cap \text{Cof}$ is the set of all trivial cofibrations.

Moreover, (Fib, Cof) is known as the phantom structure on $\mathcal{C}$.

We have a remark about examples and our work so far.

Remark 3.20. In general, it is very difficult to provide examples of model categories because the verification that the appropriate datum on a category actually provides a model structure can be arduous. There are three classes of examples that are most heavily prevalent in the literature, partly due to their classical nature but also because they get used in modern work in homotopy theory. These classes are listed as follows:

1. Topological spaces.
2. Simplicial sets.
3. Chain complexes.

Actually, [59] also gives other interesting examples of model categories, though complete verifications of such categories actually having a model structure are not provided. With regards to the three examples above, there are now many resources which give a systematic overview of various interesting model structures related to the objects listed above. Therefore, we will not repeat any of that over here, especially since we are barely going to need any of it. Instead, we refer the reader to the very detailed webpages [4], [5] and [3], which also contain more detailed references.

One may ask where operator algebras stand with respect to these developments. As we described in Remark 2.12, it was already noticed by Schochet that one might reasonably want to bring the homotopy theory of C^* -algebras into the framework of Quillen's model categories. Indeed, the point was to find a model structure on $C^*\text{Alg}$ such that the W would consist of the homotopy equivalences and Fib would consist of the Schochet fibrations. It turns out that no such model structure exists. This was first noticed in the unpublished work ([1]) of (Kasper) Andersen and (Jesper) Grodal.

Theorem 3.21

Let $C^*\text{Alg}$ be viewed as a category of fibrant objects as in Proposition 4.1. There exists no model structure on $C^*\text{Alg}$ such that the full subcategory of fibrant objects coincides with the category of fibrant objects above.

Let us discuss this theorem and literature related to it in the next remark.

Remark 3.22. A variant of this theorem was proved by (Kasper) Andersen and (Jesper) Grodal in the unpublished preprint [1]; they showed that $C^*\text{Alg}$ can be equipped with the structure of a Baues fibration category and demonstrated the analogous impossibility result. The theorem above, however, was shown by Uuye in [72, Theorem A.1, p.15-16]; he proves it by showing that the loop functor on the homotopy category of commutative C^* -algebras does not admit a left adjoint. Nevertheless, recent work by (Michael) Joachim and (Anupam) Datta suggests that it is possible to place a model structure on the category of locally multiplicative convex C^* -algebras; see [21]. Similar work of this type has been done in the past involving projective systems of C^* -algebras; see [8]. However, the upshot of Joachim and Datta's work is that it addresses the homotopical nature of the Cuntz q -construction. Indeed, it tells us that the Cuntz q -construction should be viewed, up to stabilization, as a cofibrant replacement functor; see Corollary 8.10 on page 37 of their paper. This seems to address a question of (Ralf) Meyer as stated in [49, p.19]. In this reference, he asserts that the q -construction might be fruitfully viewed as a cofibrant replacement functor in some model category associated to KK-theory and references some older work done by Joachim in this direction.

As mentioned in Remark 3.20, model structures are not commonly found among the operator algebraic categories of interest; we will eventually develop the language to make this precise for $C^*\text{Alg}$. So, we have to be content with studying homotopical structures that are weaker but, in general, encapsulate the two major examples we have discussed thoroughly in Section 2. It was first recognized by (Otgonbayar) Uuye in [72] that the appropriate structures are the *categories of fibrant objects*. We have mentioned these before and in Section 3.2, we will present a rather extensive discussion of their basic theory & our contributions to it. We will use this to resolve Question 2.11 in Section 4.

3.2 Properties of Categories of Fibrant Objects

Ultimately, we wish to study categories $\mathcal{C}$ which are equipped with two full subcategories:

$$W, \text{Fib},$$

of $\text{Arr}(\mathcal{C})$; the first will be a category of *weak equivalences* and the second will be a category of *fibrations*. The literature with respect to such categories is a little sparse and we aim to extend some results about these categories. With that general aim in mind, let us introduce some terminology that will only be relevant later in this subsection.

Terminology 3.23. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. W satisfies the 2-out-of-3 property iff for any diagram of two composable arrows:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

where any two of f, g and $g \circ f$ lies in W , then the third lies in W .

2. W satisfies the 2-out-of-6 property iff for a diagram of three composable arrows:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies in W , we have that f, g, h and $h \circ g \circ f$ lie in W .

If $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization of $\mathcal{C}$ with respect to W , then we say that W is saturated in $\mathcal{C}$ iff for any morphism $f \in \text{Mor}(\mathcal{C})$ with $\gamma(f)$ being an isomorphism, we have that $f \in W$.

Here are some basic (but useful facts) which are strewn across the literature.

Claim 3.2. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. The isomorphisms of $\mathcal{C}$, viewed as a full subcategory of $\text{Arr}(\mathcal{C})$, satisfy the 2-out-of-6 property.
2. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it is 2-out-of-3.
3. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it contains all isomorphisms.

This is proved on Page 98.

Claim 3.3. Let $\mathcal{C}, \mathcal{D}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ and let W be a full subcategory of $\text{Arr}(\mathcal{C})$ consisting of all morphisms whose images under F become isomorphisms in $\mathcal{D}$. Then, W satisfies the 2-out-of-6 property^a.

^aThis is proved on Page 99.

Claim 3.4. Let $\mathcal{C}$ be a category and let W be a saturated full subcategory of $\text{Arr}(\mathcal{C})$. Then^a, W is 2-out-of-6.

^aThis is proved on Page 99.

We will work our way up to the definition of a category of fibrant objects, starting with the notion of a fibration category.

Definition 3.24

A fibration category is a pair:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}),$$

of a pointed category $\mathcal{C}$ with a full subcategory $\text{Fib}_{\mathcal{C}}$ of *fibrations* such that:

1. The isomorphisms in $\mathcal{C}$ are in $\text{Fib}_{\mathcal{C}}$.
2. If $x \in \mathcal{C}$, then the unique morphism $x \rightarrow 0$ lies in $\text{Fib}_{\mathcal{C}}$.
3. Fibrations are preserved by base extensions (recall Terminology 2.29).

A cofibration category is a category with cofibrations as defined on [73, p.320], being exactly dual to fibration categories.

Here are two simple results.

Claim 3.5. Let $\mathcal{C}$ be a category.

1. If $\mathcal{C}$ is a fibration category, then $\mathcal{C}^{\text{op}}$ is a cofibration category when paired with $\text{Fib}_{\mathcal{C}}^{\text{op}}$.
2. If $\mathcal{C}$ is a cofibration category, then $\mathcal{C}^{\text{op}}$ is a fibration category when paired with $\text{Cof}_{\mathcal{C}}^{\text{op}}$.

This is proved on Page 99.

Claim 3.6. Let $\mathcal{C}$ be a fibration category. Then, $\mathcal{C}$ admits all finite products^a.

^aThis is proved on Page 99.

Of course, one can also define structure-preserving functors, even though we will not use this notion.

Terminology 3.25. Let $\mathcal{C}, \mathcal{D}$ be fibration categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. Then, F is exact iff it preserves the zero object, if it preserves fibrations and if it preserves base extensions along a fibration.

Here are two nice constructions of fibration categories.

Construction 3.26. Let $\mathcal{C}$ be a fibration category and let I be a small category. We define:

$$\text{Fib}_{\text{Fun}(I, \mathcal{C})},$$

as follows: if $F, G \in \text{Fun}(I, \mathcal{C})$ and $\alpha \in \text{Nat}(F, G)$, then α is a fibration iff its components are fibrations in $\mathcal{C}$.

Claim. $\text{Fun}(I, \mathcal{C})$ becomes a fibration category with the data of $\text{Fib}_{\text{Fun}(I, \mathcal{C})}$.

We will show each axiom of a fibration category one-by-one.

1. We have that $\text{Fib}_{\text{Fun}(I, \mathcal{C})}$ is clearly a full subcategory because $\text{Fib}_{\mathcal{C}}$ is a full subcategory.
2. If α is an isomorphism in the category $\text{Fun}(I, \mathcal{C})$, then its components are isomorphisms in $\mathcal{C}$ and these are certainly fibrations in $\mathcal{C}$. So, it follows that α is a fibration.
3. Let $F \in \text{Fun}(I, \mathcal{C})$ and let $\alpha : F \Rightarrow 0$, where the zero functor in $\text{Fun}(I, \mathcal{C})$ is a zero object which takes every object in I to the zero object in $\mathcal{C}$. Now, the components of α are the unique morphisms from each $F(i)$ to the

zero object of $\mathcal{C}$ so these are fibrations in $\mathcal{C}$. But this means that α is a fibration in $\mathcal{C}$.

4. Limits in $\text{Fun}(I, \mathcal{C})$ are computed point-wise, so fibrations are preserved by base extensions.

This completes the proof.

Construction 3.27. Let $\mathcal{C}$ be a model category.

Claim. $(\mathcal{C}, \text{Fib}_{\mathcal{C}})$ forms a fibration category.

With the definition of a model category that we have given, this is not obvious. This fact, while stated in references which discuss categories of fibrant objects, has never been explicitly shown so we will show it:

- 1.
- 2.
- 3.

This proves the claim.

Finally, we have two examples coming from operator algebra theory.

Example 3.28. We have the following claim.

Claim. TFSH:

1. $G\text{-}C^*\text{Alg}$ is a fibration category when equipped with the Schochet fibrations.
2. $G\text{-}BanAlg$ is a fibration category when equipped with the Schochet fibrations.

We'll verify each of the axioms as listed in [Definition 3.24](#) one-by-one.

1. Clearly, $G\text{-}C^*\text{Alg}$ and $G\text{-}BanAlg$ have zero objects; it's just $\{0\}$.
2. In both cases, axioms (2) and (3) are taken care of by the statements of [Proposition 2.28](#).

This proves the result.

For the next example, we will need the notion of a semisplit surjection as introduced in [Terminology 2.31](#)

Example 3.29. Let $G \in \text{TopGrp}$ and define:

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Semisplit surjections in } G\text{-}C^*\text{Alg}\}.$$

Here's the main claim.

Claim. The pair $(G\text{-}C^*\text{Alg}, \text{Fib})$ forms a fibration category.

The verification that Fib is a full subcategory of $\text{Arr}(G\text{-}C^*\text{Alg})$ is a direct computation with the definition. The more important point is showing the fibration category structure and this is done essentially in the proof of [\[15, Proposition 2.10, p.26-27\]](#). That result is stated for discrete, countable groups but the same proof will work for LCH groups.

Once we have a fibration category, we also would want to add an additional bit of data about *weak equivalences*; this is a second subcategory of the arrow category that we include in our data.

Definition 3.30

A weak fibration category is a triple:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}}),$$

such that:

1. $(\mathcal{C}, \text{Fib}_{\mathcal{C}})$ is a fibration category.
2. $W_{\mathcal{C}}$ is a full subcategory of $\text{Arr}(\mathcal{C})$; this is known as the subcategory of weak equivalences.
3. All isomorphisms in $\mathcal{C}$ are contained in $W_{\mathcal{C}}$.
4. Suppose we have a commutative diagram in $\mathcal{C}$:

$$\begin{array}{ccccc} b & \xrightarrow{\text{red}} & a & \xleftarrow{\text{blue}} & c \\ \downarrow & & \downarrow & & \downarrow \\ b' & \xrightarrow{\text{red}} & a' & \xleftarrow{\text{blue}} & c' \end{array}$$

where the red arrows are fibrations and the vertical arrows are weak equivalences. Then, the induced map:

$$b \times_a c \rightarrow b' \times_{a'} c',$$

is in $W_{\mathcal{C}}$. This requirement is known as the Gluing Lemma.

A weak cofibration category is a category with cofibrations and weak equivalences as defined in [\[73, p.326\]](#).

Since our notation for the two subcategories of a weak fibration category involve the notation for the category itself, we will find it reasonable to mostly omit explicit mention of the subcategories when referring to a weak fibration category. Here's some further terminology that will be repeated later for categories of fibrant objects.

Terminology 3.31. Let $\mathcal{C}$ be a category equipped with two full subcategories:

$$\text{Fib}_{\mathcal{C}}, W_{\mathcal{C}},$$

of $\text{Arr}(\mathcal{C})$. Then, the morphisms in the subcategory $\text{Fib}_{\mathcal{C}} \cap W_{\mathcal{C}}$ are known as trivial fibrations.

Before we give examples, let us develop some theory associated to this definition since it is rather interesting. In particular, the Gluing Lemma is ideally something we'd like to establish, rather than postulate. We have the following result, which is an amalgamation of facts that are scattered throughout the literature but are useful to have in a single place.

Proposition 3.32

Let $\mathcal{C}$ be a category and suppose that there are two full subcategories Fib, W of $\text{Arr}(\mathcal{C})$ so that:

1. All isomorphisms of $\mathcal{C}$ are in W .
2. W satisfies the 2-out-of-3 property.
3. If $f \in \text{Mor}(\mathcal{C})$, then $f = p \circ i$, where $p \in \text{Fib}$ and $i \in W$.
4. Fibrations are preserved by base extensions.

Then, the following statements hold:

1. If $f \in \text{Mor}(\mathcal{C})$ is a section of a morphism in $W \cap \text{Fib}$, then $f \in W$.
2. If $f \in \text{Mor}(\mathcal{C})$ with factorization:

$$f = p \circ i,$$

then $f \in W$ iff $p \in \text{Fib} \cap W$.

3. If $\mathcal{C}$ is closed under pullbacks, then TFAE:

- (a) The Gluing Lemma holds.
- (b) A pullback of a morphism in W along a morphism in Fib is in W .

Proof. We'll prove each statement one-by-one.

1. Let $g \in W \cap \text{Fib}$ so that $g \circ f = \text{Id}$. Since Id is an isomorphism, it is in W . On the other hand, $g \in W$ so the 2-out-of-3 property implies that $f \in W$.
2. We'll prove each implication separately.
 - $\Rightarrow$: If $f \in W$, then the 2-out-of-3 property implies that $p \in W$.
 - $\Leftarrow$: If $p \in W$, then W being a subcategory means that $f \in W$.

This completes the proof.

3. This is the dual of the result proved in [48]. ■

Here are some examples.

Example 3.33. Let $\mathcal{C}$ be a fibration category & let $W_{\mathcal{C}}$ be the full subcategory of $\text{Arr}(\mathcal{C})$ of all isomorphisms.

Claim. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is known as the minimal weak fibration structure on a fibration category.

Example 3.34. Let $\mathcal{C}$ be a fibration category. We define $W_{\mathcal{C}}$ to be all morphisms in $\mathcal{C}$.

Claim. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is known as the maximal weak fibration structure on a fibration category.

Our interest is mainly in categories of fibrant objects. In fact, these are closely related to weak fibration categories in a manner that will be deduced later. For now, let us work towards defining categories of fibrant objects. To do this, we will need some notation, followed by some terminology.

Notation 3.35. Let $\mathcal{C}$ be a category and suppose we have the following commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\beta} & y \\ \alpha \downarrow & & \downarrow f \\ x' & \xrightarrow{g} & y' \end{array}$$

If the fibre product of the diagram above exists, then the commutative diagram above is a pullback cone. So, there exists a unique morphism $(\alpha, \beta) : x \rightarrow x' \times_{y'} y$ such that the resulting diagram formed is commutative. In particular, let $f : x \rightarrow y$ be a morphism in $\mathcal{C}$. Then, the diagonal induced by f is defined by:

$$\Delta_f \stackrel{\text{def}}{=} (\text{Id}_x, \text{Id}_x).$$

Diagram-wise, the one to imagine is the following:

$$\begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ \text{Id}_x \downarrow & & \downarrow f \\ x & \xrightarrow{f} & y \end{array}$$

commutative diagram.

Terminology 3.36. Let $\mathcal{C}$ be a category with finite products, final object e and two full subcategories:

$$\text{Fib}, W \subseteq \text{Arr}(\mathcal{C}).$$

If $x \in \mathcal{C}$, then the path space of x is an object, usually called the path object, x^I together with morphisms:

$$x \xrightarrow{s} x^I \xrightarrow{(d_0, d_1)} x \times x$$

such that $s \in W$, $(d_0, d_1) \in \text{Fib}$ and the composite is the diagonal morphism^a.

^aThe notation (s, x^I, d_0, d_1) will be used for the path space of x .

Here's some notation involving path spaces.

Notation 3.37. Let $\mathcal{C}$ is a category as in [Terminology 3.36](#) and let $x \in \mathcal{C}$. From now on, if (x^I, s, d_0, d_1) is a path space for x , then we will simply denote this by x^I unless it is absolutely necessary to specify the other morphism. Moreover, if there is one path space being considered, then the notation as a quadruple above will be valid. So, we may say “Let x^I be the path space for x ” and later in a given proof, we might use d_0, d_1 whenever necessary.

With this preparation at hand, we can define categories of fibrant objects.

Definition 3.38

Let $\mathcal{C}$ be a pointed category and two full subcategories of $\text{Arr}(\mathcal{C})$:

$$\text{Fib}, W,$$

called *fibrations* and *weak equivalences* respectively so that the following are satisfied:

1. $(\mathcal{C}, \text{Fib})$ is a fibration category.
2. Any isomorphism in $\mathcal{C}$ is in W .
3. W satisfies the 2-out-of-3 property.
4. Morphisms in $\text{Fib} \cap W$ are preserved by base extensions.
5. If $x \in \mathcal{C}$, there exists a path space (s, x^I, d_0, d_1) .

This is known as a category of fibrant objects for a homotopy theory.

We have a remark about the definition.

Remark 3.39. In the original definition of Brown, the category $\mathcal{C}$ is not required to be pointed but simply has a final object. Also, the way we've stated the definition above is much less explicit than the one given by Brown. Nevertheless, we think that this is a cleaner presentation that relates more obviously back to Waldhausen's work.

One may notice that categories of fibrant objects satisfy assumptions (1), (2) and (4) of [Proposition 3.32](#). Our first goal will be to show that it satisfies (3) as well. Indeed, this result ends up being a standard tool of the theory.

Proposition 3.40. Brown's Factorization Lemma

Let $\mathcal{C}$ be a category of fibrant objects and let $f \in \text{Mor}(\mathcal{C})$. Then:

$$\exists p \in \text{Fib} : i \in W : f = p \circ i.$$

In particular, f is a product of a fibration and a weak equivalence^a.

^aSuch a factorization of f will be called a Brown factorization of f .

The proof of this result is not so difficult but it does require a bit of preparation. The first bit of preparation involves the following lemma, where we say a bit more about the evaluation maps in a path space.

Lemma 3.41 Let $\mathcal{C}$ be a category of fibrant objects, let $x \in \mathcal{C}$ and select a path space:

$$(s, x^I, d_0, d_1).$$

Then, $d_0, d_1 : x^I \rightarrow x$ are trivial fibrations.

This lemma was originally found in [\[59, Lemma 2, p. 6-7\]](#); that result is about cofibrant objects which have cylinders in a model category. However, the structure of a category of fibrant objects is rich enough to yield the result.

Proof of Lemma 3.41. Let 0 be the zero object. Then, there is a unique morphism $f : x \rightarrow 0$ and since $\mathcal{C}$ is a

fibration category, we see that f is a fibration. Now, we have the pullback diagram:

$$\begin{array}{ccc} x \times_0 x & \xrightarrow{\pi_2} & x \\ \pi_1 \downarrow & & \downarrow f \\ x & \xrightarrow{f} & 0 \end{array}$$

and we note that since fibrations are preserved by base extensions, it follows that π_1 and π_2 are fibrations.

| **Claim.** d_0 is a fibration.

This follows from the fact that (d_0, d_1) is a fibration so composing it with π_1 produces a fibration. However, this composition is just d_0 so the claim follows, since fibrations compose to form fibrations.

| **Claim.** d_0 is a weak equivalence.

The identity map is a trivial fibration, so diagonal maps must also be trivial fibrations. Now, $(d_0, d_1) \circ s$ is a weak equivalence because it is a diagonal map and since s is a weak equivalence, we conclude by Axiom 1 of Definition 3.38 that (d_0, d_1) is a weak equivalence. But this means that d_0 is a weak equivalence^a and we're done. ■

^aThe analogous proof for d_1 will go through as well.

A reasonably big component to the proof of the factorization lemma is the so-called mapping cylinder construction.

Construction 3.42. Let $\mathcal{C}$ be a category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and let:

$$(s, y^I, d_0^y, d_1^y),$$

be a path space for y . We consider the following pullback^a:

$$\begin{array}{ccc} \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I \\ \bar{d}_f \downarrow & & \downarrow d_0^y \\ x & \xrightarrow{f} & y \end{array}$$

$\text{cyl}(f)$ is called the mapping cylinder of f relative to the given path space. For the specific categories of fibrant objects, there will always be a canonical choice of path space for any given object and in those cases, one computes the mapping cylinder with respect to that canonical choice of path space.

^aThe pullback exists because d_0^y is a fibration.

Here's another lemma where we record some basic properties of the construction.

Lemma 3.43 In the notation of Construction 3.42, TFSH:

1. $\bar{d}_f$ is a trivial fibration.
2. If f is a fibration, then $d_1^y \circ \bar{f}$ is a fibration.

Proof. We will prove each statement one-by-one.

1. This follows from the fact that d_0^y is a trivial fibration and that trivial fibrations are preserved by base extensions.
2. If f is a fibration, then $\bar{f}$ is a fibration. Since d_1^y is a fibration, the result holds. ■

We're ready to prove Brown's Factorization Lemma.

Proof of Proposition 3.40. The proof is conceptually simple. We will divide it into two parts: we'll first construct the decomposition of f and then show that it has the required properties. With that said, let us proceed.

1. Let $f : x \rightarrow y$ be a morphism and choose a path space:

$$(s, y^I, d_0^y, d_1^y).$$

Let us consider the mapping cylinder $\text{cyl}(f)$ relative to this path space; we will use the notation of Construction 3.42. We will define the morphism i from $x \rightarrow \text{cyl}(f)$ and p from $\text{cyl}(f) \rightarrow y$. For p , this is easy:

$$p \stackrel{\text{def}}{=} d_0^y \circ \pi_2.$$

Before we define i , we have a small calculation:

$$\begin{aligned} d_0^y \circ (s \circ f) &= \pi_y^1 \circ (d_0^y \times d_1^y) \circ (s \circ f) \\ &= \pi_y^1 \circ \Delta_y \circ f \\ &= \text{Id}_y \circ f \\ &= f \\ &= f \circ \text{Id}_x, \end{aligned}$$

where $\pi_y^1 : y \times y \rightarrow y$ is a canonical projection onto the first coordinate. This gives us a diagram:

$$\begin{array}{ccccc}
 & & & & \\
 & & & & \\
 x & \xrightarrow{s \circ f} & & & y^I \\
 & \searrow \text{Id}_x & \text{cyl}(f) & \xrightarrow{\pi_2} & y^I \\
 & & \downarrow \pi_1 & & \downarrow d_0^y \\
 & & x & \xrightarrow{f} & y
 \end{array}$$

which commutes. By universal property of the pullback, we see that there is a unique map:

$$i : x \rightarrow \text{cyl}(f),$$

which makes the entire diagram above commute. In particular:

$$\begin{aligned}
 p \circ i &= (d_0^y \circ \pi_2) \circ i \\
 &= d_0^y \circ (\pi_2 \circ i) \\
 &= d_0^y \circ (s \circ f) \\
 &= f \circ \text{Id}_x \\
 &= f,
 \end{aligned}$$

as was desired.

2. Let's deal with the properties. So, the pullback diagram above tells us that:

$$\text{Id}_x = \pi_1 \circ i.$$

But π_1 is a trivial fibration, so that proves statement (2). The fact that i is a weak equivalence follows from the fact that it is right inverse to a trivial fibration, as proved in Proposition 3.32. Now, we have a claim.

| Claim. p is a fibration.

So, the idea is that we can write p as the following composition:

$$z \xrightarrow{g} x \times y \longrightarrow y$$

where the second morphism is a projection. The first morphism is formed by the following pullback:

$$\begin{array}{ccc}
 (x \times y) \times_{y \times y} y^I \simeq x \times_y y^I & \longrightarrow & x \times y \\
 \downarrow & & \downarrow f \times \text{Id}_y \\
 y^I & \xrightarrow{(d_0, d_1)} & y \times y
 \end{array}$$

while the second one is formed by the following pullback:

$$\begin{array}{ccc}
 x \times y & \longrightarrow & y \\
 \downarrow & & \downarrow \\
 x & \longrightarrow & 0
 \end{array}$$

where $x \rightarrow e$ and $y \rightarrow e$ are fibrations. So, our projection $x \times y \rightarrow y$ is a fibration. Now, the first map is also a fibration because (d_0, d_1) is a fibration. It remains to be shown that the composition actually gives us what we want but this is also clear based on previous calculations.

This completes the proof. ■

This result is interesting in its own right. The next result is referenced casually in several places in the literature but as we have never seen an explicit proof, we take this opportunity to fill in the gap.

Proposition 3.44

Let $\mathcal{C}$ be a pointed category and suppose that there are two full subcategories:

$$\text{Fib}_{\mathcal{C}}, W_{\mathcal{C}},$$

satisfying axioms (1) – (4) in Definition 3.38. TFAE:

1. This structure satisfies axiom (5).
2. If $f \in \text{Mor}(\mathcal{C})$, then there is a $p \in \text{Fib}_{\mathcal{C}}$ and $i \in W$ such that $f = p \circ i$.

Proof. (1) $\Rightarrow$ (2) is simply Brown's Factorization Lemma. For (2) $\Rightarrow$ (1), let $x \in \mathcal{C}$ and consider the diagonal map:

$$\Delta_x : x \rightarrow x \times x,$$

where this is well-defined because $\mathcal{C}$ is closed under finite products. Now, we factorize this as follows:

$$\Delta_x = p_x \circ i_x,$$

where $p_x \in \text{Fib}_{\mathcal{C}}$ and $i_x \in W$. But now, we see that the quadruple:

$$(i_x, x, \pi_1 \circ p_x, \pi_2 \circ p_x),$$

is a path space for x , where $\pi_1, \pi_2 : x \times x \rightarrow x$ are the canonical projections. ■

Let us discuss some corollaries of Brown's factorization result. We'll start with the ones which appear directly in the literature (specifically in [11]). The first two below are mentioned in Brown's paper with one-line indications towards proofs. We have included them with the details further fleshed out for the convenience of the reader.

Claim 3.7. Let $\mathcal{C}$ be a category of fibrant objects. Then, any commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & y \\ i \downarrow & & \downarrow p \\ x' & \xrightarrow{\quad} & y' \end{array}$$

with i being a weak equivalence and p being a fibration, can be imbedded in a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{\quad} & z & \xrightarrow{\quad} & y \\ & \searrow i & \downarrow t & & \downarrow p \\ & & x' & \xrightarrow{\quad} & y' \end{array}$$

where t is a trivial fibration. This is proved on Page 99.

Claim 3.8. Let $\mathcal{C}$ be a category of fibrant objects, let $x, y \in \mathcal{C}$, let $u : x \rightarrow y$ be a morphism and suppose that we have two path spaces $(x^I, s, d_0, d_1), (y^I, s^y, d_0^y, d_1^y)$. Then, there is a path space (x^J, s^x, d_0^x, d_1^x) , a trivial fibration $t : x^J \rightarrow x^I$ and a morphism $\bar{u} : x^J \rightarrow y^I$ such that the following diagram:

$$\begin{array}{ccccc} & & & & y \\ & & & & \downarrow s^y \\ x & \xrightarrow{s^x} & x^J & \xrightarrow{\bar{u}} & y^I \\ \downarrow s & & \downarrow (d_0^x, d_1^x) & & \downarrow (d_0^y, d_1^y) \\ x^I & \xrightarrow{(d_0, d_1)} & x \times x & \xrightarrow{u \times u} & y \times y \end{array}$$

(Note: The diagram above is a simplified representation of the complex commutative diagram in the image, which includes additional dashed arrows and labels like t and $\bar{u}$.)

commutes. This is proved on Page 99.

The next result is a neat corollary of the factorization lemma.

Corollary 3.45. Ken Brown's Lemma

Let $\mathcal{C}$ be a category of fibrant objects, let $\mathcal{D}$ be a category with a set:

$$W \subseteq \text{Mor}(\mathcal{D}),$$

satisfying the 2-out-of-3 property and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W$. Then, $F(W_{\mathcal{C}}) \subseteq W$.

Proof. Let $f \in \text{Mor}(\mathcal{C})$ be a weak equivalence. We write:

$$f = p \circ i,$$

where p is a trivial fibration and i is a weak equivalence. So, $F(p) \in W$ and it would be sufficient to show that $F(i) \in W$. To do this, we note that i is right-inverse to a trivial fibration q so we must have that:

$$q \circ i = \text{Id}.$$

Since Id is an isomorphism, it is a trivial fibration so we see that $F(q) \circ F(i) = F(\text{Id})$ with $F(q)$ and $F(\text{Id})$ being in W . It follows that $F(i) \in W$. But this means that $F(f) \in W$ and we're done. ■

The next result seems to have first been proved in the C^* -algebra context in [67, Proposition 1.13, p.412-413]. However, the proof actually generalizes so that the analogue of that result holds in any category of fibrant objects.

Corollary 3.46

Let $\mathcal{C}$ be a category of fibrant objects and suppose we have a pullback diagram in $\mathcal{C}$:

$$\begin{array}{ccc} u & \xrightarrow{g_1} & x \\ g_2 \downarrow & & \downarrow f_1 \\ y & \xrightarrow{f_2} & z \end{array}$$

where f_1 is a fibration and f_2 is a weak equivalence. Then, g_1 is a weak equivalence.

Proof. Let us proceed in two steps.

1. We'll prove this result assuming that f_2 is also a fibration.
2. We'll extend the result to f_2 just being a weak equivalence.

Again, we deal with these two steps separately:

1. If we assume that f_2 is also a fibration, then we have a pullback of a trivial fibration and this means that g_1 is a trivial fibration. In particular, g_1 is a weak equivalence.
2. We factorize f_2 as $j \circ r$, with j being a fibration and r being a weak equivalence. Now, we have a diagram:

$$\begin{array}{ccccc} u & \xrightarrow{g_1} & x & & \\ & \searrow \bar{r} & \nearrow \bar{j} & & \\ & & m' & & \\ g_2 \downarrow & & \downarrow & & \downarrow f_1 \\ y & \xrightarrow{r} & m & \xrightarrow{j} & z \end{array}$$

Let us explain what's going on in the diagram:

- (a) m' is the pullback $x \times_z m$.
- (b) The maps g_1 and $r \circ g_2$ give rise to a commutative square and $\bar{r}$ is the morphism formed from the commutative square via the universal property of the pullback.

Having explained the diagram, we can now present it more cleanly:

$$\begin{array}{ccccc} u & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & x \\ g_2 \downarrow & & \downarrow & & \downarrow f_1 \\ y & \xrightarrow{r} & m & \xrightarrow{j} & z \end{array}$$

Since f_2 is a weak equivalence and since r is a weak equivalence, we use the 2-out-of-3 property to conclude that j is a weak equivalence. So, j is a trivial fibration and we use (1) above to conclude that $\bar{j}$ is a weak equivalence. But this means that $\bar{j} \circ \bar{r} = g_1$ is a weak equivalence.

This completes the proof. ■

As a consequence, we have the following corollary.

Corollary 3.47. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects. Then, it is a weak fibration category.

Proof. Since a pullback of a morphism in $W_{\mathcal{C}}$ along a morphism in Fib is in W , we see that $W_{\mathcal{C}}$ satisfies the Gluing Lemma. It follows that our triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration subcategory. ■

We have a remark about these results so far.

Remark 3.48. Of course, it's reasonable for one to wonder when a weak fibration category is a category of fibrant objects. Indeed, let $\mathcal{C}$ be a weak fibration category. Then, we need to satisfy the following additional requirements:

1. $W_{\mathcal{C}}$ needs to satisfy the 2-out-of-3 property.
2. Morphisms in $\text{Fib}_{\mathcal{C}} \cap W_{\mathcal{C}}$ are preserved by base extensions.
3. The existence of a path space for each object in $\mathcal{C}$.

Statements (1) and (2) depend, more or less, on the ambient choice of the category. The existence of a path space can, as we have seen, be deduced from the existence of an appropriate factorization.

With the results above, we can also say more about the mapping cylinder.

Proposition 3.49. A. Vats, 2026

In the notation of [Construction 3.42](#), TFSH:

1. f is a weak equivalence iff $\bar{f}$ is a weak equivalence.

2. A Brown factorization of f produces a Brown factorization of $\bar{f}$.
3. If there is a morphism $\phi : x \rightarrow y^I$ such that:

$$f = d_0^y \circ \phi,$$

then $\bar{d}_f$ is a split epi.

4. The morphism ϕ in statement (5) exists, so $\bar{d}_f$ is always split epi.

Proof. We will prove each statement one-by-one.

1. We'll prove each implication separately.

$\Rightarrow$: This is an application of Corollary 3.46.

$\Leftarrow$: We consider the pullback diagram defining the mapping cylinder and note that $d_0^y \circ \bar{f}$ is a weak equivalence since each factor is a weak equivalence. Hence, we see that $f \circ \bar{d}_f$ is a weak equivalence and since $\bar{d}_f$ is a weak equivalence, it follows by the 2-out-of-3 property that f is a weak equivalence.

This proves the result.

2. We will use an argument analogous to that of Corollary 3.46. Let us factorize f as $j \circ r$, where $j : m \rightarrow y$ is a fibration and $r : x \rightarrow m$ is a weak equivalence. The process described in the proof of Corollary 3.46 allows us to put all of this in a large commutative diagram:

$$\begin{array}{ccccc} \text{cyl}(f) & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & y^I \\ \bar{d}_f \downarrow & & \downarrow & & \downarrow d_0^y \\ x & \xrightarrow{r} & m & \xrightarrow{j} & y \end{array}$$

The key observation is that the right square above is a pullback, so the entire rectangle is a pullback iff the left square is a pullback^a. Since the entire rectangle is a pullback, we see that the left square is a pullback. Now:

$$\bar{f} = \bar{j} \circ \bar{r}.$$

Since j is a fibration, we see that $\bar{j}$ is a fibration. So, the morphism in the middle of the diagram is a fibration because d_0^y is a fibration. Since r is a weak equivalence, we can use Corollary 3.46 to conclude that $\bar{r}$ is a weak equivalence. This produces a Brown factorization for $\bar{f}$ so we're done.

3. The point is to write the given information into the diagram of Construction 3.42. To be precise, we have:

$$\begin{array}{ccccc} & & \phi & & \\ & \curvearrowright & & \curvearrowleft & \\ x & \xrightarrow{\quad} & \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I \\ & \searrow r & \downarrow \bar{d}_f & & \downarrow d_0^y \\ & & x & \xrightarrow{f} & y \\ & \nearrow \text{Id}_x & & & \end{array}$$

So, the given condition on ϕ ensures that the outer diagram above commutes. By the universal property of the pullback, we obtain a morphism $r : x \rightarrow \text{cyl}(f)$ such that the entire diagram above commutes. We see that:

$$\bar{d}_f \circ r = \text{Id}_x,$$

as was desired.

4. Let $s : y \rightarrow y^I$ be the morphism into the path object which comes as part of the data of the path space. Then:

$$d_0^y \circ s = \text{Id}_y,$$

so we see that:

$$d_0^y \circ s \circ f = \text{Id}_y \circ f = f.$$

So, we simply define $\phi \stackrel{\text{def}}{=} s \circ f$ and this works, so $\bar{d}_f$ is split epi. ■

^aWe are using the Pasting Law for pullbacks here; see [6].

This corollary seems a little strange so let us provide some context.

Remark 3.50. The first two statements just give us additional information about the maps that appear in the pullback diagram for the mapping cylinder of a morphism. The third statement is a formal analogue of [67, Proposition 1.8, p.410] which was proved concretely in $C^*\text{Alg}$ (in fact, we prove a bit more). As it turns out, the proof that we give here will specialize to that result once we have shown that $C^*\text{Alg}$ is a category of fibrant objects.

Of course, one would hope that we can construct categories of fibrant objects by looking at special subcategories. This

leads us to the notion of *fibrant subcategory* which we introduce as a definition. This seems to not be in the literature.

Definition 3.51. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$ so that:

1. $\mathcal{D}$ retains the zero object of $\mathcal{C}$.
2. $W_{\mathcal{D}}$ is a full subcategory of $W_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
3. $\text{Fib}_{\mathcal{D}}$ is a full subcategory of $\text{Fib}_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
4. $\text{Fib}_{\mathcal{D}}$ is closed under pullbacks by fibrations in $\text{Fib}_{\mathcal{D}}$.
5. If $x \in \mathcal{D}$, then there is a path space of x which lies in $\mathcal{D}$.

In that case, $(\mathcal{D}, W_{\mathcal{D}}, \text{Fib}_{\mathcal{D}})$ is known as a fibrant subcategory.

In fact, it is the case that a fibrant subcategory is actually a category of fibrant objects.

Proposition 3.52. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects. Any fibrant subcategory of $\mathcal{C}$ is a category of fibrant objects^a.

^aOf course, equipped with the specified structure in Definition 3.51.

Proof. Let $\mathcal{D}$ be a fibrant subcategory of $\mathcal{C}$. We will verify each of the axioms one-by-one.

1. Well, $\mathcal{D}$ is a full subcategory which retains the final object of $\mathcal{C}$ so it is a category with a final object.
2. Let $f, g \in \text{Mor}(\mathcal{D})$ so that $g \circ f$ is well-defined and suppose that two out of three of these are in $W_{\mathcal{D}}$. This means that two out of three of these are in W , so we conclude that the third must be in W . Hence, the third must also be in $W_{\mathcal{D}}$.
3. Let f be an isomorphism in $\mathcal{D}$. Then, it's an isomorphism in $\mathcal{C}$ so $f \in W \cap \text{Fib}$. But this means that $f \in W_{\mathcal{D}}$ and $f \in \text{Fib}_{\mathcal{D}}$. It follows that f is a trivial fibration in $\mathcal{D}$.
4. Let $f, g \in \text{Fib}_{\mathcal{D}}$ such that $g \circ f$ exists. Then, $f, g \in \text{Fib}$ so $g \circ f \in \text{Fib}$. But this means that $g \circ f \in \text{Fib}_{\mathcal{D}}$.
5. Consider the following pullback diagram in $\mathcal{D}$:

$$\begin{array}{ccc} P & \longrightarrow & B \\ \bar{p} \downarrow & & \downarrow p \\ A & \longrightarrow & C \end{array}$$

where $p \in \text{Fib}$. This is also a pullback diagram in $\mathcal{C}$, so we see that $\bar{p} \in \text{Fib}$. But this means that $\bar{p} \in \text{Fib}_{\mathcal{D}}$; the same argument holds if we start by assuming that $p \in \text{Fib} \cap W$.

6. Let $x \in \mathcal{D}$. Since $\mathcal{D}$ inherits the final object of $\mathcal{C}$, we have a unique morphism $x \rightarrow e$, where e is the final object, and this morphism lies in $\mathcal{D}$. But it is also in Fib , so it follows that it's in $\text{Fib}_{\mathcal{D}}$. This implies, of course, that fibrant subcategories are closed under finite products.
7. Let $x \in \mathcal{D}$. Then, $x^I \in \mathcal{D}$ so we obtain our quadruple (s, x^I, d_0, d_1) . Now, $s \in W$ so we see that $s \in W_{\mathcal{D}}$. Next, $(d_0, d_1) \in \text{Fib}$. On the other hand, $(d_0, d_1) \in \text{Mor}(\mathcal{D})$ because $\mathcal{D}$ is closed under finite products. This implies that $(d_0, d_1) \in \text{Fib}_{\mathcal{D}}$.

This proves the desired result. ■

Here's a remark about our work so far.

Remark 3.53. Definition 3.51 is just about the best that we can do with the structure available. Indeed, the main problem is to find an appropriate path space. Any and all conditions needed to ensure that we can pick a path space in the subcategory are entirely dependent on the structure of the ambient category of fibrant objects.

Whenever we have a functor between categories of fibrant objects and this functor preserves the weak equivalences of both categories, we can always induce a functor between the respective homotopy categories using the universal property of the localization. Let us formalize the preceding sentence in the next construction.

Construction 3.54. Let $\mathcal{C}, \mathcal{D}$ be categories of fibrant objects and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor such that:

$$F(W_{\mathcal{C}}) \subseteq W_{\mathcal{D}}.$$

Then, we note that $\gamma_{\mathcal{D}} \circ F : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{D})$ is a functor which inverts morphisms in $W_{\mathcal{C}}$. So, the universal property of the localization yields that there exists a unique functor $\mathbf{Ho}(\mathcal{C}) : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{D})$ such that the following diagram:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \gamma_{\mathcal{C}} \downarrow & & \downarrow \gamma_{\mathcal{D}} \\ \mathbf{Ho}(\mathcal{C}) & \xrightarrow{\mathbf{Ho}(F)} & \mathbf{Ho}(\mathcal{D}) \end{array}$$

commutes. This is known as the homotopy functor of F between homotopy categories. In fact, we can prove more.

| **Claim.** For the construction above to go through, it is sufficient to assume that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W_{\mathcal{D}}$.

This follows immediately from Corollary 3.45 and the fact that $W_{\mathcal{D}}$ satisfies the 2-out-of-3 property.

It is notable that categories of fibrant objects have enough structure where we can define homotopies between morphisms. The goal of this section is to study the two different notions of homotopy between morphisms that one can define in a category of fibrant objects. Moreover, we will extend some basic results available in the literature in this generality. Let us start by simply defining the two different ways in which two parallel morphisms can be homotopic.

Terminology 3.55. Let $\mathcal{C}$ be a category of fibrant objects and let $f, g : x \rightarrow y$ be morphisms.

1. f is right-homotopic to g iff there exists a path space^a (y^I, s, d_0, d_1) and a morphism $h : x \rightarrow y^I$ such that:

$$\begin{aligned} f &= d_0 \circ h, \\ g &= d_1 \circ h, \end{aligned}$$

We denote this by writing $f \simeq_{h,r} g$ and we say that h is a right-homotopy from f to g .

2. f is homotopic to g iff there exists a weak equivalence $t : z \rightarrow x$ such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

We denote this by writing $f \simeq_h g$ and we say that f is homotopic to g .

Clearly, if two morphisms are right-homotopic, then they are homotopic^b.

^aMany of our examples of categories of fibrant objects come with canonical path spaces. In those cases, we will simply define these notions with respect to those canonical choices of path spaces.

^bWe simply use the identity morphism as the required weak equivalence.

The next result was essentially mentioned as a comment in [11, p.422]. Nevertheless, we furnish it with a detailed proof.

Claim 3.9. Let $\mathcal{C}$ be a category of fibrant objects, let $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ be the localization and let $f, g \in \mathcal{C}(x, y)$ so:

$$f \simeq_{h,r} g,$$

in $\mathcal{C}$. Then, $\gamma(f) = \gamma(g)$. In particular, this is true when f and g are homotopic. This is proved on Page 100.

The following somewhat technical result is also essential.

Proposition 3.56. K. Brown, 1973, [11, Proposition 1, p.423]

Let $\mathcal{C}$ be a category of fibrant objects and let $f, g \in \mathcal{C}(x, y)$ so that:

$$f \simeq_{h,r} g.$$

TFSH:

1. If $u \in \mathcal{C}(z, x)$, then $f \circ u \simeq_{h,r} g \circ u$.
2. If $u \in \mathcal{C}(y, z)$, then there is a trivial fibration $t : x' \rightarrow x$ so that $u \circ f \circ t \simeq_{h,r} u \circ g \circ t$.

As a consequence, both $\simeq_{h,r}$ and $\simeq_h$ are equivalence relations^a on $\text{Mor}(\mathcal{C})$.

^aFormally, what we mean is that this forms an equivalence relation on each hom-class.

With the result above in hand, we can set some notation which also doubles up as terminology.

Notation 3.57. Let $\mathcal{C}$ be a category of fibrant objects.

1. A right-homotopy equivalence in $\mathcal{C}$ is a morphism f whose image is an isomorphism in $\mathcal{C}/\simeq_{h,r}$.
2. A homotopy equivalence in $\mathcal{C}$ is a morphism f whose image is an isomorphism in $\mathcal{C}/\simeq_h$.

The full subcategory of right-homotopy equivalences in $\text{Arr}(\mathcal{C})$ is denoted by:

$$W_{\mathcal{C},h,r},$$

while the full subcategory of homotopy equivalences in $\text{Arr}(\mathcal{C})$ is denoted by:

$$W_{\mathcal{C},h}.$$

We define $\pi(\mathcal{C})$ to be:

$$\mathcal{C}/\simeq_h,$$

the quotient category formed via $\simeq_h$. This is the π -category of $\mathcal{C}$. Moreover, this category comes equipped with a canonical functor $\pi : \mathcal{C} \rightarrow \pi(\mathcal{C})$ which we refer to as the projection.

Here is a notable result.

Claim 3.10. Let $\mathcal{C}$ be a category of fibrant objects. Then:

$$W_{\mathcal{C},h,r}, W_{\mathcal{C},h},$$

satisfy the 2-out-of-6-property. This is proved on Page 100.

We have a remark about this simple result.

Remark 3.58. So, even though we do not necessarily assume that the weak equivalences in a category of fibrant objects is 2-out-of-6, it is the case that it produces two subcategories of morphisms which are 2-out-of-6.

One basic result concerning categories of fibrant objects involves information about the mapping spaces of the homotopy

category. We will see that the morphisms in the mapping spaces can be described by a calculus-of-fractions formula.

Theorem 3.59. K. Brown, [11, Theorem 1, p.425]

Let $\mathcal{C}$ be a category of fibrant objects. TFSH:

1. $\pi(W) \subseteq \pi(\mathcal{C})$ admits a calculus of right fractions.
2. $\forall x, y \in \mathcal{C} : [x, y]_{\mathcal{C}} \simeq \operatorname{colim}_{x' \xrightarrow{\sim} x} \pi\mathcal{C}(x', y)$.

Moreover, if $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization functor, then TFSH:

1. If $x, y \in \mathbf{Ho}(\mathcal{C})$, then any morphism in $[x, y]_{\mathcal{C}}$ can be written as a right fraction:

$$x \xleftarrow{\gamma(t)} x' \xrightarrow{\gamma(f)} y$$

where $t \in W$ is a weak equivalence.

2. If $x, y \in \mathcal{C}$ and $f_0, f_1 \in \mathcal{C}(x, y)$, then $\gamma(f_0) = \gamma(f_1)$ iff f_0 and f_1 are right-homotopic.

We have a remark about this theorem.

Remark 3.60. This result is a slightly weaker version of a standard theorem which explains why model categories were of interest in the first place; they help us to control the size of the localization with respect to the weak equivalences and give us useful homotopical information about the set of weak equivalences. The result above has a similar flavor.

It is rather unappealing that Proposition 3.56 does not allow for compatibility with compositions to the highest degree. Indeed, precompositions are compatible with right-homotopies while post-compositions are not as compatible. Over here, we take some inspiration from our work with homotopy theory on categories of equivariant operator algebras and introduce the following structures that will allow us to “fix” the issues above.

Definition 3.61. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects and suppose that we have further data of:

1. a functor $T : \mathcal{C} \rightarrow \mathcal{C}$ such that for each $x \in \mathcal{C}$, $T(x)$ gives us a path space (the canonical path space) and;
2. two natural transformations $\alpha_0, \alpha_1 : T \Rightarrow \operatorname{Id}_{\mathcal{C}}$ such that for each $x \in \mathcal{C}$, the morphisms:

$$\alpha_0^x, \alpha_1^x : T(x) \rightarrow x,$$

are the evaluation morphisms in the path space of $T(x)$.

Then, the quadruple $(\mathcal{C}, T, \alpha_0, \alpha_1)$ forms a h-category of fibrant objects^a.

^aAll constructions involving path spaces will, then, occur with respect to the canonical path space.

In such a category, we see that right-homotopy is fully compatible with compositions.

Proposition 3.62. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h-category of fibrant objects, let $x, y, z \in \mathcal{C}$, let $f_1, f_2 \in \mathcal{C}(x, y)$ and let $g_1, g_2 \in \mathcal{C}(y, z)$ so that:

$$f_1 \simeq_{h,r} f_2,$$

$$g_1 \simeq_{h,r} g_2.$$

Then, $g_1 \circ f_1 \simeq_{h,r} g_2 \circ f_2$.

Proof. The key is to proceed exactly as in the proof of Proposition 2.24. The first step of the argument is simply statement (1) of Proposition 3.56, while the last step is purely formal and proceeds as written. So, what we need to do is to clarify step (2). So, let $g_1 = g_2 = g$ and let $h : x \rightarrow y^I$ be a homotopy such that:

$$f_1 = \alpha_0^y \circ h,$$

$$f_2 = \alpha_1^y \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} T(g) \circ h.$$

This is a morphism from x to z^I . Now, let us show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$ by a computation:

$$\alpha_0^z \circ H = \alpha_0^z \circ T(g) \circ h = g \circ \alpha_0^y \circ h = g \circ f_1,$$

$$\alpha_1^z \circ H = \alpha_1^z \circ T(g) \circ h = g \circ \alpha_1^y \circ h = g \circ f_2,$$

which completes the proof. ■

It is the case that a h-category of fibrant objects admits an internal notion of homotopy which agrees with the notion of homotopy defined in concrete categories. It is also the case that such a category will admit an internal notion of fibration. To define this, we will utilize statement (2) of Proposition 2.26 as inspiration.

Terminology 3.63. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h-category of fibrant objects and let $p \in \mathcal{C}(x, y)$. p is an internal fibration (int-fibration) iff for any $z \in \mathcal{C}$, any $f \in \mathcal{C}(z, x)$ and for any $h \in \mathcal{C}(z, T(y))$ which is a homotopy of $p \circ f$:

$$\alpha_0^y \circ h = p \circ f,$$

there exists a $H \in \mathcal{C}(z, T(x))$ such that the following diagram:

$$\begin{array}{ccc}
 & & x \\
 & \nearrow f & \uparrow \alpha_0^x \\
 z & \text{---} H \text{---} & T(x) \\
 & \searrow h & \downarrow T(p) \\
 & & T(y)
 \end{array}$$

commutes.

With a h -category of fibrant objects, we can also characterize fibrations in a different way. The discussion here was inspired by [67, Proposition 1.10, p.410-411]. We will, first of all, require a simple construction.

Construction 3.64. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and consider the diagram:

$$\begin{array}{ccccc}
 & & & \alpha_0^x & \\
 & & & \curvearrowright & \\
 T(x) & & & & x \\
 & \searrow v & \xrightarrow{\bar{d}_f} & & \\
 & \text{cyl}(f) & & & \\
 & \downarrow \bar{f} & & & \downarrow f \\
 & T(y) & \xrightarrow{\alpha_0^y} & & y \\
 & \nearrow T(f) & & &
 \end{array}$$

So, the key point is that we have the pullback square for the mapping cylinder and the outer diagram commutes due to the naturality of α_0 . By the universal property of the pullback, we induce a unique morphism $v : T(x) \rightarrow \text{cyl}(f)$ such that the entire diagram commutes. We refer to this as the *cyl-mapcyl* morphism.

We should collect some properties of internal fibrations. Actually, these properties will show that a suitable h -category of fibrant objects actually admits **another** fibration category structure.

Proposition 3.65. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects so that T preserves pullbacks. TFSH:

1. The internal fibrations are closed under composition.
2. If $x \in \mathcal{C}$, then the unique map $x \rightarrow 0$ is an internal fibration.
3. Let $f \in \text{Mor}(\mathcal{C})$ be an isomorphism. Then, it is an internal fibration.
4. Internal fibrations are preserved by base extensions.

In particular, $\mathcal{C}$ is a fibration category when equipped with its internal fibrations.

Proof. Due to the work we have already put into making the proof of Proposition 2.28 as formal as possible. So, in fact, the proof of this result is simply the proof of that result with the cylinder replaced with T and with the time-evaluation maps replaced with the components of α_0 and α_1 . The additional requirement on T to preserve pullbacks ensures that the proof of statement (4) in the aforementioned result also carries over here. ■

Let us characterize internal fibrations in a h -category of fibrant objects among the morphisms. This result was proved in C^*Alg by Schochet in [67, Proposition 1.10, p.410-412] and we were able to adapt the proof to this setting.

Proposition 3.66

In the notation of Construction 3.64 & assuming that T preserves pullbacks, TFAE:

1. f is an internal fibration.
2. The *cyl-mapcyl* morphism v is a split epi.

Proof. We will use the notation of Construction 3.64 throughout the proof.

(1) $\Rightarrow$ (2): Observe that:

$$\alpha_0^y \circ \bar{f} = f \circ \bar{d}_f.$$

So, this means that $\bar{f}$ is a homotopy of $\bar{d} \circ f$. Now, we consider the following homotopy lifting problem:

$$\begin{array}{ccc}
 & & x \\
 & \nearrow \bar{d}_f & \uparrow \alpha_0^x \\
 \text{cyl}(f) & \dashrightarrow w & T(x) \\
 & \searrow \bar{f} & \downarrow T(f) \\
 & & T(y)
 \end{array}$$

Since f is an internal fibration, we see there exists a homotopy lift $w : \text{cyl}(f) \rightarrow T(x)$. Let us check that this is, in fact, a right inverse to the map v . Observe that we have the following equations:

$$\begin{aligned}
 \bar{d}_f &= \alpha_0^x \circ w = \bar{d}_f \circ (v \circ w), \\
 \bar{f} &= T(f) \circ w = \bar{f} \circ (v \circ w).
 \end{aligned}$$

Now, we apply the universal property of the pullback to conclude that $v \circ w = \text{Id}_{\text{cyl}(f)}$.

(2) $\Rightarrow$ (1): Let $w : \text{cyl}(f) \rightarrow T(x)$ be a split for w and consider the following lifting problem:

$$\begin{array}{ccc}
 & & x \\
 & \nearrow g & \uparrow \alpha_0^x \\
 z & & T(x) \\
 & \searrow h & \downarrow T(f) \\
 & & T(y)
 \end{array}$$

where h is a homotopy of $f \circ g$. Now, we can write this out in a pullback square:

$$\begin{array}{ccccc}
 & & & & g \\
 & & & & \nearrow \\
 z & \xrightarrow{\quad \hat{h} \quad} & \text{cyl}(f) & \xrightarrow{\quad \bar{d}_f \quad} & x \\
 & \searrow h & \downarrow \bar{f} & & \downarrow f \\
 & & T(y) & \xrightarrow{\quad \alpha_0^y \quad} & y
 \end{array}$$

where the diagram commutes precisely because h is a homotopy of $f \circ g$. By the universal property of the pullback, we have a morphism $\hat{h} : z \rightarrow \text{cyl}(f)$. Now, we define:

$$k \stackrel{\text{def}}{=} w \circ \hat{h}.$$

This is a morphism from z to $T(x)$, so it is a candidate homotopy lift. Let's show that it works:

$$\begin{aligned}
 \alpha_0^x \circ k &= \alpha_0^x \circ (w \circ \hat{h}) \\
 &= (\alpha_0^x \circ w) \circ \hat{h} \\
 &= \bar{d}_f \circ \hat{h} \\
 &= g.
 \end{aligned}$$

This shows that k is a homotopy of g . Next, we have the following computation:

$$\begin{aligned}
 T(f) \circ k &= T(f) \circ (w \circ \hat{h}) \\
 &= (T(f) \circ w) \circ \hat{h} \\
 &= \bar{f} \circ \hat{h} \\
 &= h,
 \end{aligned}$$

as was desired. So, f is an internal fibration. ■

It is reasonable to spend some effort in taking the results above as far as possible; we will leave this to future work. Instead, let us turn towards a result about categories of fibrant objects which deserves some attention. Introducing this result ([Theorem 3.77](#)) requires some preparation and in fact, we will not discuss all of the terminology used within the theorem within this subsection, seeing as it is too much to cover. We will mainly use the remainder of this subsection to work towards the statement of that theorem and provide motivation for why it is interesting. Then, we will prove it in

Section 4.2. The first thing we need to do is discuss loops. For this, we will need some terminology.

Terminology 3.67. Let $\mathcal{C}$ be a category and let $f \in \mathcal{C}(x, y)$. Then, we define:

$$\text{Fibre}(f) \stackrel{\text{def}}{=} \text{Eq}(f, 0),$$

the fibre of f . Similarly, we define:

$$\text{Cofibre}(f) \stackrel{\text{def}}{=} \text{CoEq}(f, 0),$$

the cofibre of f . In general, if we have a sequence:

$$x \xrightarrow{i} y \xrightarrow{p} z,$$

of morphisms in $\mathcal{C}$, then:

1. It is a fibre sequence iff $p \circ i = 0$ and $i = \text{Fibre}(p)$.
2. It is a cofibre sequence iff $p \circ i = 0$ and $p = \text{Cofibre}(i)$

Finally, it is fibre-cofibre sequence iff it is both a fibre sequence and a cofibre sequence.

Now, we use this to construct the loop object of an object in a category of fibrant objects.

Construction 3.68. Let $\mathcal{C}$ be a category of fibrant objects and let $x \in \mathcal{C}$. The loop-object of x is the fibre:

$$\Omega(x) \stackrel{\text{def}}{=} \text{Fibre}((d_0, d_1)),$$

where (s, x^I, d_0, d_1) is a path-space of x in $\mathcal{C}$.

We can construct a functor from the loop and this functor satisfies some neat formal properties. The next result details some key properties of this functor and the loop construction as a whole. Note that we use some terms which have not been explicitly defined here; we will not need those notions in an essential way for our work later.

Proposition 3.69. [11, Section 4, p. 428-434]

Let $\mathcal{C}$ be category of fibrant objects. Then, Ω defines a functor:

$$\Omega : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{C}),$$

called the loop-object functor. This functor satisfies the following properties:

1. If $x \in \mathcal{C}$, then $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$ and $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$.
2. If $p : x \rightarrow y$ is a fibration with fibre^a F , then there is a natural right action:

$$F \times \Omega(y) \rightarrow F,$$

in $\mathbf{Ho}(\mathcal{C})$. In particular, we have a natural morphism $\Omega(y) \rightarrow F$ in $\mathbf{Ho}(\mathcal{C})$.

^aThis is the domain of the fibre morphism.

We have underlined the undefined terms and will define them momentarily. Here is a remark.

Remark 3.70. This result can be found, distributed in two results which have been given complete proofs, in [11, Section 4, p. 428-434]. To be precise, the first statement of the proposition is covered by Theorem 3 (p. 430-431) in the aforementioned paper, while the second statement is covered by Proposition 3 (p. 431-432).

Now, we will define (abelian) group objects in a category. The idea is that the properties of (abelian) groups can be represented by the commutativity of some diagrams and this can be mimicked in a fairly general way within a category. Our treatment of these notions is based on the excellent exposition in [24, Part I, Section 2, p.3-4].

Definition 3.71

Let $\mathcal{C}$ be a pointed category (the zero object is denoted by 1) with finite products. A group object in $\mathcal{C}$ is a quadruple:

$$(G, m, e, i),$$

consisting of $G \in \mathcal{C}$ and three morphisms:

$$m : G \times G \rightarrow G,$$

$$e : 1 \rightarrow G,$$

$$i : G \rightarrow G,$$

such that the following diagrams:

$$\begin{array}{ccc}
 G \times (G \times G) & \simeq & (G \times G) \times G \xrightarrow{m \times \text{Id}_G} G \times G \\
 \downarrow \text{Id}_G \times m & & \downarrow m \\
 G \times G & \xrightarrow{m} & G
 \end{array}$$

$$\begin{array}{ccccc}
 1 \times G & \xrightarrow{e \times \text{Id}_G} & G \times G & \xleftarrow{\text{Id}_G \times e} & G \times 1 \\
 & \searrow \simeq & \downarrow m & \swarrow \simeq & \\
 & & G & &
 \end{array}$$

$$\begin{array}{ccc}
 G & \xrightarrow{(1,i)} & G \times G \\
 \downarrow & & \downarrow m \\
 1 & \xrightarrow{e} & G
 \end{array}$$

commute. This group object is an abelian group object iff the following diagram:

$$\begin{array}{ccc}
 G \times G & \xrightarrow{T} & G \times G \\
 & \searrow m & \downarrow m \\
 & & G
 \end{array}$$

commutes, where T is the interchange map^a.

^aThis is constructed by the universal property of the product on the canonical projections.

Of course, the (abelian) group objects in Set are just (abelian) groups, because the diagrams above precisely encode the defining axioms of (abelian) groups. Now, we should also explain the notion of an action of a group object on another object in a category. Again, group actions of usual groups on sets are encoded by maps and equations, the latter of which are encoded by diagrams. So, we simply adapt those diagrams to a category.

Definition 3.72

Let $\mathcal{C}$ be a category with finite products and a final object 1 , let G be a group object in $\mathcal{C}$ and let $x \in \mathcal{C}$. A (right) action of G on x is a map $\nu : x \times G \rightarrow x$ such that the following diagram:

$$\begin{array}{ccccc}
 (x \times G) \times G & \simeq & x \times (G \times G) & \xrightarrow{\text{Id}_x \times m} & x \times G \xleftarrow{\text{Id}_x \times e} x \times 1 \\
 \downarrow \nu \times \text{Id}_G & & \downarrow \nu & & \downarrow \nu \\
 x \times G & \xrightarrow{\nu} & x & &
 \end{array}$$

commutes^a.

^aThe two subdiagrams, which do commute, reflect the axioms for group actions.

Now, there is a classical result of model category theory which says that if $\mathcal{M}$ is a model category and $x, y \in \mathcal{M}$, then $[y, \Omega(x)]$ has the structure of a group and $[y, \Omega^2(x)]$ has the structure of an abelian group. In that setting, this result is difficult to prove. For instance, it is essentially shown within [33, Corollary 6.1.6, p.151] and makes use of the full structure available on model categories. However, it seems that a significant simplification of that proof is available and it just uses Proposition 3.69 and a formal characterization of (abelian) group objects. Let us sketch the latter first.

Proposition 3.73

Let $\mathcal{C}$ be a locally small category with a final object and have finite products & let $x \in \mathcal{C}$. TFAE:

1. x has a group structure in $\mathcal{C}$.
2. The functor $\mathcal{C}(-, x)$ has a group structure in $\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$.

If either (1) or (2) holds, then x is an abelian group object iff $\mathcal{C}(-, x)$ is an abelian group object in $\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$.

Proof Sketch. This is explained in [47, Proposition 1, p.75-76]. However, the basic idea is that we can translate between morphisms in $\mathcal{C}$ and morphisms between hom-sets by using the Yoneda Lemma. For example, let $m : x \times x \rightarrow x$ be the multiplication morphism if x is a group object and let $y \in \mathcal{C}$. Then, we have the following composition:

$$\mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x),$$

where the first map is a natural isomorphism because representable functors preserve products while the second map is obtained by applying $\mathcal{C}(y, -)$ to the morphism m . So, the above forms a “multiplication” map for the hom-set $\mathcal{C}(y, x)$ and one checks that this multiplication actually satisfies the relevant properties. Going backwards, suppose

that $\mathcal{C}(y, x)$ is a group for each $y \in \mathcal{C}$. So, this has a multiplication map denoted as follows:

$$\bar{m} : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x).$$

We precompose this with the natural isomorphism $p : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \simeq \mathcal{C}(y, x \times x)$ and we get:

$$\bar{m} \circ p : \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x).$$

Due to the naturality of $\bar{m}$ and the naturality of p , we see that the above is a component of a natural transformation $\mathcal{C}(-, x \times x) \Rightarrow \mathcal{C}(-, x)$. Now, we apply the Yoneda Lemma and note that this natural transformation corresponds precisely to a morphism in $\mathcal{C}(x \times x, x)$ and this is our multiplication map. ■

Now, we can state and prove our result.

Theorem 3.74

Let $\mathcal{C}$ be a category of fibrant objects and let $x, y \in \mathcal{C}$.

1. $[y, \Omega(x)]$ is a group.
2. $[y, \Omega^2(x)]$ is an abelian group.

Proof. Observe that $\mathbf{Ho}(\mathcal{C})$ is a locally small category with a final object and admits finite products. Since $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$, we see that the functor $\mathbf{Ho}(\mathcal{C})(-, \Omega(x))$ is a group object in $\mathbf{Fun}(\mathbf{Ho}(\mathcal{C})^{\text{op}}, \mathbf{Set})$. So, $\mathbf{Ho}(\mathcal{C})(y, \Omega(x)) = [y, \Omega(x)]$ is a group, which proves statement (1). For statement (2), the argument is precisely analogous, except that we use the fact that $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$. ■

As is the case with homotopy theory on spaces, there is a long-exact homotopy sequence obtained from a fibration.

Proposition 3.75. Barratt-Puppe Fibration Sequence, [59, Proposition 1.3.4, p.45-46]

Let $\mathcal{C}$ be a category of fibrant objects and let $p : x \rightarrow y$ be a fibration with fibre F . There is an exact sequence:

$$\dots \rightarrow [z, \Omega^2(y)]_{\mathcal{C}} \rightarrow [z, \Omega(F)]_{\mathcal{C}} \rightarrow [z, \Omega(x)]_{\mathcal{C}} \rightarrow [z, \Omega(y)]_{\mathcal{C}} \rightarrow [z, F]_{\mathcal{C}} \rightarrow [z, x]_{\mathcal{C}} \rightarrow [z, y]_{\mathcal{C}},$$

for any $z \in \mathcal{C}$.

With the results above, we can construct the stable homotopy category of a category of fibrant objects. We note that this is the main category of interest in Theorem 3.77 and we take its construction from [72, Definition 1.23, p.5-6].

Construction 3.76. Let $\mathcal{C}$ be a category of fibrant objects. We define the stable homotopy category of $\mathcal{C}$:

$$\mathbf{SHo}(\mathcal{C}) \stackrel{\text{def}}{=} \mathbf{Ho}(\mathcal{C})[\Omega^{-1}],$$

obtained from $\mathbf{Ho}(\mathcal{C})$ by inverting the endofunctor Ω . To be precise:

1. The objects are pairs:

$$(x, n),$$

where $x \in \mathbf{Ho}(\mathcal{C})$ and $n \in \mathbb{Z}$.

2. If $(x, n), (y, m) \in \mathbf{SHo}(\mathcal{C})$, then we define:

$$\mathbf{SHo}(\mathcal{C})((x, n), (y, m)) \stackrel{\text{def}}{=} \text{colim}_{k \rightarrow \infty} [\Omega^{n+k}(x), \Omega^{m+k}(y)]_{\mathcal{C}},$$

where the colimit on the right-hand side is taken in the category of sets.

For $n \in \mathbb{Z}$, we have a natural functor:

$$\Omega^n : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{SHo}(\mathcal{C}),$$

which takes an object $x \in \mathbf{Ho}(\mathcal{C})$ and maps it to the pair (x, n) .

This leads us into the major theorem that we wanted to discuss.

Theorem 3.77. The Spanier-Whitehead Construction

Let $\mathcal{C}$ be a category of fibrant objects. Then, the stable homotopy category:

$$\mathbf{SHo}(\mathcal{C}),$$

is a triangulated category with shift:

$$\Sigma = \Omega^{-1} : \mathbf{SHo}(\mathcal{C}) \rightarrow \mathbf{SHo}(\mathcal{C}),$$

given by $(x, n) \mapsto (x, n - 1)$ and the distinguished triangles given by triangles isomorphic to triangles of the form:

$$(\Omega(y), n) \rightarrow (F, n) \rightarrow (x, n) \rightarrow (y, n),$$

where $n \in \mathbb{Z}$ and $x \rightarrow y$ a fibration, $F \rightarrow x$ a fibre inclusion & $\Omega(y) \rightarrow F$ is the morphism in Proposition 3.69.

For the rest of this subsection, we will be focused on understanding this variation of a rather classical, analogous result from stable homotopy theory. The proof is surprisingly quite tricky; let us speak about that in the next remark.

Remark 3.78. This theorem was stated in the form above in [72, Theorem 1.24, p.6] and is one of the main results of the paper. Nevertheless, it is not furnished with an explicit proof. The author seems to simply recourse to citing three different resources, with only one seeming to be remotely useful in recovering a full proof of this result; see [30, Section 9, p.42-43]. That paper, due to (Alex) Heller, does not seem to provide the correct formalism for proving the theorem above; we will give a brief argument for this later. One may expect that this result does hold for pointed

model categories and this is, indeed, the case, as is shown by Ivo Dell’Ambrogio in [24]. However, what he does is to show that the homotopy categories of pointed model categories satisfy a collection of 6 properties which guarantee that their Spanier-Whitehead Construction exists & that the theorem above holds.

This verification seems to use the explicit fact that the homotopy category of a pointed model category is equipped with a closed (right) action of the homotopy category of pointed simplicial sets (with the Joyal model structure), a fact that we were not able to establish for categories of fibrant objects. One might hope to deduce the theorem from the one for pointed model categories but the natural way to do this would be to find an “enveloping pointed model category” of our category of fibrant objects. Since this does not need to exist, that hope seems to not pan out.

The key construction that’s needed to translate our situation to that of Heller’s paper is the loop in a category of fibrant objects. Before we construct this, it is useful to introduce some terminology.

To prove the theorem above regarding the stable homotopy category, Heller proceeds via the notion of a *h-c-category*; this stands for a “homotopy-cofibration category” and it is meant to axiomatize the properties of homotopy. In our case, we will dualize those axioms and we will give a brief argument for why the axioms of a category of fibrant objects do not satisfy the axioms of Heller. Before we do this, we need the following result.

Proposition 3.79

Let $\mathcal{C}$ be a cofibration category. Then, $\mathcal{C}$ is a category with a system of cofibrations in the sense of Heller^a.

^aSee [30, p.34].

Proof. This follows literally by dualizing the axioms of a fibration category and comparing them to the axioms C0 and C1 as laid out in Heller’s paper cited above. In particular, one needs the fact that cofibrations are preserved by pushouts in a cofibration category but this follows by the axioms of Waldhausen. ■

With this, we can appropriately dualize the axioms of Heller.

Definition 3.80

Let $\mathcal{C}$ be a fibration category equipped with an equivalence relation:

$$\simeq,$$

on all morphism classes. We define:

$$\mathcal{C}_{\square} \stackrel{\text{def}}{=} \mathcal{C} / \simeq,$$

and set aside some terminology:

1. A morphism in $\mathcal{C}$ is a homotopy-equivalence iff its image in $\mathcal{C}_{\square}$ is an isomorphism.
2. If $x \in \mathcal{C}$, then x is contractible iff the morphism $x \rightarrow 0$ is a homotopy equivalence.
3. A morphism in $\mathcal{C}$ is a hom-fibration^a iff it is a fibration and a homotopy equivalence.

Now, suppose that $\simeq$ satisfied the following axioms:

1. If $x, y, z \in \mathcal{C}$ and $f, f' \in \mathcal{C}(z, x)$ and $g, g' \in \mathcal{C}(z, y)$ so that:

$$f \simeq f',$$

$$g \simeq g',$$

then the natural maps $(f, g) \simeq (f', g')$, where these morphisms are defined from^b $z \rightarrow x \times y$.

2. If we have a morphism (f', f, f'') in $\text{Fib}_{\mathcal{C}}$ and there is a morphism g'' in $\mathcal{C}$ such that $f'' \simeq g''$, then there exist morphisms g', g in $\mathcal{C}$ such that $g' \simeq f'$ and $g \simeq f$, with the triple (g', g, g'') being a morphism in $\text{Fib}_{\mathcal{C}}$.
3. Hom-fibrations are preserved by base extensions.
4. For every morphism f in $\mathcal{C}$, there is a factorization:

$$f = p \circ i,$$

where p is a fibration and i is a homotopy equivalence.

This is known as a h-f-category^c. The set of all homotopy equivalences is denoted by W_h .

^aThe reason for this name is that Heller uses the term “homotopy cofibration” for something else in his paper, so we avoid using the obvious analogue of that piece of terminology just to prevent confusion. Also, we do not use “trivial fibration” for obvious reasons.

^bThis makes sense because fibration categories admit finite products.

^cThat is, a “homotopy-fibration-category”.

In fact, we can quite easily show that h-f categories are weak fibration categories.

Proposition 3.81. A. Vats, 2026

Let $\mathcal{C}$ be a h-f category. Then, $\mathcal{C}$ is a category of fibrant objects where:

1. the weak equivalences are homotopy equivalences and;
2. the fibrations are just the given one on $\mathcal{C}$ as a h-f category.

In fact, only axioms (3) and (4) are needed to ensure we have the structure of a category of fibrant objects. Moreover,

the full subcategory W_h actually satisfies the 2-out-of-6 property.

Proof. Let's start by noting that the homotopy equivalences do form a full subcategory because functors map identity morphisms to identity morphisms and isomorphisms to isomorphisms. In particular, the canonical functor from $\mathcal{C}$ to $\mathcal{C}_\square$ does this so we do have full subcategories of $\text{Arr}(\mathcal{C})$ in our hands. Now, we will show that the axioms listed in Definition 3.38 are satisfied. We will do this by verifying them one-by-one.

1. By definition, a h-f category is a fibration category.
2. Again, isomorphisms are sent to isomorphisms by functors so an isomorphism is a homotopy equivalence.
3. In fact, the 2-out-of-3 property will follow if we show the 2-out-of-6 property; this is just statement (2) of Claim 3.2. On the other hand, the 2-out-of-6 property follows from Claim 3.4.
4. This follows because hom-fibrations are preserved by base extensions.
5. This follows from Proposition 3.44.

This proves the result. ■

In the next remark, we will explain why Heller's formalism seems to not be sufficient for proving our theorem.

Remark 3.82. The explanation ultimately boils down to the fact that not all categories of fibrant objects are h-f categories. Indeed, the right way to approach the above would be to use the internal notion of homotopy which is available in a category of fibrant objects and to show that we have a h-f category with respect to these homotopy equivalences & fibrations. However, it is not clear that we should expect the homotopy equivalences to behave well enough with respect to the fibrations so that all of the axioms above are satisfied. Indeed, it is not clear that the factorization axiom would be satisfied, for example. Even if we switch to the weak equivalences, it is not clear that the path space object would behave well with respect to products; this would be needed to verify Axiom (1) of Definition 3.80. Finally, Heller's paper does not even show that the Spanier-Whitehead construction applied to h-f categories gives triangulated categories; he shows that it gives a pre-triangulated category and establishing the octahedral axiom does not seem to be an easy task^a. It seems to not be the case that proceeding via h-f categories will give us the desired result, even if it might be desirable for us to obtain our construction in that way.

^aWe will define these terms in just a moment.

At this stage, we have discussed these categorical structures to an acceptable degree. Before we develop further theory about them, we really do need to cook up some examples and also address the earlier issue about "admissible" subcategories of $C^\star$ -algebras. This will be the topic for the next section.

4 Admissibility of Subcategories & The Spanier-Whitehead Construction

Summary

In this section, we finally address Question 2.11. In particular, we will use the work we have put into studying categories of fibrant objects & related structures to cook up a reasonable definition of “admissible subcategory”. We will see that it does answer the aforementioned question in the positive; this will be demonstrated by showing that Cuntz’s proof of the Bott Periodicity Theorem does go through for functors (with certain nice properties) defined on admissible subcategories.

Here are the (less common) notational conventions we’ll follow for this section.

Conventions

There are no special conventions for this section.

4.1 Admissible Subcategories & Applications Towards Axiomatic (Co)Homology

We will start with the following proposition which is known in the non-equivariant case for C^* -algebras.

Proposition 4.1. O. Uuye (2013), [72], A. Vats (2026)

Let $\mathcal{C}$ be either $G\text{-}C^*\text{Alg}$ or $G\text{-}BanAlg$ and define:

$$W \stackrel{\text{def}}{=} \{\text{homotopy equivalences in } \mathcal{C}\},$$

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Schochet fibrations in } \mathcal{C}\}.$$

Then, $(\mathcal{C}, W, \text{Fib})$ forms a category of fibrant objects.

Proof. We just need to verify the axioms one-by-one, in the order presented within Definition 3.38.

1. $G\text{-}C^*\text{Alg}$ has a final object; it's just $\{0\}$.
2. $(\mathcal{C}, \text{Fib})$ is a fibration category, as established in Example 3.28.
3. Every isomorphism in $\mathcal{C}$ is in W by Claim 2.6.
4. W satisfies the 2-out-of-3 property by statement (3) of Claim 2.8.
5. Trivial fibrations are preserved by base extensions, by Theorem 2.37.
6. Let $A \in G\text{-}C^*\text{Alg}$. We define:

$$A^I \stackrel{\text{def}}{=} \text{Cyl}_G(A).$$

Now, we define three maps:

$$d_0 : A^I \rightarrow A, \phi \mapsto d_0(\phi) \stackrel{\text{def}}{=} \phi(0),$$

$$d_1 : A^I \rightarrow A, \phi \mapsto d_1(\phi) \stackrel{\text{def}}{=} \phi(1)$$

For each $a \in A$, let $s(a) : [0, 1] \rightarrow A$ denote the constant function with value a . Define:

$$s : A \rightarrow A^I, a \mapsto s(a).$$

So, we have a quadruple (s, A^I, d_0, d_1) and now, we have to show that this forms a path space for A . It's clear that the composition $(d_0, d_1) \circ s$ gives the diagonal map, so we need to show that s is a homotopy equivalence and that (d_0, d_1) is a Schochet fibration. We will do this separately:

(a) We claim that d_0 is a homotopy inverse of s . Let's note that:

$$\forall \phi \in A^I : (s \circ d_0)(\phi) = s(d_0(\phi)) = s(\phi(0)),$$

$$\forall a \in A : (d_0 \circ s)(a) = d_0(s(a)) = s(a)(0) = a.$$

So, we see that $d_0 \circ s = \text{Id}_A$ and now, we need to construct a homotopy:

$$(h_t : A^I \rightarrow A^I)_{t \in [0, 1]},$$

as follows:

$$\forall t \in [0, 1] : \forall \phi \in A^I : \forall s \in [0, 1] : h_t^1(\phi)(s) \stackrel{\text{def}}{=} \phi(st)$$

It follows that s is a homotopy equivalence.

(b) We need to prove that (d_0, d_1) is a fibration. Consider the following lifting problem:

$$\begin{array}{ccc} & & A^I \\ & \nearrow f & \downarrow (d_0, d_1) \\ D & \xrightarrow{h_t} & A \times A \end{array}$$

where the diagram commutes when $t = 0$. We define:

$$\forall t \in [0, 1] : \forall x \in D : \forall s \in [0, 1] : H_t(x)(s) \stackrel{\text{def}}{=} (\pi_1 \circ h_t)(x),$$

where $\pi_1 : A \times A \rightarrow A$ is the canonical projection onto the first factor. This clearly gives a homotopy lift for $(h_t : D \rightarrow A \times A)_{t \in [0, 1]}$ so this shows that (d_0, d_1) is a fibration.

This shows that the quadruple forms a path space.

This proves the claim. ■

We have two remarks about this result.

Remark 4.2. It was established by Uuye, in [72, Theorem 2.19, p.10], that $C^*\text{Alg}$ with the structure above forms a category of fibrant objects. However, as far as we can tell, there are a few reasons as to why the approach used by him in the aforementioned reference does not directly generalize to the equivariant case:

1. The proof relies on purely topological characterizations of homotopy equivalences and Schochet fibrations in terms of weak homotopy equivalences and Serre fibrations between spaces. For the latter, a necessary component of the proof seems to be to use the fact that the maximal tensor product of C^* -algebras preserves pullbacks. This does not seem to be true for the projective tensor product of Banach algebras and there seems to be no easy fix for this defect in the setting of Banach algebras. One might consider using the injective tensor product of Banach algebras but it is not clear that the resulting tensor product is actually an algebra.

2. The proof does not account for equivariance. Indeed, it is not clear that these characterizations hold when one needs to account for equivariance. In fact, to work with the model structures available for categories of equivariant spaces, it is necessary that even stronger characterizations hold^a. It is not clear that this is the case.

Our proof, on the other hand, has a few distinct advantages:

1. It does not rely on these topological characterizations. Indeed, it is elementary.
2. It works for both equivariant C^* -algebras and Banach algebras.

Indeed, there seems to be an indication that this proof (or some variation of it) would also work for more sophisticated categories of equivariant topological algebras. Since such categories are becoming more important in modern work on K-theory, we plan to investigate these matters more systematically in future work.

^aSee [2].

Remark 4.3. In Proposition 4.1, if we fix $\mathcal{C}$ to be $G\text{-}C^*\text{Alg}$ and take the weak equivalences to be KK^G -equivalences^a & the fibrations to be Schochet fibrations, then we still do get a category of fibrant objects. Moreover, its homotopy category will be Kasparov's equivariant KK-category.

^aWe will discuss KK-theory in ??.

We can switch the fibrations in the example above to semisplit surjections.

Example 4.4. Consider $G\text{-}C^*\text{Alg}$ and define:

$$W \stackrel{\text{def}}{=} \{\text{homotopy equivalences in } G\text{-}C^*\text{Alg}\},$$

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Semisplit surjections in } G\text{-}C^*\text{Alg}\}.$$

Now, we have the following claim.

Claim. $(G\text{-}C^*\text{Alg}, W, F)$ is a category of fibrant objects.

This is proved in Bunke's paper (give the exact reference later).

Now, we can address the question of what is an admissible subcategory of $C^*\text{Alg}$. We will eventually want to do this for the equivariant case as well but it is better to start from the non-equivariant setting.

Definition 4.5

Let $\mathcal{C}$ be a full subcategory of $C^*\text{Alg}$. We say that $\mathcal{C}$ is admissible iff:

1. $\mathcal{C}$ is closed under the cylinder.
2. $\mathcal{C}$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for each $n \in \mathbb{N}$.
3. $\mathcal{C}$ is closed under sequential inductive limits.
4. $\mathcal{C}$ is closed under ideals, quotients and extensions.

Moreover, we require that $\mathcal{C}$ also contains 0 and $\mathbb{C}$.

In fact, we can show that any admissible subcategory of $C^*\text{Alg}$ will be closed under such constructions. Before we prove this, let us introduce another construction which will be relevant for later work in this subsection.

Construction 4.6. Let $A, B \in C^*\text{Alg}$ and let $f \in C^*\text{Alg}(A, B)$. We define:

$$\text{cone}(f),$$

to be a C^* -algebra that fits into the following pullback:

$$\begin{array}{ccc} \text{cone}(f) & \longrightarrow & \text{Cone}(B) \\ \downarrow & & \downarrow p_0^B \\ A & \xrightarrow{f} & B \end{array}$$

We refer to this as the mapping cone of f .

Now, we are led to the following result.

Proposition 4.7. A. Vats, 2026

Let $\mathcal{A}$ be an admissible subcategory of $C^*\text{Alg}$.

1. $\mathcal{A}$ is closed under pullbacks.
2. $\mathcal{A}$ is closed under the cone and suspensions.
3. $\mathcal{A}$ is closed under the mapping cone and mapping cylinder.
4. $\mathcal{C}(S^1) \in \mathcal{A}$.
5. $\mathcal{K} \in \mathcal{A}$.
6. $\mathcal{T} \in \mathcal{A}$; this is the Toeplitz C^* -algebra.

Moreover, $\mathcal{A}$ is a fibrant subcategory of C^*Alg .

Proof. Let's prove each statement one-by-one.

1. This is [55, Proposition 3.4, p.259]. It is only implied by Axiom 4 of Definition 4.5.
2. Let $A \in \mathcal{A}$. Then, we have the following pullback diagrams:

$$\begin{array}{ccc} \text{Cone}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{p_1^A} & A \end{array} \quad \begin{array}{ccc} \text{Sus}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{(p_0^A, p_1^A)} & A \oplus A \end{array}$$

Since these pullback diagrams live in $\mathcal{A}$, it follows that $\text{Cone}(A)$ and $\text{Sus}(A) \in \mathcal{A}$.

3. These are defined by pullback diagrams which are, themselves, in $\mathcal{A}$ so they must lie in $\mathcal{A}$.
4. We consider the following commutative diagram:

$$\begin{array}{ccc} \mathcal{C}(S^1) & \longrightarrow & \mathcal{C}([0, 1]) \\ \downarrow & & \downarrow \\ \mathbb{C} & \xrightarrow{\Delta_{\mathbb{C}}} & \mathbb{C} \oplus \mathbb{C} \end{array}$$

where $\Delta_{\mathbb{C}}$ is the diagonal map and the vertical map takes a function $f \in \mathcal{C}([0, 1])$ & maps it to $(f(0), f(1))$. Since $\mathbb{C} \in \mathcal{A}$ and since $\mathcal{A}$ is closed under cylinders, we know that $\mathcal{C}([0, 1]) = \text{Cyl}(\mathbb{C}) \in \mathcal{A}$. By closure under pullbacks, we see that $\mathbb{C} \oplus \mathbb{C} \in \mathcal{A}$. This implies, by closure under pullbacks, that $\mathcal{C}(S^1) \in \mathcal{A}$.

5. $\mathcal{K}$ can be written as an inductive limit of full matrix algebras and these belong to $\mathcal{A}$, so $\mathcal{K}$ belongs to $\mathcal{A}$.
6. The Topelitz C^* -algebra fits into the following exact sequence:

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{T} \rightarrow \mathcal{C}(S^1) \rightarrow 0,$$

and since $\mathcal{K}, \mathcal{C}(S^1)$ lie in $\mathcal{A}$, we see that $\mathcal{T} \in \mathcal{A}$.

Finally, the last statement holds because $\mathcal{A}$ is closed under pullbacks. ■

We have a remark about this result.

Remark 4.8. Technically, the pullback diagrams above regarding the homotopical constructions have at least one leg of the pullback cone being a Schochet fibration. So, we could require admissible subcategories of C^*Alg to be closed only under pullbacks by Schochet fibrations. There seems to be a characterization of this in [67, Remark 2.12, p.420-421] but it is left unproved and no further comments on it seem to have been made.

Moreover, we also have the following neat result.

Claim 4.1. Let $\mathcal{C}, \mathcal{D}$ be admissible subcategories of C^*Alg . Then, the full subcategory with objects:

$$\text{Obj}(\mathcal{C}) \cap \text{Obj}(\mathcal{D}),$$

is also admissible.

Of course, our definition also captures the usual examples.

Example 4.9. C^*Alg is certainly an admissible subcategory of C^*Alg .

Example 4.10. We have the following claim regarding ${}_{\omega}C^*Alg$.

Claim. ${}_{\omega}C^*Alg$ is an admissible subcategory.

We'll have to verify each of the required conditions one-by-one.

1. Clearly, 0 and $\mathbb{C}$ lie in ${}_{\omega}C^*Alg$.
2. ${}_{\omega}C^*Alg$ is closed under the cylinder because the tensor product of two separable C^* -algebras is separable^a.
3. ${}_{\omega}C^*Alg$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for the same reasons as in point^b (2).
4. Let $A \in C^*Alg$ be the inductive limit of a sequence of separable C^* -algebras. Then, A is the closure of the directed countable union of a sequence of separable C^* -subalgebras.
5. ${}_{\omega}C^*Alg$ is clearly closed under ideals, quotients and extensions.

This proves the claim.

^aOver here, we don't have to fuss over which tensor product is used since $\mathcal{C}[0, 1]$ is commutative and, therefore, nuclear.

^bAgain, no need to fuss over the tensor products used because $M_{n \times n}(\mathbb{C})$ is nuclear.

Example 4.11. We have the following claim regarding $\text{Nuc}C^*Alg$, the full subcategory of nuclear C^* -algebras.

Claim. $\text{Nuc}C^*Alg$ is an admissible subcategory of C^*Alg .

We'll just verify each property one-by-one.

1. 0 and $\mathbb{C}$ are nuclear C^* -algebras.
2. Let $A \in \text{Nuc}C^*\text{Alg}$. Then, its tensor product with any other nuclear C^* -algebra will be nuclear; see [14, Proposition 10.1.7, p.302]. So, we see that $\text{Cyl}(A) \in \text{Nuc}C^*\text{Alg}$.
3. Since $M_{2 \times 2}(A) \simeq M_{n \times n}(\mathbb{C}) \otimes A$ for any $A \in C^*\text{Alg}$, the result follows by the statement of the previous point.
4. Let's show that $\text{Nuc}C^*\text{Alg}$ is closed under ideals, quotients and extensions. This is the most difficult bit because it is a consequence of deep theorems. For closure under ideals, see [14, Exercise 2.3.4, p.36]. For closure under extensions, see [14, Exercise 3.8.1, p.105]. For closure under quotients, see [14, Corollary 9.4.4, p. 297].

This proves the claim.

Using **Claim 4.1** and **Example 4.10**, give us that the category of separable nuclear C^* -algebras is also admissible.

For the next result, we will need the following preparatory lemma.

Lemma 4.12 Let $A, B \in C^*\text{Alg}$ be σ -unital. Then, $A \otimes_{\max} B$ is σ -unital.

Proof. Let $(e_n)_{n \in \mathbb{N}}$ and $(f_n)_{n \in \mathbb{N}}$ be approximate units of A and B respectively. Then:

$$\begin{aligned}
 \forall n \in \mathbb{N} : \forall a \in A : \forall b \in B : \|(a \otimes b)(e_n \otimes f_n) - a \otimes b\| &= \|ae_n \otimes bf_n - a \otimes b\| \\
 &= \|ae_n \otimes bf_n - ae_n \otimes b + ae_n \otimes b - a \otimes b\| \\
 &= \|ae_n \otimes (bf_n - f) + (ae_n - a) \otimes b\| \\
 &\leq \|ae_n \otimes (bf_n - f)\| + \|(ae_n - a) \otimes b\| \\
 &= \|ae_n\| \cdot \|bf_n - f\| + \|ae_n - a\| \cdot \|b\| \\
 &\leq \|a\| \cdot \|bf_n - f\| + \|ae_n - a\| \cdot \|b\|.
 \end{aligned}$$

Taking the limit, we see that the sequence $(e_n \otimes f_n)_{n \in \mathbb{N}}$ acts as an approximate identity for the pure tensors. For the mixed tensors, we deduce this from the preceding fact and the triangle inequality for norms. Finally, we deduce it for arbitrary elements of $A \otimes_{\max} B$ by a standard $\frac{\epsilon}{3}$ -argument. ■

Now, we can present the example.

Example 4.13. We have the following claim regarding ${}^\sigma C^*\text{Alg}$, the full subcategory of σ -unital^a C^* -algebras.

Claim. ${}^\sigma C^*\text{Alg}$ is an admissible subcategory of $C^*\text{Alg}$.

We will just verify each property one-by-one.

1. 0 and $\mathbb{C}$ are clearly σ -unital.
2. Let $A \in {}^\sigma C^*\text{Alg}$. Then, $\text{Cyl}(A) \simeq \mathcal{C}([0, 1]) \otimes A$ is σ -unital since $\mathcal{C}([0, 1])$ is σ -unital.
3. Matrix stability follows in exactly the same way as above.
4. σ -unitality is extension-closed.

This proves the claim.

Note that ${}^\sigma C^*\text{Alg}$ contains ${}_\omega C^*\text{Alg}$.

^aThese are exactly the C^* -algebras which admit a sequential approximate identity.

In particular, we see that there are three other examples of categories of fibrant objects added to the list of such examples in the literature. At this stage, we are ready to move on to a more explicit set of applications of the ideas introduced above. In particular, we want to demonstrate that it is possible to do axiomatic (co)homology on fibrant subcategories of $C^*\text{Alg}$. This provides an enhancement of some of the basic results found in [67]. Since that paper is quite lengthy, our treatment will not be complete. Instead, we wish to simply convince the reader that this should be possible by showing that a number of basic results follow without trouble & that even the Bott Periodicity Theorem follows without any trouble. We will mainly focus on homology; since we are dealing with functors valued in abelian categories, these will simply encode the same basic results for cohomology theories by duality. Before we describe these so-called C^* -homology theories, let us set some notation for functors defined on admissible subcategories.

Notation 4.14. Let $\mathcal{C}$ be a category. Whenever we use the following notation for a functor:

$$F : C^*\text{Alg} \rightarrow \mathcal{C},$$

we mean that F is defined on some admissible subcategory of $C^*\text{Alg}$.

Now, we shall give the definition of an equivariant C^* -homology theory.

Definition 4.15

Let $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors^a. It is a C^* -homology theory iff:

1. Each h_n is a homotopy-invariant functor.

2. For any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $\text{dom}(h_n)$, there exists a sequence $(\partial : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{N}}$ of morphisms and a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

such that ∂ is natural with respect to morphisms of exact sequences.

The maps ∂ are known as boundary homomorphisms.

^aThe domain of each is the same, so it does not vary with n !

We have a remark about this definition.

Remark 4.16. Let's be more precise about what it means for ∂ to be natural with respect to morphisms of exact sequences. Indeed, suppose we have the following commutative diagram in $C^*\text{Alg}$ with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

then, the resulting squares:

$$\begin{array}{ccc} h_n(C) & \longrightarrow & h_{n-1}(A) \\ \downarrow & & \downarrow \\ h_n(C') & \longrightarrow & h_{n-1}(A') \end{array}$$

commute for every $n \in \mathbb{Z}$.

Sometimes, one assumes that a homology theory is also additive; this means that it should respect direct sums of equivariant C^* -algebras. In order to introduce this bit of terminology, we will carry out a construction.

Construction 4.17. Let $(A_i)_{i \in I}$ be a family in an admissible subcategory of $C^*\text{Alg}$. Then, we know that:

$$A = \bigoplus_{i \in I} A_i \in C^*\text{Alg},$$

simply by defining the action to be induced on each component. Now, we have inclusions^a:

$$\iota_j : A_j \rightarrow A.$$

If $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ is a sequence of functors^b, then we get maps:

$$h_n(\iota_j) : h_n(A_j) \rightarrow h_n(A).$$

If $\mathcal{A}$ is closed under arbitrary coproducts, then the universal property of the coproduct in $\mathcal{A}$ gives a morphism:

$$\bigoplus_{i \in I} h_n(A_i) \rightarrow h_n(A),$$

for each $n \in \mathbb{Z}$.

^aOver here, we implicitly will just assume that this belongs in our admissible subcategory. Note that we are not assuming that our admissible subcategory to be closed under arbitrary products. In the case of finite products, there is nothing to fret over.

^bThe domains are all the same.

This construction leads us to the following bit of terminology.

Terminology 4.18. Let $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors^a and let $(A_i)_{i \in I}$ be a family in $C^*\text{Alg}$. In the following, we study the morphism in **Construction 4.17** for different cardinalities of I .

1. The given sequence is finitely additive iff I is finite and our morphism is an isomorphism for each $n \in \mathbb{Z}$.
2. The given sequence is σ -additive iff I is countable and our morphism is an isomorphism for each $n \in \mathbb{Z}$.
3. The given sequence is additive iff I is arbitrary and our morphism is an isomorphism for each $n \in \mathbb{Z}$.

An additive sequence of functors is also known as a completely additive sequence of functors.

^aThe domains are all the same.

Here are some basic properties that we can already derive.

Proposition 4.19

Let $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory.

1. The boundary homomorphisms obtained from a split exact sequence are zero.
2. Each h_n is split-exact.

3. Each h_n preserves finite coproducts.
4. $h_n(0) = 0$.
5. If $n \in \mathbb{Z}$, $A \in C^*Alg$ and $h_n(\mathbb{C}) \simeq 0$, then $h_n(\iota)$ is an isomorphism^a.

^a $\iota : A \rightarrow A^\sim$ is the canonical inclusion.

Proof. We'll prove each statement one-by-one.

1. Consider a split-exact sequence in C^*Alg .

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

This implies that $\psi \circ \alpha = \text{Id}_C$. Now, let $n \in \mathbb{Z}$ be fixed. By functoriality, we conclude that $h_n(\psi)$ is surjective. But this means that its range is just $h_n(C)$. On the other hand, the split-exact sequence above gives us a long-exact sequence with boundary homomorphisms $(\partial_n : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{Z}}$. Since ∂_n connects to $h_n(\psi)$ and the long-exact sequence is exact at $h_n(C)$, it follows that $\ker(\partial_n) = h_n(C)$. That is, $\partial_n = 0$.

2. Consider a split-exact sequence in C^*Alg .

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

This induces a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

By (1), we know that $\partial = 0$ and we know that $h_n(\psi)$ is a split epimorphism by functoriality. So, the homology theory takes the split-exact sequence above to a split-exact one.

3. Let $A, B \in C^*Alg$ and let us be given the following maps:

$$\begin{aligned} \pi_A : A \oplus B &\rightarrow A, (a, b) \mapsto \pi_A(a, b) \stackrel{\text{def}}{=} a, \\ \pi_B : A \oplus B &\rightarrow B, (a, b) \mapsto \pi_B(a, b) \stackrel{\text{def}}{=} b, \\ \iota_A : A &\rightarrow A \oplus B, a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0), \\ \iota_B : B &\rightarrow A \oplus B, b \mapsto \iota_B(b) \stackrel{\text{def}}{=} (0, b). \end{aligned}$$

With these maps, we have two split-exact sequences:

$$0 \longrightarrow A \xrightarrow{\iota_A} A \oplus B \xrightarrow{\pi_B} B \longrightarrow 0$$

$$0 \longrightarrow B \xrightarrow{\iota_B} A \oplus B \xrightarrow{\pi_A} A \longrightarrow 0$$

Both of these induce split-exact sequences in $\mathcal{A}$ and this implies the result by the splitting lemma.

4. By statement (3), we see that $h_n(0 \oplus 0) \simeq h_n(0) \oplus h_n(0) \simeq h_n(0)$ so we're done.
5. Consider the split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\iota} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

associated to the unitization. For each $n \in \mathbb{Z}$, we obtain a sequence:

$$0 \longrightarrow h_n(A) \xrightarrow{h_n(\iota)} h_n(A^\sim) \xrightarrow{h_n(p)} h_n(\mathbb{C}) \longrightarrow 0$$

which is split-exact. But note that $h_n(\mathbb{C}) \simeq 0$ so we see that $h_n(\iota)$ is injective & surjective ■

Our next goal is to establish that homology theories are “essentially” formed by just one functor and the suspension. In other words, if we have h_n , then we can calculate it using h_{n-1} . Before we state this precisely, here's a construction.

Construction 4.20. Let $(h_n)_{n \in \mathbb{Z}}$ be a C^* -homology theory, let $A \in C^*Alg$ & consider the sequence:

$$0 \longrightarrow \text{Sus}(A) \longrightarrow \text{Cone}(A) \longrightarrow A \longrightarrow 0$$

as introduced in [Construction 2.10](#). This gives us the following data:

1. A sequence $(\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)))_{n \in \mathbb{Z}}$ of morphisms and;
2. A long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

such that the boundary homomorphisms are natural with respect to morphisms of exact sequences.

We refer to each σ_A as a suspension map.

With this construction at hand, we have the following key result about the structure of C^* -homology theories.

Proposition 4.21

Let $(h_n : C^*Alg \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory & let $A \in \text{dom}(h_0)$.

1. If $B \in \text{dom}(h_0)$ and $f \in C^*Alg(A, B)$ so that $f = 0$, then $h_n(f) = 0$ for each $n \in \mathbb{Z}$.
2. $\forall n \in \mathbb{Z} : h_n(\text{Cone}(A)) = 0$.

In particular, the suspension map σ_A is an isomorphism.

Proof. We'll prove each of these statements one-by-one.

1. Observe that we can factorize f as follows:

$$A \rightarrow 0 \rightarrow B.$$

Applying each h_n to f , we get that:

$$h_n(f) = h_n(0 \rightarrow B) \circ h_n(A \rightarrow 0),$$

and since $h_n(0) = 0$, it follows that $h_n(f) = 0$.

2. $\text{Cone}(A)$ is contractible by **Claim 2.7**, so $\text{Id}_{\text{Cone}(A)}$ is homotopic to 0. This means that:

$$\begin{aligned} 0 &= h_n(0) \\ &= h_n(\text{Id}_{\text{Cone}(A)}) \\ &= \text{Id}_{h_n(\text{Cone}(A))}, \end{aligned}$$

and this only happens when $h_n(\text{Cone}(A)) = 0$.

Now, let $n \in \mathbb{N}$ and consider the suspension map:

$$\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)),$$

within the following long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

Now, we'll show that σ_A is an isomorphism by showing that it is bijective.

1. We see that $h_n(\text{Cone}(A)) = 0$, so σ_A is injective since the sequence above is exact.
2. We see that $h_{n-1}(\text{Cone}(A)) = 0$, so σ_A is surjective since the sequence above is exact.

It follows that σ_A is bijective so we're done. ■

So, the result of Proposition 4.21 motivates the following bit of notation.

Notation 4.22. Let $F : C^*Alg \rightarrow \mathcal{A}$ be a functor and define:

$$\forall n \in \mathbb{N}_0 : F_n \stackrel{\text{def}}{=} F \circ \text{Sus}^n.$$

This is known as the n -th iterant of F . We refer to this as the suspension spectrum of F .

We have a remark about Notation 4.22.

Remark 4.23. For this to make sense, we need to ensure that Sus lands inside the domain of F . This is easy to ensure by restricting Sus to the domain of F , since admissible subcategories are closed under suspension^a.

^aSee statement (2) of Proposition 4.7.

We should, now, address the following natural question.

Question 4.24. Let $F : C^*Alg \rightarrow \mathcal{A}$ be a functor. When is its suspension spectrum a C^* -homology theory?

We will address this question through the proof of Proposition 4.26. That result was known for functors defined on C^*Alg and we extend to functors defined on admissible subcategories. To prove it, we will need a technical lemma.

Lemma 4.25. [14, Corollary 3.7.4, p.93] Consider the following exact sequence of C^* -algebras:

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

and let $B \in C^*Alg$. If A/I is nuclear or B is nuclear, then the tensored sequence:

$$0 \longrightarrow I \otimes_{\star} B \longrightarrow A \otimes_{\star} B \longrightarrow (A/I) \otimes_{\star} B \longrightarrow 0$$

where $\star \in \{\min, \max\}$, is exact.

We are ready to discuss the desired proposition.

Proposition 4.26. The Barratt-Puppe Sequence

Let $F : C^*Alg \rightarrow \mathcal{A}$ be homotopy-invariant & half-exact. Its suspension spectrum is a C^* -homology theory.

For functors defined on all of C^*Alg , this result can be found in [63, Proposition 8.4, p.34-35] with a reasonably detailed proof. We provide more details and explicitly show that this result can be adapted to our requirements. As it turns out, we have to take care of many more details but we will ensure that the proof remains conceptual and readable.

Proof of Proposition 4.26. Suppose we have our extension of C^* -algebras in $\text{dom}(F)$:

$$0 \longrightarrow I \xrightarrow{\iota_1} A \xrightarrow{\pi_1} A/I \longrightarrow 0$$

The proof is predicated around the following claim.

Claim. Take the given extension and suppose that we can construct an extension:

$$0 \longrightarrow \text{Sus}(A/I) \xrightarrow{\iota'_2} \text{cone}(\pi_1) \xrightarrow{\pi_2} A \longrightarrow 0$$

with the following properties:

1. There exists a $\star$ -homomorphism $\alpha_1 : I \rightarrow \text{cone}(\pi_1)$ which induces an isomorphism on F_n for all $n \in \mathbb{N}$.
2. The composite $\pi_2 \circ \alpha_1$ induces the same map on F_n as the inclusion ι for each $n \in \mathbb{N}$.

Then, the given long-exact sequence exists.

Let's start by noting that this claim makes sense in $\text{dom}(F)$. Indeed, since $A/I \in \text{dom}(F)$, we see that $\text{Sus}(A/I) \in \text{dom}(F)$. So, if the extension in the claim exists, then $\text{cone}(\pi_1) \in \text{dom}(F)$ since $\text{dom}(F)$ is extension-closed. So, the properties also make sense. At this point, the proof splits into two parts. In the first part, we'll check that we actually obtain the given long-exact sequence provided that the claim holds. In the second part, we'll prove the claim.

1. We obtain a sequence $(G_n)_{n \in \mathbb{N}}$ of exact sequences by the following specification:

$$G_1 : \quad 0 \longrightarrow I \xrightarrow{\iota_1} A \xrightarrow{\pi_1} A/I \longrightarrow 0$$

$$G_2 : \quad 0 \longrightarrow \text{Sus}(A/I) \xrightarrow{\iota_2} \text{cone}(\pi_1) \xrightarrow{\pi_2} A \longrightarrow 0$$

$$G_3 : \quad 0 \longrightarrow \text{Sus}(A) \xrightarrow{\iota_3} \text{cone}(\pi_2) \xrightarrow{\pi_3} \text{cone}(\pi_1) \longrightarrow 0$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

That is, G_1 is the extension we started with and for $n \in \mathbb{N}$, we have that the extension G_{n+1} is obtained from the extension G_n by the mapping cone construction in the claim. Now, we apply F to each of the sequences above and utilize the fact that F is a half-exact functor to obtain the following exact sequences:

$$F(G_1) : \quad F(I) \xrightarrow{F(\iota_1)} F(A) \xrightarrow{F(\pi_1)} F(A/I)$$

$$F(G_2) : \quad F(\text{Sus}(A/I)) \xrightarrow{F(\iota_2)} F(\text{cone}(\pi_1)) \xrightarrow{F(\pi_2)} F(A)$$

$$F(G_3) : \quad F(\text{Sus}(A)) \xrightarrow{F(\iota_3)} F(\text{cone}(\pi_2)) \xrightarrow{F(\pi_3)} F(\text{cone}(\pi_1))$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

Now, the idea is to just do the following “replacements”:

$$F(\text{cone}(\pi_1)) \rightsquigarrow F(I)$$

$$F(\text{cone}(\pi_2)) \rightsquigarrow F(\text{Sus}(A/I)) = F_1(A/I)$$

and glue the resulting sequences once we've done the replacements. This will actually yield the first part of the given long-exact sequence and continuing the process as we have done above will yield the rest of it. So, let's illustrate why we can carry out the replacements above. Indeed, we obtain a commutative diagram:

$$\begin{array}{ccccccc} F(\text{Sus}(A/I)) & \xrightarrow{F(\iota_2)} & F(\text{cone}(\pi_1)) & \xrightarrow{F(\pi_2)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \\ \parallel & & \downarrow F(\alpha_1)^{-1} & & \parallel & & \parallel \\ F(\text{Sus}(A/I)) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\iota_2)} & F(I) & \xrightarrow{F(\iota_1)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \end{array}$$

where the sequence on top is exact, so the sequence on the bottom is also exact. Now, we need to play a similar

game with $F(G_3)$. Indeed, we have the following commutative diagram:

$$\begin{array}{ccccc}
F(\text{Sus}(A)) & \xrightarrow{F(\iota_3)} & F(\text{cone}(\pi_2)) & \xrightarrow{F(\pi_3)} & F(\text{cone}(\pi_1)) \\
\parallel & & \downarrow F(\alpha_2)^{-1} & & \downarrow F(\alpha_1)^{-1} \\
F_1(A) & \xrightarrow{F(\alpha_2)^{-1} \circ F(\iota_3)} & F_1(A/I) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2)} & F(I)
\end{array}$$

Well, the top sequence is just $F(G_3)$ and this is exact, so the sequence on the bottom is exact. On the other hand, we can only sew this to the sequence we obtained from $F(G_1)$ and $F(G_2)$ if both morphisms from $F_1(A/I)$ to $F(I)$ coincide. Well, them coinciding means precisely that the following equality is satisfied:

$$F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2) = F(\alpha_1)^{-1} \circ F(\iota_2),$$

which is equivalent to the statement that:

$$F(\pi_3) \circ F(\alpha_2) = F(\iota_2).$$

By functoriality, the left-hand side is just $F(\pi_3 \circ \alpha_2)$ and this translates to property (2) of the claim above. So, we can sew this short exact sequence to the one obtained from $F(G_1)$ and $F(G_2)$.

2. For this, we use the explicit form of the mapping cone defined in [Construction 4.6](#). We have that:

$$\text{cone}(\pi_1) = \{(f, a) \in \text{Cone}(A/I) \times A : f(0) = 0 \wedge f(1) = \pi_1(a)\}.$$

So, we define the three maps as follows:

$$\iota_2 : \text{Sus}(A/I) \rightarrow \text{cone}(\pi_1), \quad f \mapsto \iota_2(f) \stackrel{\text{def}}{=} (f, 0),$$

$$\pi_2 : \text{cone}(\pi_1) \rightarrow A, \quad (f, a) \mapsto \pi_2(f, a) \stackrel{\text{def}}{=} a,$$

$$\alpha_1 : I \rightarrow \text{cone}(\pi_1), \quad i \mapsto \alpha_1(i) \stackrel{\text{def}}{=} (0, i).$$

The fact that $\pi_2 \circ \alpha_1$ induces the same map as the inclusion is clear. Now, we also have an extension:

$$0 \longrightarrow I \xrightarrow{\alpha_1} \text{cone}(\pi_1) \longrightarrow \text{Cone}(A/I) \longrightarrow 0$$

with the second map being the projection to $\text{Cone}(A/I)$. By [Claim 2.7](#), we know that $\text{Cone}(A/I)$ is contractible. This means that $F(\alpha_1)$ is surjective. To show injectivity, we will replace I with a C^* -algebra Q so that I and Q are homotopy-equivalent and Q fits into an exact sequence with $\text{Cone}(I)$ on the left. So, we define:

$$Q \stackrel{\text{def}}{=} \{f \in \text{Cone}(A) : f(0) \in I\}.$$

We have the following claim.

| **Claim 4.2.** Q and I are homotopy equivalent. This is proved on Page 101.

Let us consider the following exact sequence:

$$0 \longrightarrow \text{Cone}(I) \longrightarrow Q \longrightarrow \text{cone}(\pi_1) \longrightarrow 0$$

where the first map is just an inclusion of functions and the second map is defined as:

$$k : f \mapsto (\pi_1 \circ f, f(1)).$$

The fact that these are $\star$ -homomorphisms will follow by an explicit computation and it also uses the fact that π_1 is a $\star$ -homomorphism. Since $\text{Cone}(I)$ is contractible, we can conclude that $F(k)$ is injective. Let us establish a formula which connects the map k to α_1 . If we define, as in the proof of [Claim 4.2](#), the following map:

$$\psi : I \rightarrow Q, \quad j \mapsto \psi(j),$$

where $\psi(j)$ is the constant function j on $[0, 1]$. Composing this with k , we see that:

$$\begin{aligned}
\forall j \in I : (k \circ \psi)(i) &= k(\psi(i)) \\
&= (\pi_1 \circ \psi(i), \psi(i)(1)) \\
&= (0, i) \\
&= \alpha_1(i).
\end{aligned}$$

So, we conclude that $k \circ \psi = \alpha_1$ and this means that:

$$F(\alpha_1) = F(k \circ \psi) = F(k) \circ F(\psi).$$

Since $F(k)$ is injective and $F(\psi)$ is an isomorphism, we conclude that $F(\alpha_1)$ is injective and we're done.

This completes the proof. ■

Having formulated these basic results, we now come to the main test result; the Bott Periodicity Theorem. Indeed, our goal is to present a unified version of Cuntz's proof of the Bott Periodicity Theorem using the language above.

Theorem 4.27. The Bott Periodicity Theorem

Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a homotopy-invariant, half-exact and C^* -stable functor^a. Then, $F \simeq F \circ \text{Sus}^2$.

^aFrom now on, Notation 4.22 is in effect.

We have a remark about this result.

Remark 4.28. This theorem was established by Cuntz in [7, Volume 1046, Theorem 4.4, p.63] for functors satisfying the listed properties and defined on all of $C^*\text{Alg}$. A more extensive proof with details is provided also in (Nigel) Higson's PhD thesis [31, Theorem 2.1.14, p.28]. One issue with all of these proofs in the literature is that it was originally established for functors defined on all of $C^*\text{Alg}$. However, this result is typically cited when working with other homology theories, like KK-theory or E-theory or others, to conclude that they also satisfy Bott Periodicity. These other homology theories are usually defined on full subcategories of $C^*\text{Alg}$. Indeed, the selection of subcategories of Question 2.11 was chosen by Schochet with precisely this in mind. So, it would be desirable to have a unified proof of the Bott Periodicity Theorem which deals with all of these cases at once and this was our motivation for Theorem 4.27. The main contribution here is to go through Cuntz's proof & keep track of the auxiliary algebras introduced along the way, making sure that they are always contained in the domain of the functor F . For this reason, we will present a complete proof from scratch which accounts for the issue above.

Let us start working towards Cuntz's proof of this theorem. A major step in that construction is to work with the so-called *reduced* Toeplitz extension. In the next construction, we will study both. The main content of the construction lies in showing that the constructed C^* -algebras are in admissible subcategories of $C^*\text{Alg}$.

Construction 4.29. Let us consider the standard Toeplitz extension as follows:

$$0 \longrightarrow \mathcal{K} \xrightarrow{i_1} \mathcal{T} \xrightarrow{p_1} \mathcal{C}(S^1) \longrightarrow 0$$

For a detailed discussion of this extension and the proof of its existence, we strongly recommend the treatment of [40, Chapter 6.8, p.76-79]. We will briefly recall the essential points for your convenience. The key is to view $\mathcal{T}$ as the universal C^* -algebra generated by an isometry and $\mathcal{C}(S^1)$ as the universal C^* -algebra generated by a unitary. In that case, we can see that the map p_1 is induced by mapping our isometry of $\mathcal{T}$ (the generator) to the unitary of $\mathcal{C}(S^1)$ (the generator). If $v \in \mathcal{T}$ is the generator, then it will become the case that:

$$\mathcal{K} \simeq \langle 1 - vv^* \rangle,$$

with the right-hand side being an ideal in $\mathcal{T}$. The “reason” for this isomorphism is explained in the aforementioned notes very well, so we will not dwell on it. Now, from the Toeplitz extension, we can construct the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

where ev_1 is the evaluation map at $1 \in S^1$. Now, we put in $\ker(\varepsilon)$ into the diagram:

$$\begin{array}{c} 0 \\ \downarrow \\ \mathcal{T}_0 \stackrel{\text{def}}{=} \ker(\varepsilon) \\ \downarrow \\ \begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array} \end{array}$$

Now, we have the following claim.

| **Claim.** A copy of $\mathcal{K}$ is contained in $\mathcal{T}_0$, which we will call the reduced Toeplitz C^* -algebra. This is nuclear.

Let $T \in \mathcal{K}$. Then, we need to show that $i_1(T) \in \mathcal{T}_0 = \ker(\varepsilon)$. For this to work, we need to show that:

$$\varepsilon(i_1(T)) = 0.$$

But now, we see that:

$$\text{ev}_1(p_1(i_1(T))) = 0,$$

which proves the result. Finally, nuclearity follows from the fact that it sits between two C^* -algebras which are known to be nuclear; $\mathcal{K}$ is an AF algebra so it is nuclear^a and $\mathcal{C}_0(S^1 \setminus \{0\})$ is commutative so it is nuclear^b.

In light of this result, we can place the reduced Toeplitz C^* -algebra in an extension as follows:

$$\begin{array}{ccccccc}
& & & 0 & & & \\
& & & \downarrow & & & \\
0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_2} & \mathcal{T}_0 & \xrightarrow{p_2} & \mathcal{C}_0(S^1 \setminus \{1\}) \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\
& & & & \downarrow \varepsilon & & \downarrow \text{ev}_1 \\
& & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C}
\end{array}$$

where the morphisms are chosen so that the given squares commute. The extension obtained involving $\mathcal{T}_0$ is known as the reduced Toeplitz extension. There are three facts worth noting:

1. i_2 is just i_1 .
2. p_2 is just p_1 restricted to $\mathcal{T}_0$.
3. $\mathcal{C}(S^1 \setminus \{1\})$ is just $\mathcal{C}_0((0, 1))$.

Now, the key point lies in the following results.

Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$. Then, every C^* -algebra in the diagram above lies in $\mathcal{C}$.

We just need to check this for $\mathcal{T}_0$ and $\mathcal{C}(S^1 \setminus \{1\})$, since every other C^* -algebra above has been shown to lie in admissible subcategories. We note that $\mathcal{C}(S^1 \setminus \{1\}) \simeq \mathcal{C}_0(\mathbb{R}) = \text{Sus}(\mathbb{C})$ so this certainly lies in $\mathcal{C}$. Since $\mathcal{T}_0$ fits into the middle of an exact sequence where the other algebras belong to $\mathcal{C}$, we see that $\mathcal{T}_0 \in \mathcal{C}$.

We also need the following result.

Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$, let $A \in \mathcal{C}$ and suppose that B is any of the C^* -algebras appearing in the diagram above. Then, $A \otimes B \in \mathcal{C}$; every B above is nuclear so the notation is clear.

We need to check this result for all of the algebras individually.

1. Let $B = \mathbb{C}$. Then, $A \otimes \mathbb{C} \simeq A$ so $A \otimes B \in \mathcal{C}$.
2. Let $B = \mathcal{K}$. We know that $M_{n \times n}(A) \in \mathcal{C}$ for each $n \in \mathbb{N}$. Now, we can write $A \otimes \mathcal{K}$ as an inductive limit of full matrix algebras over A and since $\mathcal{C}$ is closed under inductive limits, it follows that $A \otimes \mathcal{K} \in \mathcal{C}$.
3. Let $B = \mathcal{C}(S^1)$. Then, $A \otimes B \simeq \mathcal{C}(S^1, A)$ and we can fit this into a pullback diagram:

$$\begin{array}{ccc}
\mathcal{C}(S^1, A) & \longrightarrow & \text{Cyl}(A) \\
\downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\
A & \xrightarrow{\Delta_A} & A \oplus A
\end{array}$$

Now, $A \oplus A \in \mathcal{C}$ and $\text{Cyl}(A) \in \mathcal{C}$, so we conclude that $\mathcal{C}(S^1, A) \in \mathcal{C}$.

4. Let $B = \mathcal{T}$. Let us tensor the Toeplitz exact sequence on the right by A and note that by [Lemma 4.25](#), the resulting sequence is exact since $\mathcal{C}(S^1)$ is nuclear. Now, we know that $\mathcal{K} \otimes A \in \mathcal{C}$ and $\mathcal{C}(S^1, A) \in \mathcal{C}$. So, we use the extension property of nuclearity to conclude that $\mathcal{T} \otimes A \in \mathcal{C}$.
5. Let $B = \mathcal{C}_0(S^1 \setminus \{1\})$. This is just $\text{Sus}(\mathbb{C})$, so tensoring it with A produces $\text{Sus}(A)$ and this lies in $\mathcal{C}$.
6. Let $B = \mathcal{T}_0$. The same argument as we used for $\mathcal{T}$ will work.

This proves the claim.

This completes our desired construction.

^aSee [\[14, Proposition 2.4.1, p. 39, Theorem 3.8.7, p.104\]](#).

^bSee [\[14, Proposition 2.4.2, p. 38-39, Theorem 3.8.7, p.104\]](#).

Now, the next proposition will yield the Bott Periodicity Theorem.

Proposition 4.30

Suppose we are under the assumptions of [Theorem 4.27](#) and let $A \in \text{dom}(F)$. If $\iota : \mathbb{C} \rightarrow \mathcal{T}$ is the inclusion of the identity, then $\iota \otimes \text{Id}_A$ and $\varepsilon \otimes \text{Id}_A$ induce mutually inverse isomorphisms under F .

Let us use this to prove the Bott Periodicity Theorem.

Proof of Theorem 4.27. We tensor the reduced Toeplitz extension by A so we get an extension:

$$0 \longrightarrow \mathcal{K} \otimes A \longrightarrow \mathcal{T}_0 \otimes A \xrightarrow{p_2 \otimes \text{Id}_A} \text{Sus}(A) \longrightarrow 0$$

Applying the Barratt-Puppe sequence, we obtain a long-exact sequence:

$$\dots \longrightarrow F_1(\text{Sus}(A)) \longrightarrow F_0(\mathcal{K} \otimes A) \longrightarrow F_0(\mathcal{T}_0 \otimes A) \longrightarrow F_0(\text{Sus}(A))$$

where F_n is defined by Notation 4.22 for each $n \in \mathbb{N}_0$. Now, we have the boundary map:

$$\partial : F_1(\text{Sus}(A)) \rightarrow F_0(\mathcal{K} \otimes A)$$

squeezed in-between $F_0(\mathcal{T}_0 \otimes A)$ and $F_1(\mathcal{T}_0 \otimes A)$. On the other hand, we also notice that:

$$\begin{aligned} F_1(\text{Sus}(A)) &= F_0(\text{Sus}^2(A)) = F_2(A), \\ F_0(\mathcal{K} \otimes A) &\simeq F_0(A), \end{aligned}$$

So, ∂ would give rise to an isomorphism between $F_2(A)$ and $F_0(A)$ *provided* that:

$$F_0(\mathcal{T}_0 \otimes A) = F_1(\mathcal{T}_0 \otimes A) = 0.$$

In fact, we can prove a lot more, as the next claim shows.

| Claim. $\forall n \in \mathbb{N} : F_n(\mathcal{T}_0 \otimes A) = 0$.

Since taking the iterated suspension is equivalent to iterated tensoring, it suffices to show that:

$$F_0(\mathcal{T}_0 \otimes A) = F(\mathcal{T}_0 \otimes A) = 0.$$

Well, the key is that $\sigma \otimes \text{Id}_A$ vanishes only on $\mathcal{T}_0 \otimes A$ and since $F(\sigma \otimes \text{Id}_A)$ is an isomorphism by the strength of ??, the claim follows. So, we have an isomorphism from $F_2(A) \rightarrow F_0(A)$. In fact, this specific isomorphism is a natural isomorphism of the functors F_2 and F_0 precisely because boundary maps are required to be natural with respect to the morphisms of exact sequences. This completes the proof. $\blacksquare$

Through our work above, we hope that we have demonstrated that many results involving axiomatic (co)homology on C^* -algebras can be deduced formally and they can also be understood on various interesting subcategories using the notion of fibrant subcategory. Our last order of business in the main body of this thesis is to address the Spanier-Whitehead construction. In the next subsection, we will provide a proof of that result using machinery from higher category theory.

4.2 Towards The Spanier-Whitehead Construction

Let us outline the main strategy towards proving Theorem 3.77.

Remark 4.31. Let $\mathcal{C}$ be a finitely bicomplete category of fibrant objects. Then:

1. We will take the ∞ -categorical localization of $\mathcal{C}$ at its weak equivalences; call this $\mathcal{D}$.
2. We will show that $\mathcal{D}$ is a pointed ∞ -category which is finitely bicomplete.
3. We will define the Spanier-Whitehead category:

$$\mathbf{SW}(\mathcal{D}),$$

which is supposed to be the ∞ -categorical analogue of the same construction as the stable homotopy category.

4. We will show that $\mathbf{SW}(\mathcal{D})$ is a stable ∞ -category.
5. Since the homotopy category of a stable ∞ -category is a triangulated category, the theorem will follow quickly.

Our main encyclopaedic references for higher category theory are [45], [44], [46] and [17] & we will usually refer to these resources for specific results with full proofs (though we will sometimes reference the more educational resources we'll mention in just a moment if they are clearer). Since we will need a few facts about stable ∞ -categories, we will also find it reasonable to reference [43] since it is much shorter than the aforementioned references. For a very pleasant introduction to ∞ -categories from the viewpoint of quasicategories, see [61]. For actually learning this material in a synthetic and model-independent way, we recommend the (in-progress) book project [18].

Of course, before we proceed towards the explicit realization of this strategy, we need to discuss triangulated categories to some degree. In fact, all we will need is the definition of a triangulated category for this to form a complete picture. However, we do want to present some basic results about pre-triangulated categories just to illustrate how the basic theory should be used. Let us start by saying some words about the history of triangulated category theory and its connections with operator algebras.

Remark 4.32. It is impossible (and also inappropriate) for us to give a full historical account of these topics when such accounts do exist in the literature. For a detailed dive into the classical origins of triangulated categories, see [75]. The important point is that this theory was independently developed in the 1960s by (Dieter) Puppe and (Jean-Louis) Verdier. Indeed, the classical examples of triangulated categories are the stable homotopy categories of algebraic topology and the derived categories of abelian categories which were introduced for use in algebraic geometry. Since then, this theory has found applications in many other parts of mathematics such as operator algebras, modular representation theory, symplectic geometry and many more.

To our knowledge, the PhD thesis of Andreas Thom, [71], used triangulated categories for the first time in relation to operator algebras. Thom uses these to study bivariant homology theories^a on ${}_{\omega}C^*\text{Alg}$ and shows that stable homotopy for separable C^* -algebras is a triangulated homology theory. Since then, it has become known that

Kasparov’s KK-theory and Higson’s E-theory both carry a natural triangulated structure & this has played a role in recent computations of K-theory for certain C^* -algebras; see [60]. Of course, this does not summarize all of the different ways in which this theory has interacted with operator algebras but it should give the reader a nice set of reasons to engage with this framework.

^aKasparov’s KK-Theory is an example and Connes-Higson’s E-Theory is another major example.

At this point, there are a number of neat resources on triangulated category theory; the main reference on this topic is (Amnon) Neeman’s treatise [52]. Indeed, the modern theory of triangulated categories was first systematically developed in that book. With that said, the main ingredients for specifying a triangulated category are listed as follows:

1. an additive category $\mathcal{T}$,
2. an additive auto-equivalence $\Sigma : \mathcal{T} \rightarrow \mathcal{T}$,
3. a set $\mathcal{E}$ of “exact triangles”.

So, triangulated categories are good at dealing with exact sequences. What makes them different from, say, the Quillen-exact categories of algebraic K-theory (which also deal with exact sequences), is that they also encode the notion of a “long-exact sequence” by means of the additive auto-equivalence. Indeed, one should view that as a suspension functor. So, triangulated categories give us a formalism by which we can manipulate long-exact sequences. In the next construction, we will encode the notion of a long-exact sequence using *exact triangles*, which are easy to specify once one has an additive category and accompanying additive auto-equivalence in the background.

Construction 4.33. Let $\mathcal{T}$ be an additive category and let $\Sigma \in \text{Aut}(\mathcal{T})$ be additive. We define a structure:

$$\text{Ex}\Delta(\mathcal{T}, \Sigma),$$

as follows:

1. A triangle for Σ is a sequence of morphisms:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

That is, it’s a tuple (x, y, z, f, g, h) which encodes the specific data above. The objects are triangles for Σ ; we refer to x, y, z as the vertices of the triangle while the morphisms are the sides.

2. If we have two triangles for Σ , then a morphism between them is represented by the dotted arrows below:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \vdots & & \vdots & & \vdots & & \downarrow \Sigma(a) \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x[1]' \end{array}$$

such that the diagram commutes. In other words, a morphism is a triple (a, b, c) consisting of morphisms between the vertices such that the diagram above commutes. The composition of morphisms is defined exactly as one expects and clearly, the composite of two morphisms yields a morphism between triangles.

This data turns $\text{Ex}\Delta(\mathcal{T}, \Sigma)$ into a category.

We also need one more piece of terminology to facilitate our desired definition.

Terminology 4.34. Let $\mathcal{C}$ be a category and let S be a set of objects in $\mathcal{C}$. S is closed under isomorphisms iff for every $x \in S$ and for every $y \in \mathcal{C}$ which is isomorphic to x , we have that $y \in S$.

Finally, it will be useful to have one more piece of notation.

Notation 4.35. Let $\mathcal{C}$ be a category and let $T \in \text{Fun}(\mathcal{C})$. T is an auto-equivalence iff it is an equivalence of categories. Moreover, if $x \in \mathcal{C}$ and T is a piece of datum accompanying $\mathcal{C}$, then we will write:

$$x[1] \stackrel{\text{def}}{=} T(x).$$

In this case, T will also be known as a shift/suspension. Indeed, we define:

$$\forall n \in \mathbb{Z} : x[n] \stackrel{\text{def}}{=} T^n(x),$$

where this is defined for negative values of n because T is an equivalence of categories^a.

^aWe just have to fix a quasi-inverse and simply use that for the negative shift.

With this notation in mind, we are ready to define Neeman pre-triangulated categories.

Definition 4.36

A (Neeman) pre-triangulated category is a triple $(\mathcal{T}, \Sigma, \mathcal{E})$ consisting of:

1. an additive category $\mathcal{T}$,
2. an additive auto-equivalence $\Sigma : \mathcal{T} \rightarrow \mathcal{T}$ and,
3. a set $\mathcal{E}$ of triangles which is closed under isomorphisms,

such that the following properties hold:

TR1: Let $x, y \in \mathcal{T}$ and let $f \in \mathcal{T}(x, y)$. Then, the following triangle:

$$x \xrightarrow{\text{Id}_x} x \longrightarrow 0 \longrightarrow x[1]$$

lies in $\mathcal{E}$. Moreover, for any $z \in \mathcal{T}$, there exists a triangle which fits f :

$$x \xrightarrow{f} y \longrightarrow z \longrightarrow x[1]$$

within $\mathcal{E}$.

TR2: The following triangle:

$$x \xrightarrow{u} y \xrightarrow{v} z \xrightarrow{w} x[1]$$

lies in $\mathcal{E}$ iff the following triangle:

$$y \xrightarrow{v} z \xrightarrow{w} x[1] \xrightarrow{-u[1]} y[1]$$

lies in $\mathcal{E}$.

TR3: If we have a commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow a[1] \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x[1]' \end{array}$$

of solid arrows with the rows being triangles in $\mathcal{E}$, then there exists a morphism $c : z \rightarrow z'$ such that the triple (a, b, c) forms a morphism of triangles in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$.

$\mathcal{E}$ is known as the set of distinguished/exact triangles.

Since we are going to be using the axioms of a pre-triangulated category to prove some elementary statements, it will be useful for us to introduce some terminology associated to the axioms that allow us to use them less awkwardly in proofs. This is not standard in the literature but the author found them to be rather useful.

Terminology 4.37. Let $\mathcal{T}$ be a pre-triangulated category, let $x, y \in \mathcal{T}$ and let $f \in \mathcal{C}(x, y)$ fit in a exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \longrightarrow x[1]$$

Then, the pair (z, g) is the cone of f (or dually, (x, f) is a cocone of g). Suppose we have a exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

By TR2, we have another exact triangle:

$$y \xrightarrow{g} z \xrightarrow{h} x[1] \xrightarrow{-\Sigma(f)} y[1]$$

This will be known as the shifted triangle^a. Moreover, if we have a commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \\ \downarrow a & & \downarrow b & & & & \downarrow \Sigma(a) \\ x' & \xrightarrow{f'} & y' & \xrightarrow{g'} & z' & \xrightarrow{h'} & x[1]' \end{array}$$

where the rows are exact triangles, then the pair (a, b) is called a partial morphism of exact triangles.

^aOne might even say suspended triangle but “shift” works better here.

Here are two basic results which demonstrate how this theory is used.

Claim 4.3. Let $\mathcal{T}$ be a pre-triangulated category and consider the following exact triangle:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

Then, $g \circ f = 0$ and $h \circ g = 0$. This is proved on Page 101.

One may have noticed that TR3 is very specific about which morphism is excluded from the diagram and “filled in”. However, one can show that variants of TR3 hold where the other two morphisms are excluded and then “filled in”. We formulate this a bit more precisely in the next result.

Claim 4.4. Let $\mathcal{T}$ be a pre-triangulated category and suppose we have two commutative diagrams:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow & & \downarrow & & \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array} \quad \begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ & \downarrow & & \downarrow & & & \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

where the rows are exact triangles. Then, we can “fill in” the missing morphisms so we get the following:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array} \quad \begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

commutative diagrams^a.

^aWe provide all of the details so that one obtains familiarity with using triangulated categories.

With some of these results in place, we can talk about cohomological functors on pre-triangulated categories.

Terminology 4.38. Let $\mathcal{A}$ be an additive category, let $\mathcal{T}$ be a pre-triangulated category and let $F \in \text{Fun}(\mathcal{T}, \mathcal{A})$. F is cohomological iff for every exact triangle $x \rightarrow y \rightarrow z \rightarrow x[1]$, we have that the sequence:

$$F(x) \longrightarrow F(y) \longrightarrow F(z)$$

is exact in $\mathcal{A}$.

The intuition is that cohomological functors are a recipe for turning short exact sequences into long exact sequences.

Claim 4.5. Let $\mathcal{T}$ be a pre-triangulated category and let $x \in \mathcal{T}$. Then, the (co)representable functors:

$$\mathcal{T}(x, -) : \mathcal{T} \rightarrow \text{AbGrp},$$

$$\mathcal{T}(-, x) : \mathcal{T}^{\text{op}} \rightarrow \text{AbGrp},$$

are cohomological. This is the Triangulated Yoneda Lemma and it is proved on Page 101.

Claim 4.6. Let $\mathcal{T}$ be a pre-triangulated category and consider a morphism in $\text{Ex}\Delta(\mathcal{T}, \Sigma)$ as follows:

$$\begin{array}{ccccccc} x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow \Sigma(a) \\ x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x[1]' \end{array}$$

If any two of a , b or c is an isomorphism, then so is the third. This is the Triangulated Five Lemma and it is proved on Page 102.

As a corollary, we obtain a useful characterization of when a morphism is an isomorphism in a pre-triangulated category.

Claim 4.7. Let $\mathcal{T}$ be a pre-triangulated category and let $f : x \rightarrow y$ be a morphism in $\mathcal{T}$. TFAE:

1. f is an isomorphism.
2. The triangle $x \xrightarrow{f} y \rightarrow 0 \rightarrow x[1]$ is exact.

This is proved on Page 102.

We have worked with pre-triangulated categories so far. The next step is to define triangulated categories.

Definition 4.39

Let $\mathcal{T}$ be a pre-triangulated category. It is (Neeman) triangulated iff for any pair of morphisms:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

there is a commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \xrightarrow{h} & z' & \longrightarrow & x[1] \\ \downarrow = & & \downarrow g & & \downarrow & & \downarrow = \\ x & \xrightarrow{g \circ f} & z & \xrightarrow{=} & y' & \longrightarrow & x[1] \\ & & \downarrow & & \downarrow & & \downarrow f[1] \\ & & x' & \xrightarrow{=} & x' & \longrightarrow & y[1] \\ & & \downarrow & & \downarrow & & \\ & & y[1] & \xrightarrow{h[1]} & z'[1] & & \end{array}$$

where the first two rows and the two central columns are distinguished triangles^a.

^aThis additional axiom is called the Octahedral axiom or TR4.

At this stage, we will not continue further development of triangulated category theory since we will not need its finer points here. Instead, we will move straight to the original problem of interest; proving [Theorem 3.77](#). Let us start implementing the strategy in [Remark 4.31](#) and for that, we will first discuss ∞ -categorical localizations.

Definition 4.40

Let $\mathcal{C}$ be an ∞ -category and let W be a set of morphisms in $\mathcal{C}$. The Dwyer-Kan localization (DK-localization) of $\mathcal{C}$ at W is an ∞ -category $\mathcal{C}_W$ and a functor $L : \mathcal{C} \rightarrow \mathcal{C}_W$ which induces an equivalence of ∞ -categories^a:

$$\mathrm{Fun}(\mathcal{C}_W, \mathcal{D}) \rightarrow \mathrm{Fun}^W(\mathcal{C}, \mathcal{D}), \quad f \mapsto f \circ L,$$

for any ∞ -category $\mathcal{D}$. The Dwyer-Kan localization is also defined for 1-categories with respect to some set of morphisms; we simply pass to the ∞ -category associated to the given 1-category by the nerve & take the DK-localization with respect to the set of morphisms obtained after passing through the nerve.

^a $\mathrm{Fun}^W(\mathcal{C}, \mathcal{D})$ is the full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ spanned by those functors $f : \mathcal{C} \rightarrow \mathcal{D}$ which invert W .

We have deliberately chosen different notation for the Dwyer-Kan localization to distinguish it from the 1-categorical localization. These two are related to each other in a very simple way which can be described via the homotopy category construction. We will need this construction later, so it is best to discuss it now. Roughly speaking, the homotopy category “compresses” all data about higher homotopies in an ∞ -category and produces a 1-category from the compressed data.

Construction 4.41. Let $N : \mathrm{Cat}_1 \rightarrow \mathrm{Cat}_\infty$ be the Rezk nerve^a. It has a left adjoint:

$$\mathbf{Ho}(-) : \mathrm{Cat}_\infty \rightarrow \mathrm{Cat}_1,$$

which we will refer to as the homotopy category functor.

^aIn quasicategory theory, this functor takes a small 1-category and maps it to its nerve.

The “construction” above is purely synthetic; we know that such a left adjoint will exist within a model for ∞ -categories, so we can synthetically impose the existence of such a left adjoint. For the sake of being somewhat concrete in this discussion, let us describe how this functor is constructed via quasicategories, a model for ∞ -categories.

Remark 4.42. This discussion is primarily inspired by material found in [61, p.30-34]. The key point is that if the term “ ∞ -category” means “quasicategory”, then there is a concrete construction of the Rezk nerve and there is a concrete construction of the homotopy category of a quasicategory; we will assume some familiarity with the former. For the latter, the key is to note that it’s possible to define a notion of homotopy between 1-simplices (read: morphisms) in a quasicategory. Let us denote this relation by $\simeq_h$. If X is a quasicategory, then we simply define:

$$\mathbf{Ho}(X),$$

as follows:

1. The objects are the elements of X_0 .
2. If $x, y \in X_0$, then:

$$\mathbf{Ho}(X)(x, y) \stackrel{\mathrm{def}}{=} \mathrm{hom}_X(x, y) / \simeq_h.$$

The composition, which is defined as one expects, is well-defined.

Note that this is a 1-category because we have defined it to be so. More importantly, it gives us a direct construction of the homotopy category from a quasicategory^a. As is shown in the resource above, this construction can be assembled into a functor and it is this functor which forms a left adjoint to the Rezk nerve.

^aAs (Charles) Rezk shows in the aforementioned notes, one can construct a 1-category out of any simplicial set.

The next result tells us how we can obtain the 1-categorical localization from the Dwyer-Kan localization.

Proposition 4.43

Let $\mathcal{C}$ be a 1-category and let W be a set of morphisms in $\mathcal{C}$. Then, $\mathbf{Ho}(\mathcal{C}_W) \simeq \mathcal{C}[W^{-1}]$.

Proof. We first note that, in this case, we have that:

$$\mathcal{C}_W \stackrel{\mathrm{def}}{=} N(\mathcal{C})_{N(W)},$$

where the right-hand side is the Dwyer-Kan localization of the ∞ -category $N(\mathcal{C})$ with respect to the set of morphisms $N(W)$; we mentioned this in [Definition 4.40](#). We would like to make sense of the following calculation:

$$\begin{aligned} \mathrm{Fun}(\mathbf{Ho}(\mathcal{C}_W), \mathcal{D}) &\simeq \mathrm{Fun}(\mathcal{C}_W, N(\mathcal{D})) \\ &\simeq \mathrm{Fun}(N(\mathcal{C})_{N(W)}, N(\mathcal{D})) \\ &\simeq \mathrm{Fun}^{N(W)}(N(\mathcal{C}), N(\mathcal{D})) \\ &\simeq N(\mathrm{Fun}^W(\mathcal{C}, \mathcal{D})) \\ &\simeq \mathrm{Fun}^W(\mathcal{C}, \mathcal{D}) \\ &\simeq \mathrm{Fun}(\mathcal{C}[W^{-1}], \mathcal{D}). \end{aligned}$$

This calculation implies the result if it makes sense. For this to work, we will impose that $\mathcal{D}$ is a 1-category. Now, the first step is just the nerve-homotopy adjunction of [Construction 4.41](#). In the second step, we use the definition of

$\mathcal{C}_W$. In the third step, we use the universal property of $\mathcal{C}_W$. In the fourth step, we note that $\mathcal{C}$ and $\mathcal{D}$ are 1-categories so functors between them are identified with the ∞ -functors between their associated ∞ -categories; we use this also in the fifth step. Finally, we use the universal property of the 1-categorical localization. ■

Now, we have the following result.

Proposition 4.44

Let $\mathcal{C}$ be a finitely bicomplete 1-category of fibrant objects. Then, $\mathcal{C}_W$ is a pointed and finitely-bicomplete ∞ -category.

Proof. ■

So, that completes the first two steps of the strategy outlined in [Remark 4.31](#). Now, we will move on to the third step. For this step, we will present the construction of the Spanier-Whitehead category for pointed, finitely cocomplete ∞ -categories. Since this starting situation is not what we want, we will adapt it to our situation by duality.

Construction 4.45 ([46, Construction C.1.1.1, p.1943]). Let $\mathcal{C}$ be a pointed, finitely cocomplete ∞ -category. We define:

$$\mathbf{SW}(\mathcal{C}),$$

by the following colimit in^a $\mathbf{Cat}_\infty$:

$$\mathcal{C} \xrightarrow{\Sigma} \mathcal{C} \xrightarrow{\Sigma} \dots$$

In this case, we refer to $\mathbf{SW}(\mathcal{C})$ as the Spanier-Whitehead category of $\mathcal{C}$. This construction gives a canonical map:

$$\mathcal{C} \rightarrow \mathbf{SW}(\mathcal{C}),$$

which is the “inclusion” of $\mathcal{C}$ in its Spanier-Whitehead stabilization^b.

^aNote that $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$ is the suspension functor.

^bIt is not the same construction as the usual stabilization procedure that produces $\mathbf{Sp}$.

In fact, the Spanier-Whitehead category is rather nicely behaved, as the next result shows.

Proposition 4.46. [46, Proposition C.1.1.7, p.1945]

Let $\mathcal{C}$ be a pointed, finitely cocomplete ∞ -category. TFSH:

1. $\mathbf{SW}(\mathcal{C})$ is a stable ∞ -category.
2. If $\mathcal{D}$ is a stable ∞ -category, then the canonical map $\mathcal{C} \rightarrow \mathbf{SW}(\mathcal{C})$ induces an equivalence:

$$\mathbf{Fun}^{\mathrm{ex}}(\mathbf{SW}(\mathcal{C}), \mathcal{D}) \rightarrow \mathbf{Fun}^{\mathrm{rex}}(\mathcal{C}, \mathcal{D}),$$

of ∞ -categories.

Now, we can proceed towards our Spanier-Whitehead category for pointed, finitely complete ∞ -categories.

Construction 4.47. Let $\mathcal{C}$ be a pointed, finitely complete ∞ -category. We define:

$$\mathbf{SW}(\mathcal{C})_0 \stackrel{\mathrm{def}}{=} \mathbf{SW}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}},$$

to be the Spanier-Whitehead category of $\mathcal{C}$.

Here’s a remark about this construction.

Remark 4.48. There is a slightly subtle issue with the construction above. Suppose that $\mathcal{C}$ is a pointed, finitely bicomplete ∞ -category. In that case, we can form two Spanier-Whitehead categories $\mathbf{SW}(\mathcal{C})$ and $\mathbf{SW}(\mathcal{C})_0$. One reasonable question would be to determine whether one of these has precedence over the other. In fact, it turns out that both of these categories are equivalent; the proof of this is not hard and requires the universal property.

Before we build on the remark above, here is an immediate relevant corollary.

Corollary 4.49

Let $\mathcal{C}$ be a pointed, finitely complete ∞ -category. Then, $\mathbf{SW}(\mathcal{C})_0$ is stable.

Proof. Well, this is just the usual Spanier-Whitehead category of $\mathcal{C}^{\mathrm{op}}$ and we know that this is stable. ■

With these results, we have completed steps (3) and (4) of the strategy outlined in [Remark 4.31](#). We are ready to work towards the completion of the last step. This requires us to simply state a result.

Theorem 4.50. J. Lurie, 2009, [43, Theorem 3.11, p.10-12]

Let $\mathcal{C}$ be a stable ∞ -category. Then, $\mathbf{Ho}(\mathcal{C})$ has the structure of a triangulated category^a.

^aIn the cited resource, the additive structure (along with other preliminary results) is established in the same section.

The theorem requires further clarification. For instance, it is necessary to state the triangulated structure explicitly so that we know it matches up with the triangulated structure suggested in the statement of [Theorem 3.77](#).

Construction 4.51. Let $\mathcal{C}$ be a stable ∞ -category and suppose we have a diagram:

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

in $\mathbf{Ho}(\mathcal{C})$. This diagram is a distinguished triangle iff there exists a commutative^a diagram:

$$\begin{array}{ccccc} x & \xrightarrow{\tilde{f}} & y & \longrightarrow & 0 \\ \downarrow & & \downarrow \tilde{g} & & \downarrow \\ 0' & \longrightarrow & z & \xrightarrow{\tilde{h}} & w \end{array}$$

such that the following conditions are satisfied:

1. $0, 0'$ are zero objects in $\mathcal{C}$,
2. Both squares are pushouts in $\mathcal{C}$,
3. the morphisms $\tilde{f}$ and $\tilde{g}$ represent f & g respectively,
4. the map $h : z \rightarrow x[1]$ is the composition of the homotopy class of $\tilde{h}$ with the isomorphism $w \simeq x[1]$ determined by the outer rectangle.

^aIn the sense of ∞ -category theory, of course.

We are finally able to establish the Spanier-Whitehead construction for categories of fibrant objects.

Proof of Theorem 3.77. Again, the strategy for the proof has been discussed in [Remark 4.31](#). Since the strategy just gives the proof once all of the results have been laid out, we are done. ■

5 Open Questions & Further Possibilities

Summary

In this thesis, we have had the opportunity to come into contact with questions/conjectures of interest to people in the community of operator algebras. Indeed, we have even been able to address some of these questions/conjectures to varying degrees of success. However, there is still much work to be done and we wish to conclude this thesis with some thoughts about further questions that one could profitably spend time considering. Note that the list of questions here does not reflect every possible question one could ask about the topics we discuss.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

G will denote a fixed locally compact, Hausdorff group.

5.1 An Outlook on Lifting Problems

We will start by giving an outlook on lifting problems; this will primarily focus on the work done in Section 2. Let us start by noting that Corollary 2.32 gives us a way for Schochet fibrations to be semisplit surjections. This raises a question.

| **Question 5.1.** Is there a way to produce Schochet fibrations from semisplit surjections in $G\text{-}C^*\text{Alg}$?

A naive, positive answer to this is provided by Proposition 3.40; we can just factorize out semisplit surjection into a composition of a Schochet fibration and a weak equivalence. However, this is not interesting as it applies to all morphisms in $G\text{-}C^*\text{Alg}$ and it does not tell us if the entire semisplit surjection is, itself, a Schochet fibration. In the next remark, we will discuss the extent to which it is possible to conclude that a semisplit surjection is a Schochet fibration and if there are indications for the conditions that might be necessary for that to happen.

Remark 5.2. With semisplit surjections, we do obtain a “weakened” version of the Schochet fibration property. So, let $A, B \in C^*\text{Alg}$ and let $f \in C^*\text{Alg}(A, B)$ be a semisplit surjection with g being a ccp split. Now, we consider:

$$\begin{array}{ccc} & & A \\ & \nearrow p & \uparrow f \\ D & \xrightarrow{(h_t)_{t \in [0,1]}} & B \end{array}$$

where we have that $f \circ p = h_0$. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} g \circ h_t.$$

For each $t \in [0, 1]$, we see that H_t is a ccp map. We also have that:

$$\begin{aligned} \forall t \in [0, 1] : f \circ H_t &= f \circ (g \circ h_t) \\ &= (f \circ g) \circ h_t \\ &= h_t. \end{aligned}$$

Finally, we see that:

$$H_0 = g \circ h_0 = g \circ (f \circ p) = (g \circ f) \circ p.$$

So, with a semisplit surjection, we naturally obtain a kind of “ccp homotopy lift”. The semisplit condition does not “solve” the homotopy lifting problem above but there is a sense in which we do get some nice lifting behavior. It has been known for some time, due to the work of (Man-Duen) Choi, (Edward) Effros and others, that lifting problems for C^* -algebras are best studied when working with ccp maps. So, it might be reasonable to pursue homotopy lifting problems in that context as well. We intend to pursue these matters further in future work.

In Section 2.2, our discussion of lifting problems centered around the desire to better understand Schochet C^* -algebras. We showed that, in fact, there are C^* -algebras which are not Schochet and we managed to prove a result (Proposition 2.86) allows us to conclude that $\mathcal{C}(X)$ is not Schochet when X is a compact subset of $\mathbb{C}$ satisfying mostly a topological set of conditions. However, in the statement of that result, the most mysterious condition is the path-connectedness of the set of normal elements of $\mathcal{Q}(H)$ whose spectrum is X . This prompts the following question.

| **Question 5.3.** Is there a better characterization of condition (3) of Proposition 2.86?

In general, having a better understanding of the Schochet property in its own right would be very helpful, not only for addressing the question above but also for understanding the obstruction to Blackadar’s Semiprojectivity Conjecture (Conjecture 2.121) as highlighted by us in statement (11) of Proposition 2.127. In particular, it would be nice to eventually address the following question.

| **Question 5.4.** What are some other permanence properties of (approximately) Schochet, separable C^* -algebras?

It would also be desirable to gain a better understanding of the category of approximately Schochet, separable C^* -algebras. In the next two remarks, we will discuss two possible strategies towards resolving Question 2.66.

Remark 5.5. In our first strategy, the goal is to find a marked C^* -algebra (A, I) such that:

1. A/I has a normal element which is not normal-liftable.
2. Normal elements in A/I with finite spectrum are liftable to A .
3. Every normal element in A/I can be approximated by a sequence of normal elements with finite spectrum.

In fact, the second statement is known to be true generally; see [64, Proposition 4.4, p. 12]. The third statement is not generally true but [28, Corollary 3.12, p.363] provides a nice set of conditions which allow us to conclude that a given normal element is approximated by normal elements with finite spectrum. To be precise, we have the claim.

Claim. Let A be a unital C^* -algebra of real rank zero^a and stable rank one^b and let $a \in A$ be a normal element such that for every proper closed ideal $I \subseteq A$, we have that $\pi(a) - \lambda(1_A + I)$ is invertible & homotopic to $1_A + I$ in $\text{GL}(A/I)$. Then, a can be written as the limit of a sequence of finite-spectrum normal elements.

^aIf $B \in C^*\text{Alg}$, then B is of real rank zero iff the finite-spectrum self-adjoint elements of B are dense in $B_{\text{s.a.}}$.

^bIf $B \in C^*\text{Alg}$, then B is of stable rank one iff $\text{GL}(B^\sim)$ is dense in B . If B is unital, then we may forego the unitization.

Even though real rank zero and stable rank one has been verified for many C^* -algebras, the second condition seems to be difficult to check. So, with this strategy, both statements (1) and (3) seem to be significant obstructions.

Remark 5.6. In our second strategy, the goal is to use the following result.

Claim. Let $A \in C^*\text{Alg}$ have property (IR), let I be a closed ideal in A and let $x \in A/I$ be normal. For every $\epsilon > 0$, there exists a normal element $a \in A$ such that $\|\pi(a) - x\| < \epsilon$.

This was proved in [65, Corollary 4.5, p. 129] and for this discussion, the definition of the property (IR) will not matter (though the reader can find it in Definition 3.1, p.125 of the aforementioned reference). What matters is that every normal element in a quotient of A can be approximated by a sequence of elements which are normal-liftable. If there is an example of A for which there is a normal element in a quotient which does not lift, then this is a counterexample for Question 2.66. In [65, p.125], some examples of C^* -algebras with property (IR) are provided:

1. C^* -algebras of stable rank one,
2. simple von Neumann algebras,
3. simple, purely infinite unital C^* -algebras.

In light of Example 2.78, we are not going to find counterexamples among the simple C^* -algebras with property (IR)^a. So, it seems that the next option is to check for counterexamples among C^* -algebras of stable rank one.

^aIt is mentioned in the aforementioned paper that it is not known if every simple C^* -algebra has property (IR).

We aim to pursue both strategies stated in Remarks 5.5 and 5.6 within future work. It is clear that there is an obstruction which is common to both strategies; this has to do with the normal-liftability of elements in a quotient C^* -algebra. Let us address this specific obstruction in the next remark and a possible strategy for overcoming it.

Remark 5.7. Let (A, I) be a marked C^* -algebra and let $b \in A/I$ be normal. Our problem is to determine when there exists a normal $a \in A$ such that $\pi(a) = b$. In the proof of Theorem 2.59, we have already demonstrated that it is not possible to guarantee the existence of a normal lift even to a unitary in A/I (assuming it is unital, of course). The key points in that argument are summarized abstractly as follows:

1. Find a Fredholm element a of A of non-zero Fredholm index.
2. The normal Fredholm elements of A have Fredholm index 0.
3. If $\pi(a)$ had a normal lift b , then $b - a \in I$ and their Fredholm indices must be the same.
4. Since their Fredholm indices are not the same, we are done.

With both strategies in Remarks 5.5 and 5.6, we would want to find A and I with A having stable rank one so that the argument above goes through for some reasonable notion of “Fredholm Index”. The real content of our proposed argument above is in statements (1) and (2). Statement (2) is meant to be a hard restriction on the normal elements so we have complete knowledge of their Fredholm indices. Statement (1) requires an actual construction. In fact, we can modify this strategy so that it is a bit more relaxed and we obtain the following list.

1. Consider the set of all Fredholm indices of normal elements in A .
2. Find a Fredholm element a of A whose Fredholm index does not lie in the set in statement (1).
3. By the argument above, we see that $\pi(a)$ cannot have a normal lift.

With the first list, the most difficult step seems to be step (1).

Remark 5.8. In [16], it is proved that every approximately semiprojective C^* -algebra satisfies the so-called *asymptotic homotopy lifting property*; this is essentially the same property that we have been working with except that we replace the morphisms by asymptotic morphisms which are already known to play a significant role in Connes-Higson E-Theory. Indeed, if one can find an example of a C^* -algebra which does not satisfy the asymptotic homotopy lifting theorem, then Blackadar’s conjecture is false. It might even be possible to find such a C^* -algebra among those which are approximately Schochet. We plan to investigate these matters more in the future.

In fact, one reasonable place to start looking would be to establish permanence properties of approximately Schochet C^* -algebras and show that this category does not satisfy any one of the permanence properties which are satisfied by approximately semiprojective C^* -algebras².

²We plan to do more work in this direction in the future.

5.2 An Outlook on Abstract Homotopy Theory

In this subsection, we will focus on problems involving pure homotopy theory, though these questions are also motivated by natural questions in the realm of operator algebras. The focus is mainly on the contents of Section 3 and Section 4. We will start with the following question.

Question 5.9. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects. When are the evaluation morphisms:

$$\alpha_0^x, \alpha_1^x : T(x) \rightarrow x,$$

internal fibrations in $\mathcal{C}$?

To address this, it will be necessary to impose some relationship between the path space functor and the two time-evaluation natural transformations. To motivate the required condition, let us see how it works in the case of C^* -algebras.

Remark 5.10. Let $A \in C^* \text{Alg}$. We define a map:

$$\tau : [0, 1] \times [0, 1] \rightarrow [0, 1] \times [0, 1], \quad (s, t) \mapsto \tau(s, t) \stackrel{\text{def}}{=} (t, s).$$

This can be used to construct the following isomorphism:

$$\Psi_A : \text{Cyl}(\text{Cyl}(A)) \rightarrow \text{Cyl}(\text{Cyl}(A)), \quad f \mapsto \Psi_A(f),$$

where we define:

$$\forall t, s \in [0, 1] : \Psi_A(f)(t)(s) \stackrel{\text{def}}{=} f(s)(t).$$

This is a $\star$ -isomorphism essentially because τ is a homeomorphism. Bearing this in mind, if $t \in [0, 1]$, then we would like to know the relationship between $\text{Cyl}(\alpha_t^A)$ and $\alpha_t^{\text{Cyl}(A)}$. Let us simply compute them separately:

$$\forall \phi \in \text{Cyl}(\text{Cyl}(A)) : \forall s \in [0, 1] : \text{Cyl}(\alpha_t^A)(\phi)(s) = (\alpha_t^A \circ \phi)(s) = \phi(s)(t).$$

On the other hand:

$$\forall \phi \in \text{Cyl}(\text{Cyl}(A)) : \alpha_t^{\text{Cyl}(A)}(\phi)(s) = \phi(t)(s).$$

In other words, it seems to be the case that these two are the same, up to time-exchange. That is:

$$\alpha_t^{\text{Cyl}(A)} = \text{Cyl}(\alpha_t^A) \circ \Psi_A.$$

This formula, of course, motivates what we need very well.

In light of [Remark 5.10](#), we introduce the following definition.

Definition 5.11

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects. This is a good h -category of fibrant objects iff T preserves pullbacks and there exists a natural isomorphism $\beta : T^2 \Rightarrow T^2$ such that for every $x \in \mathcal{C}$ and each $i \in \{0, 1\}$, we have:

$$\alpha_i^{T(x)} = T(\alpha_i^x) \circ \beta_x.$$

We refer to β as the time-exchange isomorphism.

This leads us to the following proposition.

Proposition 5.12

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a good h -category of fibrant objects. TFSH:

1. For $i \in \{0, 1\}$ and $x \in \mathcal{C}$, we have that $\alpha_i^x \in W_{\mathcal{C}, h}$.
2. For $i \in \{0, 1\}$ and $x \in \mathcal{C}$, we have that α_i^x is an internal fibration.

Proof.

■

Section 2

Proof. 2.1. The proof that these are C^* -algebras is completely routine. It remains to be shown that we have a valid action by automorphisms. Let's deal with the cylinder because that's where everything is happening; we need to show that action above is continuous and acts via automorphisms of $\text{Cyl}_G(A)$.

1. Let $g \in G$ and consider the map:

$$\phi_g : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(A), \quad f \mapsto \phi_g(f) \stackrel{\text{def}}{=} gf.$$

Well, it's clear that ϕ_g maps into $\text{Cyl}_G(A)$. Now, we need to show that this is a $\star$ -homomorphism.

(a) Let $f_1, f_2 \in \text{Cyl}_G(A)$ and let $\lambda \in \mathbb{C}$. Then, we have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(\lambda f_1 + f_2)(t) &= g(\lambda f_1(t) + f_2(t)) \\ &= g(\lambda f_1(t)) + g f_2(t) \\ &= \lambda(g f_1(t)) + g f_2(t) \\ &= (\lambda(g f_1) + g f_2)(t), \end{aligned}$$

which proves that ϕ_g is linear.

(b) Let $f \in \text{Cyl}_G(A)$. We have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(f^*)(t) &= (g f^*)(t) \\ &= g f^*(t) \\ &= g f(t)^* \\ &= (g f(t))^* \\ &= (g f)^*(t) \\ &= (g f)^*(t), \end{aligned}$$

which proves that ϕ_g is involution-preserving.

(c) Let $f_1, f_2 \in \text{Cyl}_G(A)$. We have the following computation:

$$\begin{aligned} \forall t \in [0, 1] : \phi_g(f_1 f_2)(t) &= g(f_1 f_2)(t) \\ &= g(f_1(t) f_2(t)) \\ &= (g f_1(t))(g f_2(t)) \\ &= \phi_g(f_1)(t) \phi_g(f_2)(t) \\ &= (\phi_g(f_1) \phi_g(f_2))(t), \end{aligned}$$

which proves that ϕ_g preserves the product.

Throughout the computation above, we've just utilized the properties of the action by automorphisms on A by G . This proves that we have a map $G \rightarrow \text{Aut}(\text{Cyl}_G(A))$ and that this is a group homomorphism follows literally from a routine computation. Now, we have to show continuity.

2. Let $f \in \text{Cyl}_G(A)$. Then, we want to consider the map:

$$g \mapsto \phi_g(f) = gf.$$

To show continuity, let $(g_\delta)_{\delta \in D}$ be a net in G that converges to some g .

Proof. 2.2. This is a $\star$ -homomorphism by the explicit formula. For equivariance, we have:

$$\begin{aligned} \forall g \in G : \forall \phi \in \text{Cyl}_G(A) : p_t^A(g\phi) &= (g\phi)(t) \\ &= g\phi(t) \\ &= g p_t^A(\phi), \end{aligned}$$

as was desired. In this case, the explicit formula for the canonical inclusion and a simple estimate tells us that this is a contractive algebra homomorphism.

Proof. 2.3. Consider the following diagram:

$$\begin{array}{ccc} \text{Cyl}_G(A) & \xrightarrow{\alpha_t^A} & A \\ \text{Cyl}_G(f) \downarrow & & \downarrow f \\ \text{Cyl}_G(B) & \xrightarrow{\alpha_t^B} & B \end{array}$$

Let's show that this commutes:

$$\begin{aligned}
\forall \phi \in \text{Cyl}_G(A) : (f \circ \alpha_t^A)(\phi) &= f(\alpha_t^A(\phi)) \\
&= f(\phi(t)) \\
&= (f \circ \phi)(t) \\
&= (\text{Cyl}_G(f)(\phi))(t) \\
&= \alpha_t^B(\text{Cyl}_G(f)(\phi)) \\
&= (\alpha_t^B \circ \text{Cyl}_G(f))(\phi).
\end{aligned}$$

This computation reflects the naturality of each α_t so we're done. In the case that $A \in G\text{-BanAlg}$, the argument does not change as we only use the algebraic properties of the maps above and those do hold in this case.

Proof. 2.4. We will deal with each sequence separately.

1. Clearly, ι is injective. For the surjectivity of p_1^A , let $a \in A$ and let $f \in \mathcal{C}[0, 1]$ such that $f(1) = 1$. Define:

$$fa : [0, 1] \rightarrow A, \quad t \mapsto (fa)(t) \stackrel{\text{def}}{=} f(t)a.$$

Observe that $(fa)(1) = f(1)a = a$. But this just means that:

$$p_1^A(fa) = a,$$

so p_1^A is surjective. Finally, we need to show that $\text{ran}(\iota) = \ker(p_1^A)$. Indeed, the fact that:

$$\text{ran}(\iota) \subseteq \ker(p_1^A),$$

follows from the exactness of the sequence. On the other hand, let $\phi \in \text{Cyl}_G(A)$ such that $p_1^A(\phi) = 0$. This means that $\phi(1) = 0$. That is, $\phi \in \text{Cone}_G(A)$. So, $\phi \in \text{ran}(\iota)$ because $\iota(\phi) = 0$.

2. The proof proceeds by almost the same argument as above; one literally just has to replace specific symbols by the appropriate one to get a proof applicable to the second sequence.

This completes the proof. The same argument works also for $A \in G\text{-BanAlg}$.

Proof. 2.5. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f_1, f_2, f_3 \in G\text{-C}^*\text{Alg}(A, B)$. We define:

$$s_A : A \rightarrow \text{Cyl}_G(A),$$

which takes $a \in A$ and maps it to the constant function $s_A(a)$ on $[0, 1]$ which takes the value a .

1. We define:

$$h : A \rightarrow \text{Cyl}_G(B), \quad a \mapsto h(a) \stackrel{\text{def}}{=} s_B(f_1(a)).$$

By definition, one observes that h is a homotopy from f_1 to itself so that proves reflexivity.

2. Let $f_1 \simeq_h f_2$ and let $h : A \rightarrow \text{Cyl}_G(B)$ be the homotopy from f_1 to f_2 . We define:

$$\forall a \in A : \forall t \in [0, 1] : H(a)(t) \stackrel{\text{def}}{=} h(a)(1 - t).$$

This defines a map $H : A \rightarrow \text{Cyl}_G(B)$. For each $a \in A$, we note that $H(a)$ is simply the precomposition of $t \mapsto 1 - t$ on $[0, 1]$ and the map $t \mapsto h(a)(t)$. Since both of these maps are continuous, we see that H maps into $\text{Cyl}_G(B)$. The fact that H is an equivariant $\star$ -homomorphism follows from the fact that h is an equivariant $\star$ -homomorphism. Finally, the fact that this works follows by direct computation^a.

3. Let $f_1 \simeq_h f_2$ and $f_2 \simeq_h f_3$, with equivariant $\star$ -homomorphisms $h_1 : A \rightarrow \text{Cyl}_G(B)$ and $h_2 : A \rightarrow \text{Cyl}_G(B)$ implementing the respective homotopies. Now, let $a \in A$ and let us define the following map:

$$H(a) : [0, 1] \rightarrow A, \quad t \mapsto H(a)(t) \stackrel{\text{def}}{=} \begin{cases} h_1(a)(2t) & \text{if } 0 \leq t < \frac{1}{2}, \\ h_2(a)(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$

This is a function which is continuous everywhere except possibly at $t = \frac{1}{2}$; it is continuous even at that point because the left and right limits of $H(a)$ exist at $t = \frac{1}{2}$ & they coincide. So, we can define a map:

$$H : A \rightarrow \text{Cyl}_G(A), \quad a \mapsto H(a).$$

Due to the explicit definition of H , we see that it must be a $\star$ -homomorphism. Moreover, it gives precisely a homotopy from f_1 to f_3 which passes through f_2 ; this follows from its explicit definition.

This completes the proof.

^aOver here, we use the fact that h is a homotopy from f_1 to f_2 .

Proof. 2.6. This is clear from the definition of isomorphism and the reflexivity of homotopy.

Proof. 2.7. We need to show that $\text{Id}_{\text{Cone}_G(A)}$ is homotopic to 0. For each $t \in [0, 1]$, we an equivariant $\star$ -homomorphism:

$$\phi_t : \text{Cone}_G(A) \rightarrow \text{Cone}_G(A).$$

Let $f \in \text{Cone}_G(A)$. Then, we will define a function $\phi_t(f) \in \text{Cone}_G(A)$ as follows:

$$\forall s \in [0, 1] : \phi_t(f)(s) \stackrel{\text{def}}{=} f(ts).$$

It is clear that $\phi_t(f)$ is continuous since it is just a composition of two continuous functions and we see that:

$$\phi_t(f)(0) = f(t \cdot 0) = f(0) = 0,$$

so ϕ_t is a well-defined function into $\text{Cone}_G(A)$. Now, we need to check that it works.

1. Note that $\phi_0(f)(s) = f(s)$ for every $s \in [0, 1]$. That is, $\phi_0(f) = f$ so $\phi_0 = \text{Id}_{\text{Cone}_G(A)}$.

2. Note that $\phi_1(f)(s) = (1 - 1)f(s) = 0$. That is, $\phi_1(f) = 0$, so $\phi_1 = 0$.

3. Let us check that each ϕ_t is a $\star$ -homomorphism.

(a) Let $\lambda \in \mathbb{C}$ and $f, g \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \forall s \in [0, 1] : \phi_t(\lambda f + g)(s) &= (\lambda f + g)(ts) \\ &= (\lambda f)(ts) + g(ts) \\ &= \lambda f(ts) + g(ts) \\ &= \lambda \phi_t(f)(s) + \phi_t(g)(s) \\ &= (\lambda \phi_t(f) + \phi_t(g))(s), \end{aligned}$$

from which we conclude that ϕ_t is linear.

(b) Let $f, g \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \forall s \in [0, 1] : \phi_t(fg)(s) &= (fg)(ts) \\ &= f(ts)g(ts) \\ &= \phi_t(f)(s)\phi_t(g)(s) \\ &= (\phi_t(f)\phi_t(g))(s), \end{aligned}$$

which proves that ϕ_t preserves products.

(c) Let $f \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \forall s \in [0, 1] : \phi_t(f^*)(s) &= f^*(ts) \\ &= \overline{f(ts)} \\ &= \overline{\phi_t(f)(s)} \\ &= \phi_t(f)^*(s), \end{aligned}$$

which proves that ϕ_t preserves involutions.

If A was just a G -Banach algebra, then we would also have to show continuity of each ϕ_t . However, this is easy:

$$\begin{aligned} \forall f \in \text{Cone}_G(A) : \|\phi_t(f)\| &= \sup_{s \in [0, 1]} |\phi_t(f)(s)| \\ &= \sup_{s \in [0, 1]} (1 - t)|f(s)| \\ &= (1 - t)\|f\|, \end{aligned}$$

which proves the result.

4. Let $f \in \text{Cone}_G(A)$ and define:

$$H_f : [0, 1] \rightarrow \text{Cone}_G(A), \quad t \mapsto H_f(t) \stackrel{\text{def}}{=} \phi_t(f).$$

We need to check that this is continuous. Let $t, s \in [0, 1]$ and consider the following estimate:

$$\begin{aligned} \|H_f(t) - H_f(s)\| &= \|\phi_t(f) - \phi_s(f)\| \\ &= \sup_{u \in [0, 1]} \|(1 - t)f(u) - (1 - s)f(u)\| \\ &\leq \sup_{u \in [0, 1]} |t - s| \cdot \|f(u)\| \\ &\leq |t - s| \cdot \|f\|. \end{aligned}$$

This calculation proves continuity.

This completes the proof.

Proof. 2.8. We will prove each statement one-by-one.

1. We have the following formal calculation:

$$\begin{aligned} g_1 &\simeq_h g_1 \circ \text{Id}_B \\ &\simeq_h g_1 \circ (f \circ g_2) \\ &\simeq_h (g_1 \circ f) \circ g_2 \\ &\simeq_h \text{Id}_A \circ g_2 \\ &\simeq_h g_2, \end{aligned}$$

which proves the desired result.

2. We have the following computation:

$$\begin{aligned} g_1 &\simeq_h \text{Id}_A \circ g_1 \\ &\simeq_h (g_2 \circ f) \circ g_1 \\ &\simeq_h g_2 \circ (f \circ g_1) \\ &\simeq_h g_2 \circ \text{Id}_B \\ &\simeq_h g_2, \end{aligned}$$

as was desired. To prove the last statement, we note that:

$$\begin{aligned} g_1 \circ f &\simeq_h g_2 \circ f \\ &\simeq_h \text{Id}_A, \end{aligned}$$

so we're done.

3. There are three cases to deal with:

- (a) Let f and g be homotopy equivalences. Indeed, the point is that $g \circ f$ has a left and right homotopy inverse. So, $g \circ f$ has a homotopy inverse.
- (b) Let f and $g \circ f$ be homotopy equivalences. Let t_1 be the homotopy inverse of f and let t_2 be the homotopy inverse of $g \circ f$. This means that $f \circ t_2$ is the right homotopy inverse of g . On the other hand, $t_2 \circ g$ is the left homotopy inverse of f . So, $t_2 \circ g$ is homotopic to t_1 . Now, we have a calculation:

$$\begin{aligned} (f \circ t_2) \circ g &= f \circ (t_2 \circ g) \\ &\simeq_h f \circ t_1 \\ &\simeq_h \text{Id}, \end{aligned}$$

which means that g has a left homotopy inverse and we're done.

- (c) Let g and $g \circ f$ be homotopy equivalences. Let t_1 be the homotopy inverse of g and let t_2 be the homotopy inverse of $g \circ f$. This means that f has a left homotopy inverse given by $t_1 \circ g$. Moreover, $f \circ t_2$ is the right homotopy inverse of g , so this means that $f \circ t_2 \simeq_h t_1$. Now, we have a calculation:

$$\begin{aligned} f \circ (t_2 \circ g) &= (f \circ t_2) \circ g \\ &\simeq_h t_1 \circ g \\ &\simeq_h \text{Id}, \end{aligned}$$

so f has a right homotopy inverse and we're done.

This completes the proof of property (3).

Proof. 2.9. By construction, the unitized family of maps still is a family of $\star$ -homomorphisms so it is sufficient to show the point-norm continuity condition. Let $x = a + \lambda 1_{A^\sim} \in A^\sim$ and define the following map:

$$H_x : [0, 1] \rightarrow B^\sim, \quad t \mapsto H_x(t) \stackrel{\text{def}}{=} \phi_t^\sim(x) = \phi_t(a) + \lambda 1_{B^\sim}.$$

Now, a direct calculation which compares $H_x(t)$ with $H_x(s)$ for $t, s \in [0, 1]$ will show that the continuity of H_x reduces to the continuity of the map $t \mapsto H_t(a)$ and we know that this is continuous, so we're done.

Section 3

Proof. 3.1. This involves just reversing the arrows in the diagram associated with the retraction and that doesn't change anything.

Proof. 3.2. We'll verify each statement one-by-one.

1. Consider the following diagram of three composable arrows:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies are isomorphisms. This implies that g has a right & left inverse so it is an isomorphism. This means that both h and f are isomorphisms. This implies that $h \circ g \circ f$ is an isomorphism and we're done.

2. Consider the following digram of two composable arrows:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

where any two of f, g and $g \circ f$ are in W . Now, we have to deal with a few cases:

- (a) Suppose that $f, g \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow y \rightarrow y \rightarrow z,$$

where the first map is f , the second map is the identity and the third is g . In that case, we see that $\text{Id}_y \circ f$ and $g \circ \text{Id}_y$ belong to W , so we can use the 2-out-of-6 property to conclude that $g \circ \text{Id}_y \circ f = g \circ f \in W$.

- (b) Let $f, g \circ f \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow x \rightarrow y \rightarrow z,$$

where the first map is the identity, the second is f and the third is g . Now, we note that $f \circ \text{Id}_x \in W$ and $g \circ f \in W$ so the 2-out-of-6 property allows us to conclude that $g \in W$ (among other facts, of course).

- (c) Let $g, g \circ f \in W$. Then, we consider the following diagram of composable arrows:

$$x \rightarrow y \rightarrow z \rightarrow z,$$

where the first map is f , the second is g and the third is the identity. So, we note that $g \circ f \in W$ and $\text{Id}_z \circ g \in W$ so the 2-out-of-6 property lets us conclude that $f \in W$ and we're done.

This completes the proof of the given statement.

3. Let f be an isomorphism. Then, we consider the following diagram of composable arrows:

$$\bullet \rightarrow \bullet \rightarrow \bullet \rightarrow \bullet,$$

where the first arrow is f , the second one is f^{-1} and the third is f . Well, $f^{-1} \circ f$ and $f \circ f$ belong to W since

they are identity arrows. By the 2-out-of-6 property, we conclude $f \in W$ and so does f^{-1} . This completes the overall result.

Proof. 3.3. Let's suppose that we have a diagram:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies in W . Then, we note that we have 6 morphisms:

1. $F(f)$,
2. $F(g)$,
3. $F(h)$,
4. $F(g \circ f) = F(g) \circ F(f)$,
5. $F(h \circ g) = F(h) \circ F(g)$,
6. $F(h \circ g \circ f) = F(h) \circ F(g) \circ F(f)$,

where $F(g \circ f)$ and $F(h \circ g)$ are isomorphisms. But now, we can apply the 2-out-of-6 property for the isomorphisms in $\mathcal{D}$ to conclude that all 6 of the morphisms listed above are isomorphisms. But this implies that $f, g, h, h \circ g \circ f \in W$ so W satisfies the 2-out-of-6 property.

Proof. 3.4. The localization functor with respect to W satisfies the hypothesis of preceding result, so we're done.

Proof. 3.5. This is clear by the definition, since taking opposites just reverses the directions of the arrows.

Proof. 3.6. Since $\mathcal{C}$ has a final object 0 , we see that it admits empty products. Next, it's enough to show that $\mathcal{C}$ admits binary products so let $x, y \in \mathcal{C}$. Then, we consider the diagram:

$$x \rightarrow 0 \leftarrow y$$

If we take the pullback of this diagram, then we get $x \times y$ by the universal property of the pullback. Indeed, $\mathcal{C}$ is closed under such pullbacks because both maps in this one are fibrations.

Proof. 3.7. Consider the following pullback diagram:

$$\begin{array}{ccc} x & & y \\ \downarrow s & \searrow \pi_2 & \\ x' \times_{y'} y & \xrightarrow{\pi_2} & y \\ \downarrow \pi_1 & & \downarrow p \\ x' & \xrightarrow{\quad} & y' \end{array}$$

i (curved arrow from x to x')

Now, we want to factorize s as follows:

$$\begin{array}{ccc} x & & y \\ \downarrow j & \searrow \pi_2 & \\ z & \xrightarrow{q} & y \\ \downarrow q & & \downarrow p \\ x' \times_{y'} y & \xrightarrow{\pi_2} & y \\ \downarrow \pi_1 & & \downarrow p \\ x' & \xrightarrow{\quad} & y' \end{array}$$

i (curved arrow from x to x')

where $s = q \circ j$, with q being a fibration and j being a weak equivalence. Now, we form the following diagram:

$$\begin{array}{ccc} x & \xrightarrow{j} & z & \xrightarrow{\pi_2 \circ q} & y \\ & \searrow i & \downarrow \pi_1 \circ q & & \downarrow p \\ & & x' & \xrightarrow{\quad} & y' \end{array}$$

Define $t = \pi_1 \circ q$. Well, π_1 and q are fibrations, where π_1 is a fibration because p is a fibration & fibrations are preserved by base extensions. So, t is a fibration. On the other hand, $i = t \circ j$ is a weak equivalence and j is a weak equivalence, so it follows that t is a weak equivalence and we're done.

Proof. 3.8. The key is to use the boundary of the diagram given in the statement of this result and apply the

preceding result repeatedly until we get the desired diagram. We start with the following situation:

$$\begin{array}{ccc} x & \xrightarrow{s^y \circ u} & y^I \\ \downarrow s & & \downarrow (d_0^y, d_1^y) \\ x^I & \xrightarrow{(u \circ d_0, u \circ d_1)} & y \times y \end{array}$$

Since s is a weak equivalence and (d_0^y, d_1^y) is a fibration, we can apply the preceding result. This gives us the diagram:

$$\begin{array}{ccccc} x & \xrightarrow{s^x} & x^J & \xrightarrow{\bar{u}} & y^I \\ & \searrow s & \downarrow t & & \downarrow (d_0^y, d_1^y) \\ & & x^I & \xrightarrow{(u \circ d_0, u \circ d_1)} & y \times y \end{array}$$

where t is a trivial fibration. We re-draw this diagram as follows:

$$\begin{array}{ccccc} & & & & y \\ & & & & \downarrow s^y \\ & & & & y^I \\ x & \xrightarrow{s^x} & x^J & \xrightarrow{\bar{u}} & y^I \\ \downarrow s & & \downarrow (d_0^x, d_1^x) & & \downarrow (d_0^y, d_1^y) \\ x^I & \xrightarrow{(d_0, d_1)} & x \times x & \xrightarrow{u \times u} & y \times y \end{array}$$

(Note: In the original diagram, there is a dashed arrow t from x^J to x^I and a solid arrow u from x to y .)

where we note that (d_0^x, d_1^x) is defined as $(d_0, d_1) \circ t$. This implies that the diagram above commutes as a whole; if we were to remove (d_0^x, d_1^x) , then we simply obtain the diagrams we had previously and those certainly commute.

Proof. 3.9. If f, g are homotopic, then there is a weak equivalence t such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

So, applying the first statement, we see that:

$$\begin{aligned} \gamma(f) \circ \gamma(t) &= \gamma(f \circ t) \\ &= \gamma(g \circ t) \\ &= \gamma(g) \circ \gamma(t), \end{aligned}$$

where we can cancel $\gamma(t)$ since it is an isomorphism. So, it is sufficient for us to prove the first statement. In that case, let y^I be a path space and let $h : x \rightarrow y^I$ be a right-homotopy from f to g . Then, we see that:

$$\begin{aligned} f &= d_0 \circ h, \\ g &= d_1 \circ h. \end{aligned}$$

Now, we have a calculation^a:

$$\begin{aligned} \gamma(f) &= \gamma(d_0 \circ h) \\ &= \gamma(d_0) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1 \circ h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(g). \end{aligned}$$

Now, we need to show that:

$$\gamma(d_0) = \gamma(d_1).$$

By definition of the path space, there is a morphism $s : x \rightarrow x^I$ as part of the data of a path space such that:

$$\begin{aligned} d_0 \circ s &= \text{Id}_x, \\ d_1 \circ s &= \text{Id}_x. \end{aligned}$$

Applying γ to both sides in both equations, we have that:

$$\begin{aligned} \gamma(d_0) \circ \gamma(s) &= \gamma(d_0 \circ s) \\ &= \gamma(\text{Id}_x) \\ &= \gamma(d_1) \circ \gamma(s). \end{aligned}$$

Since γ inverts weak equivalences and s is a weak equivalence, we can cancel $\gamma(s)$ and we're done.

^aThis is justified since d_1 is a trivial fibration so its image in $\mathbf{Ho}(\mathcal{C})$ is an isomorphism.

Proof. 3.10. We simply apply **Claim 3.3** to the quotient functors.

Section 4

Proof. 4.1. This is clear, essentially because both $\mathcal{C}$ and $\mathcal{D}$ are admissible.

Proof. 4.2. We define two obvious maps:

$$\phi : Q \rightarrow I, \quad f \mapsto \phi(f) \stackrel{\text{def}}{=} f(0),$$

$$\psi : I \rightarrow Q, \quad j \mapsto \psi(j),$$

where $\psi(j)$ is the constant function j on $[0, 1]$. These maps are well-defined and we note that:

$$\forall f \in Q : (\psi \circ \phi)(f) = f(0),$$

and:

$$\forall j \in I : (\phi \circ \psi)(j) = j,$$

which means that $\phi \circ \psi = \text{Id}_I$. Now, let $f \in Q$ and define:

$$h(f) : [0, 1] \rightarrow Q, \quad t \mapsto (s \mapsto h(f)(t)(s) = f(ts)).$$

By currying, one can see that $h(f)$ is a continuous map (this can also be proved by hand). So, we have constructed a function $h : Q \rightarrow \text{Cyl}(Q)$ and this will be a $\star$ -homomorphism. In fact, it is a homotopy from $\psi \circ \phi$ to Id_Q and we're done.

Proof. 4.3. By TR2, we can always shift g and h to the left, making them the two morphisms towards the left end of a exact triangle. So, it suffices to prove that $g \circ f = 0$. Well, we have the following diagram which represents a morphism of exact triangles:

$$\begin{array}{ccccccc} x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \\ \text{Id}_x \downarrow & & \downarrow f & & \downarrow & & \downarrow \text{Id}_{x[1]} \\ x & \xrightarrow{f} & y & \xrightarrow{g} & z & \xrightarrow{h} & x[1] \end{array}$$

which implies that $g \circ f = 0$. Over here, we applied TR3 to the diagram above without the arrow from $0 \rightarrow z$.

Proof. 4.4. Let's start with the right diagram in the statement of the result. Now, we shift each row of the right diagram to the left via TR2 and we form a commutative diagram:

$$\begin{array}{ccccccc} y & \longrightarrow & z & \longrightarrow & x[1] & \longrightarrow & \Sigma(y) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ y' & \longrightarrow & z' & \longrightarrow & x[1]' & \longrightarrow & \Sigma(y)' \end{array}$$

with each row being an exact triangle. By TR3, we obtain a morphism $x[1] \rightarrow x[1]'$ such that the diagram above commutes with this extra morphism. Since Σ is an automorphism, we can find a unique morphism $x \rightarrow x'$ such that the aforementioned right diagram commutes. Now, we apply this result to the left diagram after we've shifted the rows of that one to the left by TR3.

Proof. 4.5. We'll start by proving the result for $\mathcal{T}(x, -)$. We start with an exact triangle:

$$x' \xrightarrow{f} y' \xrightarrow{g} z' \xrightarrow{h} x[1]'$$

and consider the sequence:

$$\mathcal{T}(x, x') \xrightarrow{\mathcal{T}(x, f)} \mathcal{T}(x, y') \xrightarrow{\mathcal{T}(x, g)} \mathcal{T}(x, z')$$

Well, this is a sequence of abelian groups and to check exactness, we need to check that:

$$\text{ran}(\mathcal{T}(x, f)) = \ker(\mathcal{T}(x, g)).$$

We'll prove each inclusion separately.

$\subseteq$: Since $g \circ f = 0$, it follows that:

$$0 = \mathcal{T}(x, g \circ f) = \mathcal{T}(x, g) \circ \mathcal{T}(x, f),$$

which means that $\text{ran}(\mathcal{T}(x, f)) \subseteq \ker(\mathcal{T}(x, g))$.

$\supseteq$: For the other inclusion, let $u \in \ker(\mathcal{T}(x, g))$. Then, $u \in \mathcal{T}(x, y')$ and $g \circ u = 0$. Now, we need a $v \in \mathcal{T}(x, x')$ such that $f \circ v = u$. The trick is to fit all of this data in a commutative diagram with a partial morphism between two exact triangles. In this case, the following diagram does the trick:

$$\begin{array}{ccccccc} x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \\ & & \downarrow u & & \downarrow & & \\ x' & \xrightarrow{f} & y' & \xrightarrow{g} & z' & \xrightarrow{h} & x[1]' \end{array}$$

Indeed, this diagram commutes since $g \circ u = 0$. So, now, we use **Claim 4.4** to fill in the diagram above with

a morphism $v : x \rightarrow x'$ so that the diagram commutes. But this means precisely that:

$$u = f \circ u = \mathcal{T}(x, f)(u),$$

as was desired.

This completes the proof.

Proof. 4.6. Let's start with the following claim.

| **Claim.** If a and b are isomorphisms, then c is a isomorphism.

With this last claim, the rest of the result follows easily.

| **Claim.** It's sufficient for us to have proved the previous claim.

Proof. 4.7.

(1) $\Rightarrow$ (2): Consider the following commutative diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & x[1] \\ \downarrow \text{Id}_x & & \downarrow f^{-1} & & \downarrow & & \downarrow \\ x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \end{array}$$

Note that the rows are exact triangles in $\mathcal{T}$. Since the bottom row belongs to the set of exact triangles of $\mathcal{T}$ by assumption and since the set of exact triangles is closed under isomorphisms, we see that the top row must also be an exact triangle.

(2) $\Rightarrow$ (1): Well, we have the following diagram:

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & x[1] \\ \downarrow \text{Id}_x & & & & \downarrow & & \downarrow \\ x & \xrightarrow{\text{Id}_x} & x & \longrightarrow & 0 & \longrightarrow & x[1] \end{array}$$

Now, we use **Claim 4.4** to find a morphism $g : y \rightarrow x$ such that the above is a commutative diagram with g filled into it. This implies that $g \circ f = \text{Id}_x$. Now, consider the following diagram:

$$\begin{array}{ccccccc} y & \longrightarrow & y & \longrightarrow & 0 & \longrightarrow & \Sigma y \\ & & \downarrow & & \downarrow & & \\ x & \xrightarrow{f} & y & \longrightarrow & 0 & \longrightarrow & \Sigma x \end{array}$$

By the aforementioned result, we can find a morphism $h : y \rightarrow x$ such that $f \circ h = \text{Id}_y$. But this means that f is a split epimorphism and a split epimorphism, from which it follows that it is an isomorphism.

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